37. USB 2.0 Full-Speed Module (USBFS)

37.1 Overview

The USB 2.0 Full-Speed module (USBFS) operates as a host or device controller compliant with the Universal Serial Bus (USB) specification revision 2.0. The host controller supports USB 2.0 full-speed and low-speed transfers, and the device controller supports USB 2.0 full-speed transfers. The USBFS has an internal USB transceiver and supports all of the transfer types defined in the USB 2.0 specification

The USBFS has FIFO buffer for data transfers, providing a maximum of 10 pipes. Any endpoint number can be assigned to pipes 1 to 9, based on the peripheral devices or the communication requirements for your system.

Table 37.1 lists the USBFS specifications, Figure 37.1 shows a block diagram, and Table 37.2 lists the I/O pins.

Table 37.1 USBFS specifications

Parameter	Specifications
Features	● USB Device Controller (UDC) and USB 2.0 transceiver supporting host controller, and device controller, and On-The-Go (OTG) functions ● Host and device controller can be switched by software ● Self-power or bus power mode selectable
	Host controller features: ● Full-speed transfer (12 Mbps) and low-speed transfer (1.5 Mbps) ● Automatic scheduling for SOF and packet transmissions ● Programmable intervals for isochronous and interrupt transfers ● Communications with multiple peripheral devices connected through a single hub
	Device controller features: ● Full-speed transfer (12 Mbps)*1 ● Control transfer stage control function ● Device state control function ● Auto response function for SET_ADDRESS request ● SOF interpolation
Supported transfer types	● Control transfer ● Bulk transfer ● Interrupt transfer ● Isochronous transfer
Pipe configuration	● FIFO buffer for USB communication ● Up to 10 pipes selectable, including the Default Control Pipe (DCP) ● Pipes 1 to 9 assignable to any endpoint number
	Transfer conditions specifiable for each pipe: ● Pipe 0: Control transfer with 64-byte single buffer ● Pipes 1 and 2: Selectable to bulk transfer with 64-byte double buffer or isochronous transfer with 256-byte double buffer ● Pipes 3 to 5: Bulk transfer with 64-byte double buffer ● Pipes 6 to 9: Interrupt transfer with 64-byte single buffer
Other features	● Reception end function using transaction count ● Function that changes the BRDY interrupt event notification timing (BFRE) ● Automatic clearing of the FIFO buffer after the data for the pipe specified in the DnFIFO port (n = 0, 1) is read (DCLRM) ● NAK setting function for response PID generated on transfer end (SHTNAK) ● On-chip pull-up and pull-down resistors for D+ and D–
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	Security and Privilege attribution can be set

Note 1. Low-speed transfer (1.5 Mbps) is not supported.

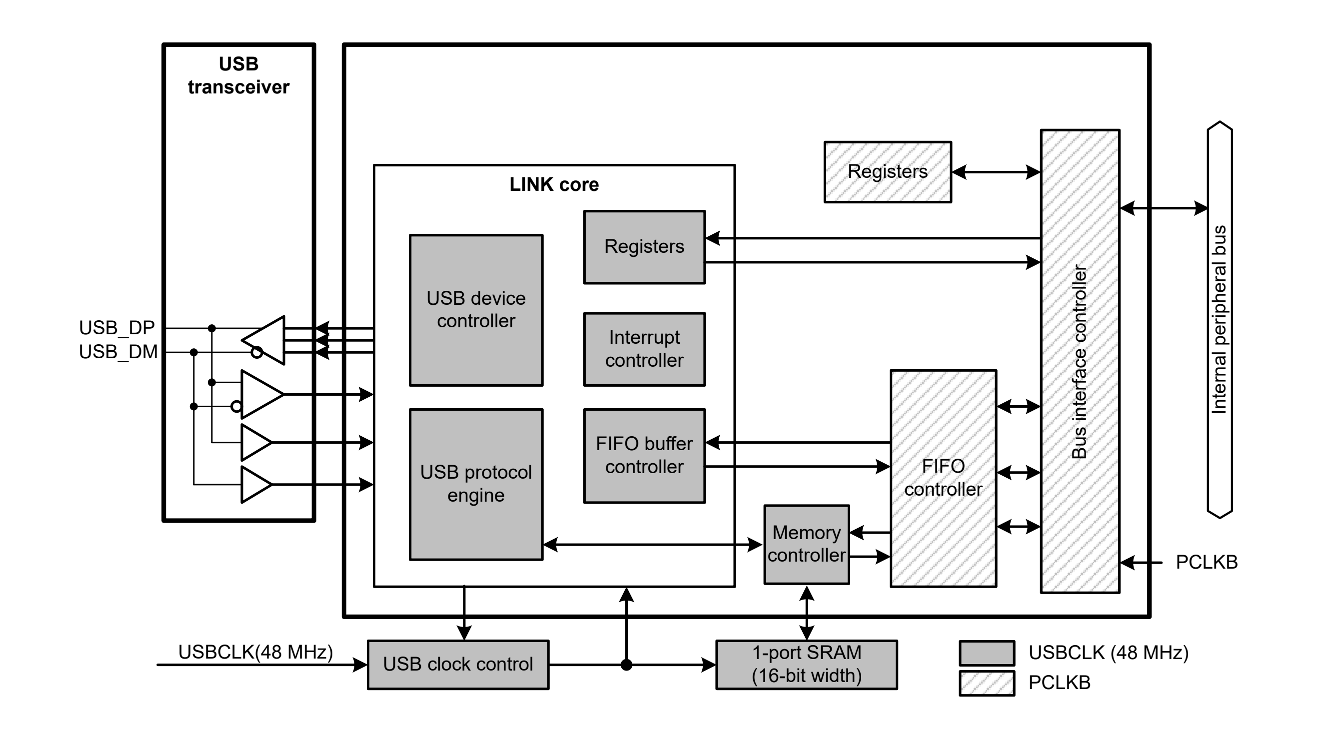


Figure 37.1 USBFS block diagram

Table 37.2 USBFS pin configuration

Function	Pin name	I/O	Description
USBFS	USB_DP	I/O	D+ I/O pin of the USB on-chip transceiver. This pin should be connected to the D+ pin of the USB bus.
	USB_DM	I/O	D– I/O pin of the USB on-chip transceiver. This pin should be connected to the D– pin of the USB bus.
	USB_VBUS	Input	USB cable connection monitor pin. This pin should be connected to VBUS of the USB bus. The VBUS pin status (connected or disconnected) can be detected when the USB module is operating as a function controller.
	USB_EXICEN	Output	Low-power control signal for external power supply (OTG) chip.
	USB_VBUSEN	Output	VBUS (5 V) supply enable signal for external power supply chip.
	USB_OVRCURA, USB_OVRCURB USB_OVRCURA-DS, USB_OVRCURB-DS	Input	External overcurrent detection signals should be connected to these pins. VBUS comparator signals should be connected to these pins when the OTG power supply chip is connected. USB_OVRCURA or USB_OVRCURB pins can be used in Software standby mode or in normal mode. USB_OVRCURA-DS, USB_OVRCURB-DS are dedicated pins that can generate interrupt in Software standby mode or in normal mode as well as for cancel Deep Software Standby mode 1.
	USB_ID	Input	MicroAB connector ID input signal should be connected to this pin during operation in OTG mode.
	VCC_USB	Input	Power supply pins.
	VSS_USB	Input	Ground pins.

37.2 Register Descriptions

37.2.1 SYSCFG : System Configuration Control Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x000

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	SCKE	—	—	—	DCFM	DRPD	DPRPU	—	—	—	USBE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	USBE	USBFS Operation Enable 0: Disable 1: Enable	R/W
2:1	—	These bits are read as 0. The write value should be 0.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	DPRPU	D+ Line Resistor Control 0: Disable line pull-up 1: Enable line pull-up	R/W
5	DRPD	D+/D– Line Resistor Control 0: Disable line pull-down 1: Enable line pull-down	R/W
6	DCFM	Controller Function Select 0: Select device controller 1: Select host controller	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
9:8	—	These bits are read as 0. The write value should be 0.	R/W
10	SCKE	USB Clock Enable 0: Stop clock supply to the USBFS 1: Enable clock supply to the USBFS	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: After writing 1 to the SCKE bit, read it to confirm that it is set to 1.

USBE bit (USBFS Operation Enable)

The USBE bit enables or disables operation of the USBFS.

Changing the USBE bit from 1 to 0 initializes the bits listed in Table 37.3. Only change this bit while the SCKE bit is 1. In host controller mode, this bit must be set to 1 after setting the DRPD bit to 1, eliminating SYSSTS0.LNST[1:0] flags chattering, and confirming that the USB bus state is stable.

Table 37.3 Registers initialized by writing 0 to the SYSCFG.USBE bit

Selected function	Register	Bit	Remarks
Device controller	SYSSTS0	LNST[1:0]	Value is saved in host controller mode
	DVSTCTR0	RHST[2:0]	—
	INTSTS0	DVSQ[2:0]	Value is saved in host controller mode
	USBADDR	USBADDR[6:0]	Value is saved in host controller mode
	USBREQ	BREQUEST[7:0], BMREQUESTTYPE[7:0]	Value is saved in host controller mode
	USBVAL	WVALUE[15:0]	Value is saved in host controller mode
	USBINDX	WINDEX[15:0]	Value is saved in host controller mode
	USBLENG	WLENTUH[15:0]	Value is saved in host controller mode
Host controller	DVSTCTR0	RHST[2:0]	—
	FRMNUM	FRNM[10:0]	Value is saved in device controller mode

DPRPU bit (D+ Line Resistor Control)

The DPRPU bit enables or disables pulling up the D+ line in device controller mode.

When the DPRPU bit is set to 1 in device controller mode, the USBFS pulls up the D+ line to notify the USB host that it attached. Changing the DPRPU bit from 1 to 0 releases the pull-up, thereby notifying the USB host that it detached.

Set this bit to 1 in device controller mode and to 0 in host controller mode.

DRPD bit (D+/D– Line Resistor Control)

TheDRPD bit enables or disables pulling down D+ and D– lines in host controller mode.

Set this bit to 1 in host controller mode and to 0 in device controller mode.

DCFM bit (Controller Function Select)

The DCFM bit selects the host or device function of the USBFS.

Only change this bit when the DPRPU and DRPD bits are both 0.

SCKE bit (USB Clock Enable)

The SCKE bit stops or enables the 48-MHz clock supply to the USBFS.

When this bit is 0, only SYSCFG is permitted to be read from and written to; the other registers related to the USB should not be read from or written to.

37.2.2 SYSSTS0 : System Configuration Status Register 0

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x004

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVCMON[1:0]		—	—	—	—	—	—	—	HTACT	SOFEA	—	—	IDMON	LNST[1:0]	
Value after reset:	x	x	0	0	0	0	0	0	0	0	0	0	0	x	0	0

Bit	Symbol	Function	R/W
1:0	LNST[1:0]	USB Data Line Status Monitor Indicates the status of the USB data lines, see Table 37.4	R
2	IDMON	External ID0 Input Pin Monitor 0: USB_ID pin is low 1: USB_ID pin is high	R
4:3	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
5	SOFEA	Active Monitor When the Host Controller Is Selected 0: SOF output stopped 1: SOF output operating	R
6	HTACT	USB Host Sequencer Status Monitor 0: Host sequencer completely stopped 1: Host sequencer not completely stopped	R
13:7	—	These bits are read as 0.	R
15:14	OVCMON[1:0]	External USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB or USB_OVRCURB-DS Input Pin Monitor OVCMON[1] indicates the USB_OVRCURA or USB_OVRCURA-DS pin status. OVCMON [0] indicates the USB_OVRCURB or USB_OVRCURB-DS pin status.	R

Note: S-TYPE-3, P-TYPE-3

Note: The values of the OVCMON[1:0] and IDMON bits depend on the status of the USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB or USB_OVRCURB-DS and USB_ID pins.

LNST[1:0] bits (USB Data Line Status Monitor)

The LNST[1:0] bits indicate the state of the USB data lines, D+ and D–. For details, see Table 37.4.

In device controller mode, read the LNST[1:0] bits after connection processing (SYSCFG.DPRPU bit = 1). In host controller mode, read them after enabling pull-down of the lines (SYSCFG.DRPD bit = 1).

Table 37.4 Status of the USB data bus lines (D+ and D–)

LNST[1:0] bits	During full-speed operation	During low-speed operation
00b	SE0	SE0
01b	J-State	K-State
10b	K-State	J-State
11b	SE1	SE1

SOFEA bit (Active Monitor When the Host Controller Is Selected)

The SOFEA bit is used in host controller mode to check whether the output of the last SOF is complete when the USBFS is suspended because of a 0 setting to the DVSTCTR0.UACT bit.

In host controller mode, check that both the HTACT and SOFEA bits are 0 before setting the SYSCFG.USBE bit to 0 to stop the USBFS or setting the SYSCFG.SCKE bit to 0 to stop the clock signal supply during communication.

HTACT bit (USB Host Sequencer Status Monitor)

The HTACT bit is set to 0 when the host sequencer of the USBFS is completely stopped.

In host controller mode, check that the HTACT bit is 0 before setting the DVSTCTR0.UACT bit to 0 to place the USBFS in the suspended state or setting the SYSCFG.SCKE bit to 0 to stop the clock signal supply during communication.

OVCMON[1:0] bits (External USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB or USB_OVRCURB-DS Input Pin Monitor)

The OVCMON[1:0] bits indicate the status of the overcurrent signals from an external power supply IC.

37.2.3 DVSTCTR0 : Device State Control Register 0

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x008

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	HNPBTOA	EXICEN	VBUSEN	WKUP	RWUPE	USBRST	RESUME	UACT	—	RHST[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	RHST[2:0]	USB Bus Reset Status 0 0 0: In host controller mode: Communication speed indeterminate (powered state or no connection) In device controller mode: Communication speed indeterminate 0 0 1: In host controller mode: Low-speed connection In device controller mode: USB bus reset in progress 0 1 0: In host controller mode: Full-speed connection In device controller mode: USB bus reset in progress or full-speed connection 0 1 1: Setting prohibited Others: In host controller mode: USB bus reset in progress In device controller mode: Setting prohibited	R
3	—	This bit is read as 0. The write value should be 0.	R/W
4	UACT	USB Bus Enable 0: Disable downstream port (disable SOF transmission) 1: Enable downstream port (enable SOF transmission)	R/W
5	RESUME	Resume Output 0: Do not output resume signal 1: Output resume signal	R/W
6	USBRST	USB Bus Reset Output 0: Do not output USB bus reset signal 1: Output USB bus reset signal	R/W
7	RWUPE	Wakeup Detection Enable 0: Disable downstream port remote wakeup 1: Enable downstream port remote wakeup	R/W
8	WKUP	Wakeup Output 0: Do not output remote wakeup signal 1: Output remote wakeup signal	R/W
9	VBUSEN	USB_VBUSEN Output Pin Control 0: Output low on external USB_VBUSEN pin 1: Output high on external USB_VBUSEN pin	R/W
10	EXICEN	USB_EXICEN Output Pin Control 0: Output low on external USB_EXICEN pin 1: Output high on external USB_EXICEN pin	R/W
11	HNPBTOA	Host Negotiation Protocol (HNP) Control Use this bit when switching from device B to device A in OTG mode. If the HNPBTOA bit is 1, the internal function control remains in the Suspend state until the HNP processing ends even if SYSCFG.DPRPU = 0 or SYSCFG.DCFM = 1.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The USBFS controller does not support low-speed connections in device controller mode. When this value is read, abnormal connection processing must be executed in higher level application software.

RHST[2:0] bits (USB Bus Reset Status)

RHST[2:0] bits indicate the status of the USB bus reset.

In host controller mode, writing 1 to the USBRST bit causes the RHST[2:0] bits to set to 100b. When 0 is written to the USBRST bit and the USBFS ends the SE0 state, the RHST[2:0]bits update to a new value.

In device controller mode, if the USBFS detects a USB bus reset, the RHST[2:0] bits indicate 010b if the DPRPU bit is 1, and a DVST interrupt is generated.

UACT bit (USB Bus Enable)

When set to 1 in host controller mode, the UACT bit enables USB bus operation by controlling SOF packet transmission to the USB bus in addition to data and reception. The USBFS starts SOF packet output within one frame period after the UACT bit is set to 1. When UACT is set to 0, the USBFS enters the idle state after the SOF packet output.

The USBFS sets the UACT bit to 0 on any of the following conditions:

- A DTCH interrupt is detected during communication (when UACT = 1)

- An EOFERR interrupt is detected during communication (when UACT = 1)

Always write 1 to the UACT bit at the end of the USB bus reset processing (writing 0 to the USBRST bit) or at the end of resume processing from the suspended state (writing 0 to the RESUME bit).

In device controller mode, always set this bit to 0.

RESUME bit (Resume Output)

The RESUME bit controls the resume signal output in host controller mode.

When this bit is set to 1, the USBFS drives the USB port to the K-state and outputs the resume signal. The USBFS sets the bit to 1 on detection of a remote wakeup signal while the RWUPE bit is 1 and in the USB Suspend state.

The USBFS continues outputting the K-state while the RESUME bit is 1, until the bit is cleared to 0 by software. The RESUME bit must be 1 (resume period) for the time defined in the USB 2.0 specification. Only set this bit to 1 while the interface is in the Suspend state. Write 1 to the UACT bit simultaneously with the end of the resume processing (writing 0 to the RESUME bit).

Always set this bit to 0 in device controller mode.

USBRST bit (USB Bus Reset Output)

The USBRST bit controls the output of the USB bus signal in host controller mode. When this bit set to 1, the USBFS drives the USB port to the SE0 state to reset the USB bus. The USBFS continues outputting SE0 while the USBRST bit is 1, until the bit is cleared to 0 by software. The USBRST bit must be 1 (USB bus reset period) for the time defined in the USB 2.0 specification. Writing 1 to the USBRST bit during communication (UACT = 1) or during resume processing (RESUME = 1) prevents the USBFS from starting USB bus reset processing until both the UACT and RESUME bits become 0. Write 1 to the UACT bit simultaneously with the end of the USB bus reset processing (writing 0 to the USBRST bit).

Always set this bit to 0 in device controller mode.

RWUPE bit (Wakeup Detection Enable)

The RWUPE bit enables or disables remote wakeup signals (resume signals) from downstream peripheral devices in host controller mode. When this bit is set to 1, the USBFS detects a remote wakeup signal (K-state for 2.5 µs) from a downstream peripheral device, and performs resume processing, driving the K-state. When the RWUPE bit is set to 0, the USBFS ignores remote wakeup signals (K-states) from peripheral devices connected to the USB port.

Do not stop the internal clock when the RWUPE bit is 1, even in the Suspend state (SYSCFG.SCKE bit must be set to 1).

Always set this bit to 0 in device controller mode.

WKUP bit (Wakeup Output)

The WKUP bit enables or disables remote wakeup signals (resume signals) to the USB bus in device controller mode.

The USBFS controls the output timing of the remote wakeup signals. When this bit is set to 1, the USBFS clears it to 0 after outputting the K-state for 10 ms. The USB 2.0 specification specifies that the USB bus idle state must be kept for 5 ms or longer before a remote wakeup signal is sent. If the USBFS writes 1 to the WKUP bit immediately after detecting the Suspend state, the K-state is output after 2 ms.

Only write 1 to the WKUP bit when the device is in the Suspend state (INTSTS0.DVSQ[2:0] = 1xxb) and the USB host enables the remote wakeup signal. Do not stop the internal clock while this bit is 1, even in the Suspend state (SYSCFG.SCKE bit must be set to 1).

Always set this bit to 0 in host controller mode.

HNPBTOA bit (Host Negotiation Protocol (HNP) Control)

TheHNPBTOA bit is used when switching from device B to device A while in OTG mode.

If the HNPBTOA bit is 1, the internal function control maintains the Suspend state until HNP processing ends, even if the SYSCFG.DPRPU bit is set to 0 or the SYSCFG.DCFM bit is set to 1. Resume interrupts (RESM) are not generated even if a falling edge of D+ is detected.

After this bit is set to 1, the HNP processing ends when a host attach event is detected, because of a pull-up by the initiating party, or the HNPBTOA bit is cleared to 0 by software because the HNP processing times out.

37.2.4 CFIFO/CFIFOL : CFIFO Port Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x014

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	FIFOPORT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	FIFOPORT[15:0]*1	FIFO Port Read receive data from the FIFO buffer or write transmit data to the FIFO buffer by accessing these bits	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The valid bits depend on the MBW settings (CFIFOSEL.MBW) and BIGEND settings (CFIFOSEL.BIGEND) in the associated port selection register. See Table 37.5 and Table 37.6.

Three FIFO ports are available:

- CFIFO
- D0FIFO
- D1FIFO

Each FIFO port is configured with:

- A port register (CFIFO, D0FIFO, or D1FIFO) that handles reading of data from the FIFO buffer and writing of data to the FIFO buffer
- A port selection register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL) that selects the pipe assigned to the FIFO port
- A port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR)

Each FIFO port has the following constraints:

- Access to the FIFO buffer for DCP control transfers is through the CFIFO port
- Access to the FIFO buffer for DMA or DTC transfers is through the D0FIO or D1FIFO port
- The D0FIFO and D1FIFO ports can also be accessed by the CPU
- When using functions specific to the FIFO port, such as the DMA or DTC transfer function, you cannot change the pipe number selected in the CURPIPE[3:0] bits of the port selection register
- Registers configuring a FIFO port do not affect other FIFO ports
- The same pipe must not be assigned to two or more FIFO ports
- There are two FIFO buffer states, one giving access rights to the CPU and the other to the serial interface engine (SIE). When the SIE has access rights, the FIFO buffer cannot be accessed by the CPU

FIFOPORT[15:0] bits (FIFO Port)

When the FIFOPORT[15:0] bit is accessed, the USBFS reads the received data from the FIFO buffer or writes the transmit data to the FIFO buffer. The FIFO port register can be accessed only when the FRDY bit in the associated port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR) is 1.

The valid bits in the FIFO port register depend on the MBW and BIGEND settings in the port selection register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL). See Table 37.5 and Table 37.6.

Table 37.5 Endian operation in 16-bit access

CFIFOSEL.BIGEND bit	Bits [15:8]	Bits [7:0]
0	N + 1 data	N + 0 data
1	N + 0 data	N + 1 data

Table 37.6 Endian operation in 8-bit access

CFIFOSEL.BIGEND bit	Bits [15:8]	Bits [7:0]
0	Access prohibited*1	N + 0 data
1	Access prohibited*1	N + 0 data

Note 1. Writing to or reading from these areas is not allowed.

37.2.5 DnFIFO/DnFIFOL : DnFIFO Port Register (n = 0, 1)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x018 + 0x4 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	FIFOPORT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	FIFOPORT[15:0]*1	FIFO Port Read receive data from the FIFO buffer or write transmit data to the FIFO buffer by accessing these bits	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The valid bits depend on the MBW settings (CFIFOSEL.MBW, D0FIFOSEL.MBW, and D1FIFOSEL.MBW) and BIGEND settings (CFIFOSEL.BIGEND, D0FIFOSEL.BIGEND, and D1FIFOSEL.BIGEND) in the associated port selection register. See Table 37.7 and Table 37.8.

Three FIFO ports are available:

- CFIFO
- D0FIFO
- D1FIFO

Each FIFO port is configured with:

- A port register (CFIFO, D0FIFO, or D1FIFO) that handles reading of data from the FIFO buffer and writing of data to the FIFO buffer
- A port selection register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL) that selects the pipe assigned to the FIFO port
- A port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR)

Each FIFO port has the following constraints:

- Access to the FIFO buffer for DCP control transfers is through the CFIFO port
- Access to the FIFO buffer for DMA or DTC transfers is through the D0FIO or D1FIFO port
- The D0FIFO and D1FIFO ports can also be accessed by the CPU
- When using functions specific to the FIFO port, such as the DMA or DTC transfer function, you cannot change the pipe number selected in the CURPIPE[3:0] bits of the port selection register
- Registers configuring a FIFO port do not affect other FIFO ports
- The same pipe must not be assigned to two or more FIFO ports
- There are two FIFO buffer states, one giving access rights to the CPU and the other to the serial interface engine (SIE). When the SIE has access rights, the FIFO buffer cannot be accessed by the CPU

FIFOPORT[15:0] bits (FIFO Port)

When the FIFOPORT bit is accessed, the USBFS reads the received data from the FIFO buffer or writes the transmit data to the FIFO buffer. The FIFO port register can be accessed only when the FRDY bit in the associated port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR) is 1.

The valid bits in the FIFO port register depend on the MBW and BIGEND settings in the port selection register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL). See Table 37.7 and Table 37.8.

Table 37.7 Endian operation in 16-bit access

CFIFOSEL.BIGEND bit D0FIFOSEL.BIGEND bit D1FIFOSEL.BIGEND bit	Bits [15:8]	Bits [7:0]
0	N + 1 data	N + 0 data
1	N + 0 data	N + 1 data

Table 37.8 Endian operation in 8-bit access

CFIFOSEL.BIGEND bit D0FIFOSEL.BIGEND bit D1FIFOSEL.BIGEND bit	Bits [15:8]	Bits [7:0]
0	Access prohibited*1	N + 0 data
1	Access prohibited*1	N + 0 data

Note 1. Writing to or reading from these areas is not allowed.

37.2.6 CFIFOSEL : CFIFO Port Select Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x020

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCNT	REW	—	—	—	MBW	—	BIGEND	—	—	ISEL	—	CURPIPE[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	CFIFO Port Access Pipe Specification 0x0: Default Control Pipe 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	ISEL	CFIFO Port Access Direction When DCP Is Selected 0: Select reading from the FIFO buffer 1: Select writing to the FIFO buffer	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	BIGEND	CFIFO Port Endian Control 0: Little endian 1: Big endian	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
10	MBW	CFIFO Port Access Bit Width 0: 8-bit width 1: 16-bit width	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
14	REW	Buffer Pointer Rewind 0: Do not rewind buffer pointer 1: Rewind buffer pointer	W*1
15	RCNT	Read Count Mode 0: The DTLN[8:0] bits (CFIFOCTR.DTLN[8:0], D0FIFOCTR.DTLN[8:0], D1FIFOCTR.DTLN[8:0]) are cleared when all receive data is read from the CFIFO. In double buffer mode, the DTLN[8:0] value is cleared when all data is read from only a single plane. 1: The DTLN[8:0] bits are decremented each time the receive data is read from the CFIFO.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only 0 can be read.

Do not specify the same pipe number in the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers. When the CURPIPE[3:0] bits in the D0FIFOSEL and D1FIFOSEL registers are set to 0000b, no pipe is selected.

Do not change the pipe number while DMA or DTC transfer is enabled.

CURPIPE[3:0] bits (CFIFO Port Access Pipe Specification)

The CURPIPE[3:0] bits specify the pipe number to use for reading or writing data through the CFIFO port. After writing to these bits, read them to check that the written value agrees with the read value before proceeding to the next process. Do not set the same pipe number to the CURPIPE[3:0] bits in CFIFOSEL, D0FIFOSEL, and D1FIFOSEL.

During FIFO buffer access, even when an attempt is made to change the CURPIPE[3:0] setting, the current access setting is retained until access is complete.

ISEL bit (CFIFO Port Access Direction When DCP Is Selected)

After writing a new value to the ISEL bit with the DCP as the selected pipe, read the ISEL bit to check that the written value agrees with the read value before proceeding to the next process. Set the ISEL and CURPIPE[3:0] bits simultaneously.

MBW bit (CFIFO Port Access Bit Width)

The MBW bit specifies the bit width for accessing the CFIFO port.

When the selected pipe is receiving, set the CURPIPE[3:0] and MBW bits simultaneously. After a write to these bits starts a data read from the FIFO buffer, do not change the MBW bit until all of the data is read.

When the selected pipe is transmitting, the bit width cannot be changed from 8-bit to 16-bit while data is being written to the FIFO buffer.

An odd number of bytes can also be written through byte-access control even when 16-bit width is selected.

REW bit (Buffer Pointer Rewind)

The REW bit specifies whether to rewind the buffer pointer.

When the selected pipe is receiving, setting this bit to 1 while the FIFO buffer is being read allows re-reading of the FIFO buffer from the first data. In double buffering, this setting enables re-reading of the currently-read FIFO buffer plane from the first entry.

Do not set this bit to 1 while simultaneously changing the CURPIPE[3:0] bits. Before setting the REW bit to 1, be sure to check that the FRDY bit is 1.

To rewrite to the FIFO buffer from the first data for the transmitting pipe, use the BCLR bit.

37.2.7 DnFIFOSEL : DnFIFO Port Select Register (n = 0, 1)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x028 + 0x4 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCNT	REW	DCLRM	DREQE	—	MBW	—	BIGEND	—	—	—	—	CURPIPE[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	FIFO Port Access Pipe Specification 0x0: Default Control Pipe 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	BIGEND	FIFO Port Endian Control 0: Little endian 1: Big endian	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
10	MBW	FIFO Port Access Bit Width 0: 8-bit width 1: 16-bit width	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
12	DREQE	DMA/DTC Transfer Request Enable 0: Disable DMA/DTC transfer request 1: Enable DMA/DTC transfer request	R/W
13	DCLRM	Auto Buffer Memory Clear Mode Accessed after Specified Pipe Data is Read 0: Disable auto buffer clear mode 1: Enable auto buffer clear mode	R/W
14	REW	Buffer Pointer Rewind 0: Do not rewind buffer pointer 1: Rewind buffer pointer	W
15	RCNT	Read Count Mode 0: Clear DTLN[8:0] bits in (CFIFOCTR.DTLN[8:0], D0FIFOCTR.DTLN[8:0], D1FIFOCTR.DTLN[8:0]) when all receive data is read from DnFIFO (after read of a single plane in double buffer mode) 1: Decrement DTLN[8:0] bits each time receive data is read from DnFIFO	R/W

Note: S-TYPE-3, P-TYPE-3

The same pipe must not be specified in the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers. When the CURPIPE[3:0] bits in the D0FIFOSEL and D1FIFOSEL registers are set to 0000b, no pipe is selected. The pipe number must not be changed while DMA or DTC transfer is enabled.

CURPIPE[3:0] bits (FIFO Port Access Pipe Specification)

The CURPIPE[3:0] bits specify the pipe number to use for reading or writing data through the DnFIFO port. After writing to these bits, read them to check that the written value agrees with the read value before proceeding to the next process. Do not set the same pipe number to the CURPIPE[3:0] bits in CFIFOSEL, D0FIFOSEL, and D1FIFOSEL.

During FIFO buffer access, even when an attempt is made to change the CURPIPE[3:0] setting, the current access setting is retained until access is complete.

MBW bit (FIFO Port Access Bit Width)

The MBW bit specifies the bit width for accessing the DnFIFO port.

When the selected pipe is receiving, after a write to these bits starts a data read from the FIFO buffer, do not change the MBW bit until all of the data is read. Set the CURPIPE[3:0] and MBW bits simultaneously.

When the selected pipe is transmitting, the bit width cannot be changed from 8-bit to 16-bit while data is being written to the FIFO buffer.

An odd number of bytes can also be written through byte-access control even when 16-bit width is selected.

DREQE bit (DMA/DTC Transfer Request Enable)

The DREQE bit enables or disables issuing of DMA or DTC transfer requests. To enable DMA or DTC transfer requests, set this bit to 1 after setting the CURPIPE[3:0] bits. To change the CURPIPE[3:0] setting, first set this bit to 0.

DCLRM bit (Auto Buffer Memory Clear Mode Accessed after Specified Pipe Data is Read)

The DCLRM bit enables or disables automatic FIFO buffer clearing after data in the selected pipe is read.

When this bit is set to 1, on receiving a zero-length packet while the FIFO buffer assigned to the selected pipe is empty, or when reading of a received short packet is complete while the PIPECFG.BFRE bit is 1, the USBFS sets the BCLR bit in the FIFO port control register to 1.

When using the USBFS with the SOFCFG.BRDYM bit set to 1, set this bit to 0.

REW bit (Buffer Pointer Rewind)

The REW bit specifies whether to rewind the buffer pointer.

When the selected pipe is receiving, setting this bit to 1 while the FIFO buffer is being read allows re-reading of the FIFO buffer from the first data. In double buffering, this setting enables re-reading of the currently-read FIFO buffer plane from the first entry.

Do not set this bit to 1 while simultaneously changing the CURPIPE[3:0] bits. Before setting the bit to 1, be sure to check that the FRDY bit is 1.

To rewrite to the FIFO buffer from the first data for the transmitting pipe, use the BCLR bit.

RCNT bit (Read Count Mode)

The RCNT bit specifies the read mode for the value in the D0FIFOCTL.DTLN bit and D1FIFOCTL.DTLN bit. When accessing DnFIFO with the PIPECFG.BFRE bit set to 1, set the RCNT bit to 0.

37.2.8 CFIFOCTR : CFIFO Port Control Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x022

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BVAL	BCLR	FRDY	—	—	—	—	DTLN[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	DTLN[8:0]	Receive Data Length Indicates the receive data length. The meaning of the values differs depending on the RCNT bit setting in the port select register. For details, see the description of the DTLN[8:0] bits.	R
12:9	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
13	FRDY	FIFO Port Ready 0: FIFO port access disabled 1: FIFO port access enabled	R
14	BCLR	CPU Buffer Clear 0: No operation 1: Clear FIFO buffer on the CPU side	W
15	BVAL	Buffer Memory Valid Flag 0: Invalid (writing 0 has no effect) 1: Writing ended	R/W

Note: S-TYPE-3, P-TYPE-3

The CFIFOCTR, D0FIFOCTR, and D1FIFOCTR registers correspond to the CFIFO, D0FIFO, and D1FIFO buffers.

DTLN[8:0] bits (Receive Data Length)

The DTLN[8:0] bits indicate the length of the receive data.

While the FIFO buffer is being read, the DTLN[8:0] bits indicate different values depending on the DnFIFOSEL.RCNT bit (n = 0, 1), as follows:

- RCNT = 0

  The USBFS sets the DTLN[8:0] bits to indicate the length of the receive data until the CPU or DMA/DTC has read all of the received data from a single FIFO buffer plane.

  While the PIPECFG.BFRE bit = 1, the USBFS retains the length of the receive data until the BCLR bit is set to 1, even after all the data is read.

- RCNT = 1

  The USBFS decrements the value indicated in the DTLN[8:0] bits each time data is read from the FIFO buffer. The value is decremented by 1 when MBW = 0, and by 2 when MBW = 1.

  The USBFS sets these bits to 0 when all the data is read from one FIFO buffer plane. In double buffer mode, if data is received in one FIFO buffer plane before all of the data is read from the other plane, the USBFS sets these bits to indicate the length of the receive data in the former plane when all of the data is read from the latter plane.

FRDY bit (FIFO Port Ready)

The FRDY bit indicates whether the FIFO port can be accessed by the CPU or DMA/DTC.

In the following cases, the USBFS sets the FRDY bit to 1 but data cannot be read through the FIFO port because there is no data to be read:

- A zero-length packet is received when the FIFO buffer assigned to the selected pipe is empty
- A short packet is received and the data is completely read while the PIPECFG.BFRE bit = 1

In these cases, set the BCLR bit to 1 to clear the FIFO buffer, and enable transmission and reception of the next data.

BCLR bit (CPU Buffer Clear)

Set the BCLR bit to 1 to clear the FIFO buffer on the CPU side for the selected pipe.

When double buffer mode is set for the FIFO buffer assigned to the selected pipe, the USBFS clears only one plane of the FIFO buffer even when both planes are read-enabled.

When the DCP is the selected pipe, setting the BCLR bit to 1 allows the USBFS to clear the FIFO buffer regardless of whether the CPU or SIE has access rights. To clear the buffer when the SIE has access rights, set the DCPCTR.PID[1:0] bits to 00b (NAK response) before setting the BCLR bit to 1.

When the selected pipe is transmitting, if 1 is written to the BVAL flag and the BCLR bit simultaneously, the USBFS clears the data that is already written, enabling transmission of a zero-length packet.

When the selected pipe is not the DCP, only write 1 to the BCLR bit while the FRDY bit in the FIFO port control register is 1 (set by the USBFS).

BVAL flag (Buffer Memory Valid Flag)

Set the BVAL flag to 1 when data is completely written to the FIFO buffer on the CPU side for the pipe selected in CURPIPE[3:0].

When the selected pipe is transmitting, set this flag to 1 in the following cases:

- To transmit a short packet, set this flag to 1 after data is written
- To transmit a zero-length packet, set this flag to 1 before data is written to the FIFO buffer

The USBFS then switches the FIFO buffer from the CPU side to the SIE side, enabling transmission.

When data of the maximum packet size is written for the pipe in continuous transfer mode, the USBFS sets the BVAL flag to 1 and switches the FIFO buffer from the CPU side to the SIE side, enabling transmission.

Only write 1 to the BVAL flag while the FRDY bit is 1 (set by the USBFS). When the selected pipe is receiving, do not set the BVAL flag to 1.

37.2.9 DnFIFOCTR : DnFIFO Port Control Register (n = 0, 1)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x02A + 0x4 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BVAL	BCLR	FRDY	—	—	—	—	DTLN[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	DTLN[8:0]	Receive Data Length Indicates the receive data length. The meaning of the values differs depending on the RCNT bit setting in the port select register. For details, see the description of the DTLN[8:0] bits.	R
12:9	—	These bits are read as 0. The write value should be 0.	R/W
13	FRDY	FIFO Port Ready 0: FIFO port access disabled 1: FIFO port access enabled	R
14	BCLR	CPU Buffer Clear 0: No operation 1: Clear FIFO buffer on the CPU side	R/W*1
15	BVAL	Buffer Memory Valid Flag 0: Invalid (writing 0 has no effect) 1: Writing ended	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Only 0 can be read.

The CFIFOCTR, D0FIFOCTR, and D1FIFOCTR registers correspond to the CFIFO, D0FIFO, and D1FIFO buffers.

DTLN[8:0] bits (Receive Data Length)

The DTLN[8:0] bits indicate the length of the receive data.

While the FIFO buffer is being read, the DTLN[8:0] bits indicate different values depending on the DnFIFOSEL.RCNT bit (n = 0, 1), as follows:

- RCNT = 0

  The USBFS sets the DTLN[8:0] bits to indicate the length of the receive data until the CPU or DMA/DTC has read all of the received data from a single FIFO buffer plane.

  While the PIPECFG.BFRE bit = 1, the USBFS retains the length of the receive data until the BCLR bit is set to 1, even after all the data is read.

- RCNT = 1

The USBFS decrements the value indicated in the DTLN[8:0] bits each time data is read from the FIFO buffer. The value is decremented by 1 when MBW = 0, and by 2 when MBW = 1.

The USBFS sets these bits to 0 when all the data is read from one FIFO buffer plane. In double buffer mode, if data is received in one FIFO buffer plane before all of the data is read from the other plane, the USBFS sets these bits to indicate the length of the receive data in the former plane when all of the data is read from the latter plane.

FRDY bit (FIFO Port Ready)

The FRDY bit indicates whether the FIFO port can be accessed by the CPU or DMA/DTC.

In the following cases, the USBFS sets the FRDY bit to 1 but data cannot be read through the FIFO port because there is no data to be read:

- A zero-length packet is received when the FIFO buffer assigned to the selected pipe is empty
- A short packet is received and the data is completely read while the PIPECFG.BFRE bit = 1

In these cases, set the BCLR bit to 1 to clear the FIFO buffer, and enable transmission and reception of the next data.

BCLR bit (CPU Buffer Clear)

Set the BCLR bit to 1 to clear the FIFO buffer on the CPU side for the selected pipe.

When double buffer mode is set for the FIFO buffer assigned to the selected pipe, the USBFS clears only one plane of the FIFO buffer even when both planes are read-enabled.

When the DCP is the selected pipe, setting the BCLR bit to 1 allows the USBFS to clear the FIFO buffer regardless of whether the CPU or SIE has access rights. To clear the buffer when the SIE has access rights, set the DCPCTR.PID[1:0] bits to 00b (NAK response) before setting the BCLR bit to 1.

When the selected pipe is transmitting, if 1 is written to the BVAL flag and the BCLR bit simultaneously, the USBFS clears the data that is already written, enabling transmission of a zero-length packet.

When the selected pipe is not the DCP, only write 1 to the BCLR bit while the FRDY bit in the FIFO port control register is 1 (set by the USBFS).

BVAL flag (Buffer Memory Valid Flag)

Set the BVAL flag to 1 when data is completely written to the FIFO buffer on the CPU side for the pipe selected in CURPIPE[3:0].

When the selected pipe is transmitting, set this flag to 1 in the following cases:

- To transmit a short packet, set this flag to 1 after data is written
- To transmit a zero-length packet, set this flag to 1 before data is written to the FIFO buffer

The USBFS then switches the FIFO buffer from the CPU side to the SIE side, enabling transmission.

When data of the maximum packet size is written for the pipe in continuous transfer mode, the USBFS sets the BVAL flag to 1 and switches the FIFO buffer from the CPU side to the SIE side, enabling transmission.

Only write 1 to the BVAL flag while the FRDY bit is 1 (set by the USBFS). When the selected pipe is receiving, do not set the BVAL flag to 1.

37.2.10 INTENB0 : Interrupt Enable Register 0

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x030

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VBSE	RSME	SOFE	DVSE	CTRE	BEMPE	NRDYE	BRDYE	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	BRDYE	Buffer Ready Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
9	NRDYE	Buffer Not Ready Response Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
10	BEMPE	Buffer Empty Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
11	CTRE	Control Transfer Stage Transition Interrupt Enable *1 0: Disable interrupt request 1: Enable interrupt request	R/W
12	DVSE	Device State Transition Interrupt Enable *1 0: Disable interrupt request 1: Enable interrupt request	R/W
13	SOFE	Frame Number Update Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
14	RSME	Resume Interrupt Enable*1 0: Disable interrupt request 1: Enable interrupt request	R/W
15	VBSE	VBUS Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The RSME, DVSE, and CTRE bits can only be set to 1 in device controller mode. Do not set these bits to 1 in host controller mode.

When a status flag in the INTSTS0 register sets to 1 and the associated interrupt request enable bit setting in theINTENB0 register is 1, the USBFS issues a USBFS interrupt request.

Regardless of the INTENB0 register setting, the status flag in the INTSTS0 register sets to 1 in response to a state change that satisfies the associated condition.

When an interrupt request enable bit in the INTENB0 register is switched from 0 to 1 while the associated status flag in the INTSTS0 register is set to 1, a USBFS interrupt is requested.

37.2.11 INTENB1 : Interrupt Enable Register 1

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x032

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRCRE	BCHGE	—	DTCHE	ATTCHE	—	—	—	—	EOFERRE	SIGNE	SACKE	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	SACKE	Setup Transaction Normal Response Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
5	SIGNE	Setup Transaction Error Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W

Bit	Symbol	Function	R/W
6	EOFERRE	EOF Error Detection Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
10:7	—	These bits are read as 0. The write value should be 0.	R/W
11	ATTCHE	Connection Detection Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
12	DTCHE	Disconnection Detection Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
13	—	This bit is read as 0. The write value should be 0.	R/W
14	BCHGE	USB Bus Change Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
15	OVRCRE	Overcurrent Input Change Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W

Note: S-TYPE-3, P-TYPE-3

Note: The bits in INTENB1 can only be set to 1 in host controller mode. Do not set these bits to 1 in device controller mode.

INTENB1 specifies the interrupt masks in host controller mode and for the setup transaction.

When a status flag in the INTSTS1 register sets to 1 and the associated interrupt request enable bit setting in theINTENB1 register is 1, the USBFS issues a USBFS interrupt request.

Regardless of the INTENB1 register setting, the status flag in the INTSTS1 register sets to 1 in response to a state change that satisfies the associated condition.

When an interrupt request enable bit in the INTENB1 register is switched from 0 to 1 while the associated status flag in the INTSTS1 register is set to 1, a USBFS interrupt is requested.

Do not enable interrupts in device controller mode.

37.2.12 BRDYENB : BRDY Interrupt Enable Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x036

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9 BRDY E	PIPE8 BRDY E	PIPE7 BRDY E	PIPE6 BRDY E	PIPE5 BRDY E	PIPE4 BRDY E	PIPE3 BRDY E	PIPE2 BRDY E	PIPE1 BRDY E	PIPE0 BRDY E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0BRDYE	BRDY Interrupt Enable for Pipe 0 0: Disable interrupt request 1: Enable interrupt request	R/W
1	PIPE1BRDYE	BRDY Interrupt Enable for Pipe 1 0: Disable interrupt request 1: Enable interrupt request	R/W
2	PIPE2BRDYE	BRDY Interrupt Enable for Pipe 2 0: Disable interrupt request 1: Enable interrupt request	R/W
3	PIPE3BRDYE	BRDY Interrupt Enable for Pipe 3 0: Disable interrupt request 1: Enable interrupt request	R/W

Bit	Symbol	Function	R/W
4	PIPE4BRDYE	BRDY Interrupt Enable for Pipe 4 0: Disable interrupt request 1: Enable interrupt request	R/W
5	PIPE5BRDYE	BRDY Interrupt Enable for Pipe 5 0: Disable interrupt request 1: Enable interrupt request	R/W
6	PIPE6BRDYE	BRDY Interrupt Enable for Pipe 6 0: Disable interrupt request 1: Enable interrupt request	R/W
7	PIPE7BRDYE	BRDY Interrupt Enable for Pipe 7 0: Disable interrupt request 1: Enable interrupt request	R/W
8	PIPE8BRDYE	BRDY Interrupt Enable for Pipe 8 0: Disable interrupt request 1: Enable interrupt request	R/W
9	PIPE9BRDYE	BRDY Interrupt Enable for Pipe 9 0: Disable interrupt request 1: Enable interrupt request	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The BRDYENB register enables or disables the INTSTS0.BRDY bit to be set to 1 when a BRDY interrupt is detected for each pipe.

When a status flag in the BRDYSTS register sets to 1 and the associated PIPEnBRDYE bit (n = 0 to 9) setting in theBRDYENB register is 1, the INTSTS0.BRDY flag sets to 1. In this case, if the BRDYE bit in INTENB0 is 1, the USBFS generates a BRDY interrupt request. While at least one PIPEnBRDY bit indicates 1, the USB generates the BRDY interrupt request when the associated interrupt request enable bit in the BRDYENB register is changed from 0 to 1 by software.

37.2.13 NRDYENB : NRDY Interrupt Enable Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x038

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9NRDYE	PIPE8NRDYE	PIPE7NRDYE	PIPE6NRDYE	PIPE5NRDYE	PIPE4NRDYE	PIPE3NRDYE	PIPE2NRDYE	PIPE1NRDYE	PIPE0NRDYE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0NRDYE	NRDY Interrupt Enable for Pipe 0 0: Disable interrupt request 1: Enable interrupt request	R/W
1	PIPE1NRDYE	NRDY Interrupt Enable for Pipe 1 0: Disable interrupt request 1: Enable interrupt request	R/W
2	PIPE2NRDYE	NRDY Interrupt Enable for Pipe 2 0: Disable interrupt request 1: Enable interrupt request	R/W
3	PIPE3NRDYE	NRDY Interrupt Enable for Pipe 3 0: Disable interrupt request 1: Enable interrupt request	R/W
4	PIPE4NRDYE	NRDY Interrupt Enable for Pipe 4 0: Disable interrupt request 1: Enable interrupt request	R/W

Bit	Symbol	Function	R/W
5	PIPE5NRDYE	NRDY Interrupt Enable for Pipe 5 0: Disable interrupt request 1: Enable interrupt request	R/W
6	PIPE6NRDYE	NRDY Interrupt Enable for Pipe 6 0: Disable interrupt request 1: Enable interrupt request	R/W
7	PIPE7NRDYE	NRDY Interrupt Enable for Pipe 7 0: Disable interrupt request 1: Enable interrupt request	R/W
8	PIPE8NRDYE	NRDY Interrupt Enable for Pipe 8 0: Disable interrupt request 1: Enable interrupt request	R/W
9	PIPE9NRDYE	NRDY Interrupt Enable for Pipe 9 0: Disable interrupt request 1: Enable interrupt request	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The NRDYENB register enables or disables the INTSTS0.NRDY bit to be set to 1 when a NRDY interrupt is detected for each pipe.

When a status flag in the NRDYSTS register sets to 1 and the associated PIPEnNRDYE (n = 0 to 9) bit setting in theNRDYENB register is 1, the INTSTS0.NRDY flag sets to 1. In this case, if the NRDYE bit in INTENB0 is 1, the USBFS generates a NRDY interrupt request. While at least one PIPEnNRDY bit indicates 1, the USBFS generates the NRDY interrupt request when the associated interrupt request enable bit in the NRDYENB register is changed from 0 to 1 by software.

37.2.14 BEMPENB : BEMP Interrupt Enable Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x03A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9BEMPE	PIPE8BEMPE	PIPE7BEMPE	PIPE6BEMPE	PIPE5BEMPE	PIPE4BEMPE	PIPE3BEMPE	PIPE2BEMPE	PIPE1BEMPE	PIPE0BEMPE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0BEMPE	BEMP Interrupt Enable for Pipe 0 0: Disable interrupt request 1: Enable interrupt request	R/W
1	PIPE1BEMPE	BEMP Interrupt Enable for Pipe 1 0: Disable interrupt request 1: Enable interrupt request	R/W
2	PIPE2BEMPE	BEMP Interrupt Enable for Pipe 2 0: Disable interrupt request 1: Enable interrupt request	R/W
3	PIPE3BEMPE	BEMP Interrupt Enable for Pipe 3 0: Disable interrupt request 1: Enable interrupt request	R/W
4	PIPE4BEMPE	BEMP Interrupt Enable for Pipe 4 0: Disable interrupt request 1: Enable interrupt request	R/W

Bit	Symbol	Function	R/W
5	PIPE5BEMPE	BEMP Interrupt Enable for Pipe 5 0: Disable interrupt request 1: Enable interrupt request	R/W
6	PIPE6BEMPE	BEMP Interrupt Enable for Pipe 6 0: Disable interrupt request 1: Enable interrupt request	R/W
7	PIPE7BEMPE	BEMP Interrupt Enable for Pipe 7 0: Disable interrupt request 1: Enable interrupt request	R/W
8	PIPE8BEMPE	BEMP Interrupt Enable for Pipe 8 0: Disable interrupt request 1: Enable interrupt request	R/W
9	PIPE9BEMPE	BEMP Interrupt Enable for Pipe 9 0: Disable interrupt request 1: Enable interrupt request	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The BEMPENB register enables or disables the INTSTS0.BEMP bit to be set to 1 when a BEMP interrupt is detected for each pipe.

When a status flag in the BEMPSTS register sets to 1 and the associated PIPEnBEMPE (n = 0 to 9) bit setting in the BEMPENB register is 1, the INTSTS0.BEMP flag sets to 1. In this case, if the BEMPE bit in INTENB0 is 1, the USBFS generates a BEMP interrupt request. While at least one PIPEnBEMP bit indicates 1, the USBFS generates the BEMP interrupt request when the associated interrupt request enable bit in the BEMPENB register is changed from 0 to 1 by software.

37.2.15 SOFCFG : SOF Output Configuration Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x03C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	TRNENSEL	—	BRDYM	—	EDGESTS	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	EDGESTS	Edge Interrupt Output Status Monitor*1 Indicates 1 during the edge processing of an edge interrupt output signal.	R
5	—	This bit is read as 0. The write value should be 0.	R/W
6	BRDYM	BRDY Interrupt Status Clear Timing 0: Clear BRDY flag by software 1: Clear BRDY flag by the USBFS through a data read from the FIFO buffer or data write to the FIFO buffer	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	TRNENSEL	Transaction-Enabled Time Select*1 0: Not low-speed communication 1: Low-speed communication	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Confirm that the EDGESTS flag is 0 before stopping the clock supply to the USBFS.

EDGESTS bit (Edge Interrupt Output Status Monitor)

The EDGESTS bit indicates 1 during the edge processing of an edge interrupt output signal. Confirm that this bit is 0 before stopping the clock supply to the USBFS.

BRDYM bit (BRDY Interrupt Status Clear Timing)

The BRDYM bit specifies how the BRDY interrupt status flags for the pipes are cleared.

TRNENSEL bit (Transaction-Enabled Time Select)

When the USB port is in use for full- or low-speed communications, the TRNENSEL bit specifies the timing with which the USBFS issues tokens in a frame (transaction-enabled time).

Set this bit to 1 when a low-speed device is connected. The bit is only valid in host controller mode. Set this bit to 0 in device controller mode.

37.2.16 INTSTS0 : Interrupt Status Register 0

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x040

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VBINT	RESM	SOFR	DVST	CTRT	BEMP	NRDY	BRDY	VBSTS	DVSQ[2:0]			VALID	CTSQ[2:0]		
Value after reset:	0	0	0	x	0	0	0	0	x	0	0	x	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CTSQ[2:0]	Control Transfer Stage 0 0 0: Idle or setup stage 0 0 1: Control read data stage 0 1 0: Control read status stage 0 1 1: Control write data stage 1 0 0: Control write status stage 1 0 1: Control write (no data) status stage 1 1 0: Control transfer sequence error	R
3	VALID	USB Request Reception 0: Setup packet not received 1: Setup packet received	R/W
6:4	DVSQ[2:0]	Device State Indicates the device state. 0 0 0: Powered state 0 0 1: Default state 0 1 0: Address state 0 1 1: Configured state Others: Suspend state	R
7	VBSTS	VBUS Input Status 0: USB_VBUS pin is low 1: USB_VBUS pin is high	R
8	BRDY	Buffer Ready Interrupt Status 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R
9	NRDY	Buffer Not Ready Interrupt Status 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R
10	BEMP	Buffer Empty Interrupt Status 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R
11	CTRT	Control Transfer Stage Transition Interrupt Status*2 0: No control transfer stage transition interrupt occurred 1: Control transfer stage transition interrupt occurred	R/W*1

Bit	Symbol	Function	R/W
12	DVST	Device State Transition Interrupt Status*2 0: No device state transition interrupt occurred 1: Device state transition interrupt occurred	R/W*1
13	SOFR	Frame Number Refresh Interrupt Status 0: No SOF interrupt occurred 1: SOF interrupt occurred	R/W*1
14	RESM	Resume Interrupt Status*2 *3 0: No resume interrupt occurred 1: Resume interrupt occurred	R/W*1
15	VBINT	VBUS Interrupt Status*3 0: No VBUS interrupt occurred 1: VBUS interrupt occurred	R/W*1

Note: S-TYPE-3, P-TYPE-3
Note: The value of the DVST bit is 0 when the MCU is reset and 1 after a USB bus reset.
Note: The value of the VBSTS bit is 1 when the USB_VBUS pin is high and 0 when the USB_VBUS pin is low.
Note: The value of the DVSQ[2:0] bits is 000b when the MCU is reset and 001b after a USB bus reset.
Note 1. To clear the VBINT, RESM, SOFR, DVST, CTRT, or VALID bits, write 0 only to the bits to be cleared. Write 1 to the other bits. Do not write 0 to the status bits indicating 0.
Note 2. The status of the RESM, DVST, and CTRT bits are changed only in device controller mode. Set the associated interrupt enable bits to 0 (disabled) in host controller mode.
Note 3. The USBFS detects a change in the status indicated in the VBINT andRESM bits even while the clock supply is stopped (SYSCFG.SCKE bit = 0), and it requests the interrupt when the associated interrupt request bit is 1. Enable the clock supply before clearing the status by software.

CTSQ[2:0] bits (Control Transfer Stage)

In host controller mode, the read value of the CTSQ[2:0] bits is invalid.

VALID bit (USB Request Reception)

In host controller mode, the read value of the VALID bit is invalid.

DVSQ[2:0] bits (Device State)

The DVSQ[2:0] bits are initialized by a USB bus reset. In host controller mode, the read value is invalid.

BRDY flag (Buffer Ready Interrupt Status)

The BRDY flag indicates the BRDY interrupt status.

The USBFS sets the BRDY bit to 1 when it detects a BRDY interrupt status (PIPEnBRDY = 1, n = 0 to 9) on at least one pipe for which BRDY interrupts are enabled (BRDYENB.PIPEnBRDYE = 1).

For the conditions that cause the PIPEnBRDY status to be asserted, see section 37.3.3.1. BRDY Interrupt.

The USBFS sets the BRDY bit to 0 when the software writes 0 to all of the PIPEnBRDY bits associated with the PIPEnBRDYE bits that are set to 1. Writing 0 to the BRDY flag in the software does not clear the flag.

NRDY flag (Buffer Not Ready Interrupt Status)

The NRDY flag indicates the NRDY interrupt status.

The USBFS sets the NRDY bit to 1 when it detects a NRDY interrupt status (PIPEnNRDY = 1, n = 0 to 9) on at least one pipe for which NRDY interrupts are enabled (NRDYENB.PIPEnNRDYE = 1).

For the conditions that cause the PIPEnNRDY status to be asserted, see section 37.3.3.2. NRDY Interrupt.

The USBFS sets the NRDY bit to 0 when the software writes 0 to all of the PIPEnNRDY bits associated with the PIPEnNRDYE bits that are set to 1. Writing 0 to the NRDY flag in the software does not clear the flag.

BEMP flag (Buffer Empty Interrupt Status)

The BEMP flag indicates the BEMP interrupt status.

The USBFS sets the BEMP bit to 1 when it detects a BEMP interrupt status (PIPEnBEMP = 1, n = 0 to 9) on at least one pipe for which BEMP interrupts are enabled (BEMPENB.PIPEnBEMPE = 1).

For the conditions that cause the PIPEnBEMP status to be asserted, see section 37.3.3.3. BEMP Interrupt.

The USBFS sets the BEMP bit to 0 when the software writes 0 to all of the PIPEnBEMP bits associated with the PIPEnBEMPE bits that are set to 1. Writing 0 to the BEMP flag in the software does not clear the flag.

CTRT flag (Control Transfer Stage Transition Interrupt Status)

In device controller mode, the USBFS updates the value of the CTSQ[2:0] bits and sets the CTRT flag to 1 on detecting a transition in the control transfer stage. When a control transfer stage transition interrupt occurs, clear the CTRT flag before the USBFS detects the next control transfer stage transition.

Values read from the CTRT flag in host controller mode are invalid.

DVST flag (Device State Transition Interrupt Status)

In device controller mode, the USBFS updates the value of the DVSQ[2:0] bits and sets the DVST flag to 1 on detecting a change in the device state. When a device state transition interrupt occurs, clear theDVST flag before the USBFS detects the next device state transition.

Values read from the DVST flag in host controller mode are invalid.

SOFR flag (Frame Number Refresh Interrupt Status)

In host controller mode, the USBFS sets the SOFR flag to 1 on updating the frame number when the DVSTCTR0.UACT bit is set to 1 by software. A SOFR interrupt is detected every 1 ms.

In device controller mode, the USBFS sets the SOFR flag to 1 on updating the frame number. A frame number refresh interrupt is detected every 1 ms.

The USBFS can detect an SOFR interrupt through the internal interpolation function even when a corrupted SOF packet is received from the USB host.

RESM flag (Resume Interrupt Status)

In device controller mode, the USBFS sets the RESM flag to 1 on detecting the falling edge of the signal on the USB_DP pin in the Suspend state (DVSQ[2:0] = 1xxb). Values read from the RESM flag in host controller mode are invalid.

VBINT flag (VBUS Interrupt Status)

The USBFS sets the VBINT flag to 1 on detecting a level change (high to low or low to high) in the USB_VBUS pin input value. The USBFS sets the VBSTS flag to indicate the USB_VBUS pin input value. When a VBUS interrupt occurs, eliminate transient elements by reading the VBSTS flag at least three times through software processing and check that the values read are the same.

37.2.17 INTSTS1 : Interrupt Status Register 1

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x042

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRCR	BCHG	—	DTCH	ATTCH	—	—	—	—	EOFERR	SIGN	SACK	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	SACK	Setup Transaction Normal Response Interrupt Status 0: No SACK interrupt occurred 1: SACK interrupt occurred	R/W*1
5	SIGN	Setup Transaction Error Interrupt Status 0: No SIGN interrupt occurred 1: SIGN interrupt occurred	R/W*1
6	EOFERR	EOF Error Detection Interrupt Status 0: No EOFERR interrupt occurred 1: EOFERR interrupt occurred	R/W*1

Bit	Symbol	Function	R/W
10:7	—	These bits are read as 0. The write value should be 0.	R/W
11	ATTCH	ATTCH Interrupt Status 0: No ATTCH interrupt occurred 1: ATTCH interrupt occurred	R/W*1
12	DTCH	USB Disconnection Detection Interrupt Status 0: No DTCH interrupt occurred 1: DTCH interrupt occurred	R/W*1
13	—	This bit is read as 0. The write value should be 0.	R/W
14	BCHG	USB Bus Change Interrupt Status*2 0: No BCHG interrupt occurred 1: BCHG interrupt occurred	R/W*1
15	OVRCR	Overcurrent Input Change Interrupt Status*2 0: No OVRCR interrupt occurred 1: OVRCR interrupt occurred	R/W*1

Note: S-TYPE-3, P-TYPE-3

Note 1. To clear the bits in INTSTS1, write 0 only to the bits to be cleared. Write 1 to the other bits.

Note 2. The USBFS detects a change in the status in the OVRCR or BCHG bit even when the clock supply is stopped (SYSCFG.SCKE = 0), and it requests the interrupt when the associated interrupt request bit is 1. Enable the clock supply (SYSCFG.SCKE = 1) before clearing the status through the software. No other interrupts can be detected while the clock supply is stopped (SYSCFG.SCKE bit = 0).

INTSTS1 is used to confirm the status of each interrupt in host controller mode. Only enable the status change interrupts indicated in the bits in INTSTS1 in host controller mode.

SACK flag (Setup Transaction Normal Response Interrupt Status)

The SACK flag indicates the status of the setup transaction normal response interrupt in host controller mode.

The USBFS detects the SACK interrupt and sets this flag to 1 when an ACK response is returned from a peripheral device during the setup transactions issued by the USBFS. If the associated interrupt enable bit is set to 1 by software, the USBFS generates the interrupt.

Values read from the SACK flag in device controller mode are invalid.

SIGN flag (Setup Transaction Error Interrupt Status)

The SIGN flag indicates the status of setup transaction error interrupts in host controller mode.

The USBFS detects the SIGN interrupt and sets this flag to 1 when an ACK response is not returned from a peripheral device three consecutive times during the setup transactions issued by the USBFS. If the associated interrupt enable bit is set to 1 by software, the USBFS generates the interrupt.

The USBFS detects the SIGN interrupt when any of the following response conditions occur for three consecutive setup transactions:

- Timeout is detected by the USBFS when the peripheral device has returned no response
- A corrupted ACK packet is received
- A handshake other than ACK (NAK, NYET, or STALL) is received

Values read from the SIGN flag in device controller mode are invalid.

EOFERR flag (EOF Error Detection Interrupt Status)

The EOFERR flag indicates the status of EOF error detection interrupts in host controller mode.

The USBFS detects the EOFERR interrupt and sets this flag to 1 on detecting that communication did not complete at the EOF2 timing defined in the USB 2.0 specification. If the associated interrupt enable bit is set to 1 by software, the USBFS generates the interrupt.

After detecting the EOFERR interrupt, the USBFS controls the hardware as follows, regardless of the associated interrupt enable bit setting:

- Sets the DVSTCTR0.UACT bit for the port in which the EOFERR interrupt was detected to 0
- Puts the port in which the EOFERR interrupt occurred into the idle state

The software must terminate all pipes in which communications are being carried out and re-enumerate the USB port.

Values read from the EOFERR flag in device controller mode are invalid.

ATTCH flag (ATTCH Interrupt Status)

The ATTCH flag indicates the status of USB attach detection interrupts in host controller mode.

The USBFS detects the ATTCH interrupt and sets this flag to 1 on detecting a J- or K-state on the full- or low-speed signal level for 2.5 µs. If the associated interrupt enable bit is set to 1 by software, the USBFS generates the interrupt.

The USBFS detects the ATTCH interrupt on any of the following conditions.

- K-state, SE0, or SE1 changes to J-state, and J-state continues for 2.5 µs
- J-state, SE0, or SE1 changes to K-state, and K-state continues for 2.5 µs

Values read from the ATTCH flag in device controller mode are invalid.

DTCH flag (USB Disconnection Detection Interrupt Status)

The DTCH flag indicates the status of USB disconnection detection interrupts in host controller mode.

The USBFS detects the DTCH interrupt and sets this flag to 1 on detecting a USB bus detach event. If the associated interrupt enable bit is set to 1 by software, the USBFS generates the interrupt.

The USBFS detects bus detach events based on the USB 2.0 specification.

After detecting the DTCH interrupt, the USBFS controls hardware as follows, regardless of the associated interrupt enable bit setting:

- Sets the DVSTCTR0.UACT bit for the port in which the DTCH interrupt was detected to 0
- Puts the port in which the DTCH interrupt occurred into the idle state

The software must terminate all pipes in which communications are being carried out and transition to a wait state for connecting to the USB port (waiting for ATTCH interrupt generation).

Values read from the DTCH flag in device controller mode are invalid.

BCHG flag (USB Bus Change Interrupt Status)

The BCHG flag indicates the status of USB bus change interrupts in host controller mode.

The USBFS detects the BCHG interrupt and sets this flag to 1 when a change in the full- or low-speed signal level occurs on the USB port. This includes any change from J-state, K-state, or SE0 to J-state, K-state, or SE0. If the associated interrupt enable bit is set to 1 by software, the USBFS generates the interrupt.

The USBFS sets the LNST[1:0] flags to indicate the input state of the USB port. When a BCHG interrupt occurs, eliminate transient elements by repeat reading the LNST[1:0] flags by software until the same value is read at least three times.

Change in the USB bus state can be detected while the internal clock is stopped.

Values read from the BCHG flag in device controller mode are invalid.

OVRCR flag (Overcurrent Input Change Interrupt Status)

The OVRCR flag indicates the status of USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB or USB_OVRCURB-DS input pin change interrupts.

The USBFS detects the OVRCR interrupt and sets this flag to 1 when a change (high to low or low to high) occurs in at least one of the input values to the USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB or USB_OVRCURB-DS pins. If the associated interrupt enable bit is set to 1 by software, the USBFS generates the interrupt.

37.2.18 BRDYSTS : BRDY Interrupt Status Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x046

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9 BRDY	PIPE8 BRDY	PIPE7 BRDY	PIPE6 BRDY	PIPE5 BRDY	PIPE4 BRDY	PIPE3 BRDY	PIPE2 BRDY	PIPE1 BRDY	PIPE0 BRDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0BRDY	BRDY Interrupt Status for Pipe 0 *2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
1	PIPE1BRDY	BRDY Interrupt Status for Pipe 1 *2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
2	PIPE2BRDY	BRDY Interrupt Status for Pipe 2*2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
3	PIPE3BRDY	BRDY Interrupt Status for Pipe 3*2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
4	PIPE4BRDY	BRDY Interrupt Status for Pipe 4 *2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
5	PIPE5BRDY	BRDY Interrupt Status for Pipe 5 *2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
6	PIPE6BRDY	BRDY Interrupt Status for Pipe 6*2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
7	PIPE7BRDY	BRDY Interrupt Status for Pipe 7 *2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
8	PIPE8BRDY	BRDY Interrupt Status for Pipe 8 *2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
9	PIPE9BRDY	BRDY Interrupt Status for Pipe 9 *2 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*1
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. When the SOFCFG.BRDYM bit is set to 0, to clear the status indicated in the bits in BRDYSTS, write 0 only to the bits to be cleared. Write 1 to the other bits.

Note 2. When the SOFCFG.BRDYM bit is set to 0, clear BRDY interrupts before accessing the FIFO.

37.2.19 NRDYSTS : NRDY Interrupt Status Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x048

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9 NRDY	PIPE8 NRDY	PIPE7 NRDY	PIPE6 NRDY	PIPE5 NRDY	PIPE4 NRDY	PIPE3 NRDY	PIPE2 NRDY	PIPE1 NRDY	PIPE0 NRDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0NRDY	NRDY Interrupt Status for Pipe 0 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
1	PIPE1NRDY	NRDY Interrupt Status for Pipe 1 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
2	PIPE2NRDY	NRDY Interrupt Status for Pipe 2 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
3	PIPE3NRDY	NRDY Interrupt Status for Pipe 3 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
4	PIPE4NRDY	NRDY Interrupt Status for Pipe 4 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
5	PIPE5NRDY	NRDY Interrupt Status for Pipe 5 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
6	PIPE6NRDY	NRDY Interrupt Status for Pipe 6 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
7	PIPE7NRDY	NRDY Interrupt Status for Pipe 7 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
8	PIPE8NRDY	NRDY Interrupt Status for Pipe 8 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
9	PIPE9NRDY	NRDY Interrupt Status for Pipe 9 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R/W*1
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. To clear the status indicated in the bits in NRDYSTS, write 0 only to the bits to be cleared. Write 1 to the other bits.

37.2.20 BEMPSTS : BEMP Interrupt Status Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x04A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPE9 BEMP	PIPE8 BEMP	PIPE7 BEMP	PIPE6 BEMP	PIPE5 BEMP	PIPE4 BEMP	PIPE3 BEMP	PIPE2 BEMP	PIPE1 BEMP	PIPE0 BEMP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIPE0BEMP	BEMP Interrupt Status for Pipe 0 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
1	PIPE1BEMP	BEMP Interrupt Status for Pipe 1 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
2	PIPE2BEMP	BEMP Interrupt Status for Pipe 2 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
3	PIPE3BEMP	BEMP Interrupt Status for Pipe 3 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
4	PIPE4BEMP	BEMP Interrupt Status for Pipe 4 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
5	PIPE5BEMP	BEMP Interrupt Status for Pipe 5 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
6	PIPE6BEMP	BEMP Interrupt Status for Pipe 6 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
7	PIPE7BEMP	BEMP Interrupt Status for Pipe 7 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
8	PIPE8BEMP	BEMP Interrupt Status for Pipe 8 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
9	PIPE9BEMP	BEMP Interrupt Status for Pipe 9 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R/W*1
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. To clear the status indicated in the bits in BEMPSTS, write 0 only to the bits to be cleared. Write 1 to the other bits.

37.2.21 FRMNUM : Frame Number Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x04C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRN	CRCE	—	—	—	FRNM[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	FRNM[10:0]	Frame Number Latest frame number.	R
13:11	—	These bits are read as 0. The write value should be 0.	R/W
14	CRCE	Receive Data Error 0: No error occurred 1: Error occurred	R/W*1
15	OVRN	Overrun/Underrun Detection Status 0: No error occurred 1: Error occurred	R/W*1

Note: S-TYPE-3, P-TYPE-3
Note 1. To clear the status, write 0 only to the bits to be cleared. Write 1 to the other bits.

FRNM[10:0] flags (Frame Number)

The USBFS sets the FRNM[10:0] flags to indicate the latest frame number, which is updated every 1 ms, when an SOF packet is issued or received.

CRCE flag (Receive Data Error)

The CRCE flag sets to 1 when a CRC error or bit stuffing error occurs during isochronous transfer. On detecting a CRC error in host controller mode, the USBFS generates an internal NRDY interrupt.

To clear the CRCE flag, write 0 to it while writing 1 to the other bits in the FRMNUM register.

OVRN flag (Overrun/Underrun Detection Status)

The OVRN flag sets to 1 when an overrun or underrun error occurs during isochronous transfer. To clear the flag, write 0 to it while writing 1 to the other bits in the FRMNUM register.

In host controller mode, the OVRN flag sets to 1 on any of the following conditions:

- For a transmitting isochronous pipe, the time to issue an OUT token comes before all of the transmit data is written to the FIFO buffer
- For a receiving isochronous pipe, the time to issue an IN token comes when no FIFO buffer planes are empty

In device controller mode, the OVRN flag sets to 1 on any of the following conditions:

- For a transmitting isochronous pipe, the IN token is received before all of the transmit data is written to the FIFO buffer
- For a receiving isochronous pipe, the OUT token is received when no FIFO buffer planes are empty

37.2.22 DVCHGR : Device State Change Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x04E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DVCHG	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	—	These bits are read as 0. The write value should be 0.	R/W
15	DVCHG	Device State Change 0: Disable writes to the USBADDR.STSRECOV[3:0] and USBADDR.USBADDR[6:0] bits 1: Enable writes to the USBADDR.STSRECOV[3:0] and USBADDR.USBADDR[6:0] bits	R/W

Note: S-TYPE-3, P-TYPE-3

For details, see section 37.3.1.5. Release from Deep Software Standby Mode 1 Because of USB Suspend/Resume Interrupts.

37.2.23 USBADDR : USB Address Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x050

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	STSRECOV[3:0]				—	USBADDR[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	USBADDR[6:0]	USB Address In device controller mode, these bits indicate the USB address assigned by the host when the USBFS processed the SET_ADDRESS request successfully.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
11:8	STSRECOV[3:0]	Status Recovery 0x4: Recovery in device controller mode: Setting prohibited Recovery in host controller mode: Return to the low-speed state (bits DVSTCTR0.RHST[2:0] = 001b) 0x8: Recovery in device controller mode: Setting prohibited Recovery in host controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b) 0x9: Recovery in device controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 001b (default state) Recovery in host controller mode: Setting prohibited 0xA: Recovery in device controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 010b (address state) Recovery in host controller mode: Setting prohibited 0xB: Recovery in device controller mode: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 011b (configured state) Recovery in host controller mode: Setting prohibited Others: Setting prohibited	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

USBADDR[6:0] bits (USB Address)

In device controller mode, the USBADDR[6:0] flags indicate the USB address received when the USBFS processed a SetAddress request successfully. The USBFS sets the USBADDR[6:0] bits to 0x00 on detecting a USB bus reset.

Writing to these bits is enabled while the DVCHGR.DVCHG bit is set to 1. On recovering from a USB power shut-off, the operation can resume from the USB address set before the software shut-off.

In host controller mode, the USBADDR[6:0] bits are invalid.

STSRECOV[3:0] bits (Status Recovery)

Use the STSRECOV[3:0] bits to resume the state of the internal sequencer on recovering from USB power shut-off. For details, see section 37.3.1.5. Release from Deep Software Standby Mode 1 Because of USB Suspend/Resume Interrupts.

Writing to these bits is enabled while the DVCHGR.DVCHG bit is set to 1.

37.2.24 USBREQ : USB Request Type Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x054

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BREQUEST[7:0]								BMREQUESTTYPE[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	BMREQUESTTYPE[7:0]	Request Type USB request bmRequestType value	R/W[*1]
15:8	BREQUEST[7:0]	Request USB request bRequest value	R/W[*1]

Note: S-TYPE-3, P-TYPE-3

Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

USBREQ stores setup requests for control transfers.

In device controller mode, the USBREQ stores the received bRequest and bmRequestType values. In host controller mode, it sets to the bRequest and bmRequestType values to be transmitted.

USBREQ is initialized by a USB bus reset.

BMREQUESTTYPE[7:0] bits (Request Type)

The BMREQUESTTYPE[7:0] bits hold the bmRequestType value of USB requests.

- In host controller mode:

  Set these bits to the value of the USB request data in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.

- In device controller mode:

  These bits indicate the value of the USB request data in reception setup transactions. Writing to the bits has no effect.

BREQUEST[7:0] bits (Request)

The BREQUEST[7:0] bits store bRequest value of the USB request.

- In host controller mode:

  Set these bits to the value of the USB request data in setup transmission transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.

- In device controller mode:

  These bits indicate the value of the USB request data in reception setup transactions. Writing to the bits has no effect.

37.2.25 USBVAL : USB Request Value Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x056

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WVALUE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WVALUE[15:0]	Value USB request wValue value	R/W*1

Note: S-TYPE-3, P-TYPE-3

Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

In device controller mode, USBVAL stores the received wValue value. In host controller mode, it sets to the wValue value to be transmitted is set.

USBVAL is initialized by a USB bus reset.

WVALUE[15:0] bits (Value)

The WVALUE[15:0] bits store wValue value of the USB request.

- In host controller mode:

  Set these bits to the value of the wValue field in USB requests of transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.

- In device controller mode:

These bits indicate the wValue value of USB requests in reception setup transactions. Writing to the bits has no effect.

37.2.26 USBINDX : USB Request Index Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x058

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WINDEX[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WINDEX[15:0]	Index USB request wIndex value	R/W*1

Note: S-TYPE-3, P-TYPE-3

Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

USBINDX stores setup requests for control transfers.

In device controller mode, it stores the received wIndex value. In host controller mode, it sets to the wIndex value to be transmitted.

USBINDX is initialized by a USB bus reset.

WINDEX[15:0] bits (Index)

The WINDEX[15:0] bits hold the wIndex value of a USB request.

- In host controller mode:

  Set these bits to the wIndex value in USB requests in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.

- In device controller mode:

  These bits indicate the wIndex value in USB requests received in reception setup transactions. Writing to the bits has no effect.

37.2.27 USBLENG : USB Request Length Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x05A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WLENTUH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WLENTUH[15:0]	Length USB request wLength value	R/W*1

Note: S-TYPE-3, P-TYPE-3

Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

USBLENG stores setup requests for control transfers.

In device controller mode, the value of wLength that is received is stored. In host controller mode, the value of wLength to be transmitted is set.

USBLENG is initialized by a USB bus reset.

WLENTUH[15:0] bits (Length)

The WLENTUH[15:0]bits hold the wLength value of a USB request.

- In host controller mode:

  Set these bits to the wLength value in USB requests in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.

- In device controller mode:

  These bits indicate the wLength value in USB requests received in reception setup transactions. Writing to the bits has no effect.

37.2.28 DCPCFG : DCP Configuration Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x05C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SHTNAK	—	—	DIR	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	DIR	Transfer Direction*1 0: Data receiving direction 1: Data transmitting direction	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	SHTNAK	Pipe Disabled at End of Transfer*1 0: Keep pipe open after transfer ends 1: Disable pipe after transfer ends	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set this bit while the PID is NAK. Before setting this bit, check that the DCPCTR.PBUSY bit is 0, and then change the DCPCTR.PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

DIR bit (Transfer Direction)

In host controller mode, the DIR bit sets the transfer direction of the data stage and status stage for control transfers. In device controller mode, set the DIR bit to 0.

SHTNAK bit (Pipe Disabled at End of Transfer)

The SHTNAK bit specifies whether to change PID to NAK on transfer end when the selected pipe is receiving. It is only valid when the selected pipe is receiving.

When the SHTNAK bit is 1, the USBFS changes the DCPCTR.PID[1:0] bits for the DCP to NAK on determining that a transfer has ended. The USBFS determines transfer end on the following condition:

- A short packet, including a zero-length packet, is successfully received.

37.2.29 DCPMAXP : DCP Maximum Packet Size Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x05E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DEVSEL[3:0]				—	—	—	—	—	MXPS[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	MXPS[6:0]	Maximum Packet Size*1 Maximum data payload specification (maximum packet size) for the DCP	R/W
11:7	—	These bits are read as 0. The write value should be 0.	R/W
15:12	DEVSEL[3:0]	Device Select*2 0x0: Address 0000b 0x1: Address 0001b 0x2: Address 0010b 0x3: Address 0011b 0x4: Address 0100b 0x5: Address 0101b Others: Setting prohibited	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the MXPS[6:0] bits while PID is NAK. Before setting these bits, check that the DCPCTR.PBUSY bit is 0, and then change the DCPCTR.PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary. After the MXPS[6:0] bits are set and the DCP is set to the CURPIPE[3:0] bits in a port select register, clear the buffer by setting the BCLR bit in the port control register to 1.

Note 2. Only set the DEVSEL[3:0] bits while PID is NAK and the DCPCTR.SUREQ bit is 0. Before setting these bits, check that the DCPCTR.PBUSY bit is 0, and then change the DCPCTR.PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

MXPS[6:0] bits (Maximum Packet Size)

The MXPS[6:0] bits specify the maximum data payload (maximum packet size) for the DCP. The initial value is 0x40 (64 bytes). Set the bits to a USB 2.0-compliant value. Do not write to the FIFO buffer or set PID = BUF whileMXPS[6:0] is set to 0.

DEVSEL[3:0] bits (Device Select)

In host controller mode, the DEVSEL[3:0] bits specify the address of the target peripheral device for a control transfer. Set up the associated DEVADDn (n = 0 to 5) register first, and then set these bits to the corresponding value. To set the DEVSEL[3:0] bits to 0010b, for example, first set the address in the DEVADD2 register.

In device controller mode, set these bits to 0000b.

37.2.30 DCPCTR : DCP Control Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x060

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	SUREQ	—	—	SUREQCLR	—	—	SQCLR	SQSET	SQMON	PBUSY	—	—	CCPL	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID 0 0: NAK response 0 1: BUF response (depends on the buffer state) 1 0: STALL response 1 1: STALL response	R/W
2	CCPL	Control Transfer End Enable 0: Disable control transfer completion 1: Enable control transfer completion	R/W
4:3	—	These bits are read as 0. The write value should be 0.	R/W
5	PBUSY	Pipe Busy 0: DCP not used for the USB bus 1: DCP in use for the USB bus	R
6	SQMON	Sequence Toggle Bit Monitor 0: DATA0 1: DATA1	R
7	SQSET	Sequence Toggle Bit Set*2 Sets the sequence toggle bit in DCP transfers. 0: Invalid (writing 0 has no effect) 1: Set the expected value for the next transaction to DATA1	R/W*1
8	SQCLR	Sequence Toggle Bit Clear*2 Clears the sequence toggle bit in DCP transfers. 0: Invalid (writing 0 has no effect) 1: Clear the expected value for the next transaction to DATA0	R/W*1
10:9	—	These bits are read as 0. The write value should be 0.	R/W
11	SUREQCLR	SUREQ Bit Clear Clears the SUREQ bit in host controller mode. 0: Invalid (writing 0 has no effect) 1: Clear SUREQ to 0	R/W*1
13:12	—	These bits are read as 0. The write value should be 0.	R/W
14	SUREQ	Setup Token Transmission Sets up token transmission in host controller mode. 0: Invalid (writing 0 has no effect) 1: Transmit setup packet	R/W
15	BSTS	Buffer Status 0: Buffer access disabled 1: Buffer access enabled	R

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is read as 0.
Note 2. Only set the SQSET and SQCLR bits while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

PID[1:0] bits (Response PID)

The PID[1:0] bits control the USB response type during control transfers.

In host controller mode, to change the PID[1:0] setting from NAK to BUF:

- When the transmitting direction is set:
  a. Write all of the transmit data to the FIFO buffer while the DVSTCTR0.UACT bit is 1 and PID is NAK.
  b. Set PID[1:0] bits to 01b (BUF).

     The USBFS then executes the OUT transaction.

- When the receiving direction is set:
  a. Check that the FIFO buffer is empty (or empty the buffer) while the DVSTCTR0.UACT bit is 1 and PID is NAK.
  b. Set PID[1:0] bits to 01b (BUF).

     The USBFS then executes the IN transaction.

The USBFS changes the PID[1:0] setting as follows:

- When the PID[1:0] bits are set to BUF (01b) by software and the USBFS has received data exceeding MaxPacketSize, the USBFS sets the PID[1:0] to STALL (11b)
- When a reception error, such as a CRC error, is detected three times consecutively, the USBFS sets thePID[1:0] bits to NAK (00b)
- On receiving the STALL handshake, the USBFS sets PID[1:0] to STALL (11b)

In device controller mode, the USBFS changes the PID[1:0] setting as follows:

- On receiving a setup packet, the USBFS sets PID[1:0] to NAK (00b). The USBFS then sets the INTSTS0.VALID flag to 1, and the PID[1:0] setting cannot be changed until the software clears the VALID flag to 0.
- When the PID[1:0] bits are set to BUF (01b) by software and the USBFS has received data exceeding MaxPacketSize, the USBFS sets PID[1:0] to STALL (11b)
- On detecting a control transfer sequence error, the USBFS sets PID[1:0] to STALL (1xb)
- On detecting a USB bus reset, the USBFS sets PID[1:0] to NAK

The USBFS does not check the PID[1:0] setting while processing a SET_ADDRESS request.

The PID[1:0] bits are initialized by a USB bus reset.

CCPL bit (Control Transfer End Enable)

In device controller mode, setting the CCPL bit to 1 enables the status stage of the control transfer to be completed. When the bit is set to 1 by software while the associated PID[1:0] bits are set to BUF, the USBFS completes the control transfer status stage.

During control read transfers, the USBFS transmits the ACK handshake in response to the OUT transaction from the USB host. During control write or no-data control transfers, it transmits the zero-length packet in response to the IN transaction from the USB host. On detecting a SET_ADDRESS request, the USBFS operates in auto response mode from the setup stage up to status stage completion regardless of the CCPL bit setting.

The USBFS changes the CCPL bit from 1 to 0 on receiving a new setup packet. The software cannot write 1 to the bit while the INTSTS0.VALID bit is 1. The bit is initialized by a USB bus reset.

In host controller mode, always write 0 to the CCPL bit.

PBUSY bit (Pipe Busy)

The PBUSY bit indicates whether DCP is used for the transaction when USBFS changes the PID[1:0] bits from BUF to NAK. The USBFS changes the PBUSY bit from 0 to 1 on start of a USB transaction for the selected pipe. It changes the PBUSY bit from 1 to 0 on completion of one transaction.

After PID is set to NAK by software, the value in the PBUSY bit indicates whether changes to pipe settings can proceed.

For details, see section 37.3.4.1. Pipe Control Register Switching Procedures.

SQMON bit (Sequence Toggle Bit Monitor)

The SQMON bit indicates the expected value of the sequence toggle bit for the next transaction during a DCP transfer.

The USBFS toggles the bit on normal completion of the transaction. It does not toggle the bit, however, when a DATAPID mismatch occurs during a transfer in the receiving direction.

In device controller mode, the USBFS sets the SQMON bit to 1 (specifies DATA1 as the expected value) on successful reception of the setup packet.

In device controller mode, the USBFS does not reference this bit during IN or OUT transactions at the status stage, and it does not toggle the bit on normal completion.

SQSET bit (Sequence Toggle Bit Set)

The SQSET bit specifies DATA1 as the expected value of the sequence toggle bit for the next transaction during a DCP transfer.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

SQCLR bit (Sequence Toggle Bit Clear)

The SQCLR bit specifies DATA0 as the expected value of the sequence toggle bit for the next transaction during a DCP transfer. It is read as 0.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

SUREQCLR bit (SUREQ Bit Clear)

In host controller mode, setting the SUREQCLR bit to 1 clears the SUREQ bit to 0. The bit is read as 0.

If transfer stops while the SUREQ bit is set to 1 in a setup transaction, set the SUREQCLR bit to 1 by software. This is not necessary at the end of a normal setup transaction, because the USBFS automatically clears the SUREQ bit to 0.

Only control the SUREQ bit through the SUREQCLR bit while the DVSTCTR0.UACT bit is 0. When UACT is 0, communication is halted or no transfer is occurring because a bus disconnection was detected.

In device controller mode, always write 0 to this bit.

SUREQ bit (Setup Token Transmission)

In host controller mode, setting the SUREQ bit to 1 triggers the USBFS to transmit the setup packet. After completing the setup transaction process, the USBFS generates either the SACK or SIGN interrupt and clears the SUREQ bit to 0. The USBFS also clears the SUREQ bit to 0 when the software sets the SUREQCLR bit to 1.

Before setting the SUREQ bit to 1, set the DCPMAXP.DEVSEL[3:0] bits, USBREQ, USBVAL, USBINDX, and USBLENG appropriately to transmit the target USB request in the setup transaction. Also check that the PID[1:0] bits for the DCP are set to NAK. After setting the SUREQ bit to 1, do not change the DCPMAXP.DEVSEL[3:0] bits, USBREQ, USBVAL, USBINDX, or USBLENG until the setup transaction is complete (SUREQ bit = 1). Write 1 to the SUREQ bit only when transmitting the setup token. Otherwise, write 0.

In device controller mode, always write 0 to this bit.

BSTS flag (Buffer Status)

The BSTS flag indicates the status of access to the DCP FIFO buffer. The meaning of this flag varies as follows depending on the CFIFOSEL.ISEL setting:

- When ISEL = 0, the bit indicates whether receive data can be read from the buffer
- When ISEL = 1, the bit indicates whether transmit data can be written to the buffer

37.2.31 PIPESEL : Pipe Window Select Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x064

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PIPESEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	PIPESEL[3:0]	Pipe Window Select 0x0: No pipe selected 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Set pipes 1 to 9 using the PIPESEL, PIPECFG, PIPEMAXP, PIPEPERI, PIPEnCTR, PIPEnTRE, and PIPEnTRN registers (n = 1 to 9).

After selecting the pipe in the PIPESEL register, pipe functions must be set in the associated PIPECFG, PIPEMAXP, and PIPEPERI registers. PIPEnCTR, PIPEnTRE, and PIPEnTRN can be set independently of the pipe selection in this register.

PIPESEL[3:0] bits (Pipe Window Select)

The PIPESEL[3:0] bits select the pipe number associated with the PIPECFG, PIPEMAXP, and PIPEPERI registers used for data writing and reading. Selecting a pipe number in the PIPESEL[3:0] bits allows writing to and reading from PIPECFG, PIPEMAXP, and PIPEPERI associated with the selected pipe number.

When PIPESEL[3:0] = 0000b, 0 is read from all of the bits in PIPECFG, PIPEMAXP, and PIPEPERI. Writing to these bits has no effect.

37.2.32 PIPECFG : Pipe Configuration Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x068

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TYPE[1:0]		—	—	—	BFRE	DBLB	—	SHTNAK	—	—	DIR	EPNUM[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	EPNUM[3:0]	Endpoint Number*1 Specifies the endpoint number for the selected pipe. Setting 0000b indicates that the pipe is not used.	R/W
4	DIR	Transfer Direction*2 *3 0: Receiving direction 1: Transmitting direction	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	SHTNAK	Pipe Disabled at End of Transfer*1 0: Continue pipe operation after transfer ends 1: Disable pipe after transfer ends	R/W
8	—	This bit is read as 0. The write value should be 0.	R/W
9	DBLB	Double Buffer Mode*2 *3 0: Single buffer 1: Double buffer	R/W
10	BFRE	BRDY Interrupt Operation Specification*2 *3 0: Generate BRDY interrupt on transmitting or receiving data 1: Generate BRDY interrupt on completion of reading data	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W
15:14	TYPE[1:0]	Transfer Type*1 0 0: Pipe not used 0 1: Pipes 1 and 2: Bulk transfer Pipes 3 to 5: Bulk transfer Pipes 6 to 9: Setting prohibited 1 0: Pipes 1 and 2: Setting prohibited Pipes 3 to 5: Setting prohibited Pipes 6 to 9: Interrupt transfer 1 1: Pipes 1 and 2: Isochronous transfer Pipes 3 to 5: Setting prohibited Pipes 6 to 9: Setting prohibited	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the TYPE[1:0], SHTNAK, and EPNUM[3:0] bits while PID is NAK. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 2. Only set the BFRE, DBLB, and DIR bits while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 3. To change the BFRE, DBLB, or DIR bits after completing USB communication on the selected pipe, in addition to the constraints described in Note 2, write 1 and 0 to the PIPEnCTR.ACLRM bit continuously through the software and clear the FIFO buffer assigned to the pipe.

PIPECFG specifies the transfer type, FIFO buffer access direction, and endpoint numbers for pipes 1 to 9. It also selects single or double buffer mode, and whether to continue or disable pipe operation at the end of transfer.

EPNUM[3:0] bits (Endpoint Number)

The EPNUM[3:0] bits specify the endpoint number for the selected pipe. Setting 0000b indicates the pipe not used.

Set these bits so that the combination of the DIR and EPNUM[3:0] settings is different from those for other pipes. The EPNUM[3:0] bits can be set to 0000b for all pipes.

DIR bit (Transfer Direction)

The DIR bit specifies the transfer direction for the selected pipe.

When the software sets this bit to 0, the USBFS uses the selected pipe for receiving. When the software sets this bit to 1, the USBFS uses the selected pipe for transmitting.

SHTNAK bit (Pipe Disabled at End of Transfer)

The SHTNAK bit specifies whether to change the PIPEnCTR.PID[1:0] bits to 00b (NAK) at the end of transfer when the selected pipe is set in the receiving direction. The bit is valid for pipes 1 to 5 in the receiving direction.

When the software sets this bit to 1 for a receiving pipe, the USBFS changes the associated PIPEnCTR.PID[1:0] bits to 00b (NAK) on determining the transfer end. The USBFS determines that the transfer has ended on the following conditions:

- A short packet data (including a zero-length packet) was successfully received
- The transaction counter is used and the number of packets specified for the transaction counter are successfully received

DBLB bit (Double Buffer Mode)

The DBLB bit selects either single or double buffer mode for the FIFO buffer used by the selected pipe. The bit is valid for pipes 1 to 5.

BFRE bit (BRDY Interrupt Operation Specification)

The BFRE bit specifies the BRDY interrupt generation timing from the USBFS to the CPU for the selected pipe.

When the software sets the BFRE bit to 1 and the selected pipe is in the receiving direction, the USBFS detects the transfer completion and generates the BRDY interrupt on reading the packet.

When a BRDY interrupt is generated with this setting, the software must write 1 to the BCLR bit in the port control register. The FIFO buffer assigned to the selected pipe is not enabled for reception until 1 is written to the BCLR bit.

When the BFRE bit is set to 1 by software and the selected pipe is in the transmitting direction, the USBFS does not generate the BRDY interrupt. For details, see section 37.3.3.1. BRDY Interrupt.

TYPE[1:0] bits (Transfer Type)

The TYPE[1:0] bits specify the transfer type for the pipe selected in the PIPESEL.PIPESEL[3:0] bits. Before setting PID to BUF and starting USB communication on the selected pipe, set the TYPE[1:0] bits to a value other than 00b.

37.2.33 PIPEMAXP : Pipe Maximum Packet Size Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x06C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DEVSEL[3:0]				—	—	—	MXPS[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	x	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	MXPS[8:0]	Maximum Packet Size*1 ● Pipes 1 and 2 1 byte (0x001) to 256 bytes (0x100) (Bit [9] not supported.) ● Pipes 3 to 5 8 bytes (0x008), 16 bytes (0x010), 32 bytes (0x020), 64 bytes (0x040) (Bits [9:7] and [2:0] not supported.) ● Pipes 6 to 9 1 byte (0x001) to 64 bytes (0x040) (Bits [9:7] not supported.)	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W
15:12	DEVSEL[3:0]	Device Select*2 0x0: Address 0000b 0x1: Address 0001b 0x2: Address 0010b 0x3: Address 0011b 0x4: Address 0100b 0x5: Address 0101b Others: Setting prohibited	R/W

Note: S-TYPE-3, P-TYPE-3

Note: The value of the MXPS[8:0] bits is 0x000 when no pipe is selected in the PIPESEL.PIPESEL[3:0] bits and 0x040 when a pipe is selected.

Note 1. Only set the MXPS[8:0] bits while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 2. Only set the DEVSEL[3:0] bits while PID is NAK. Before setting these bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00b (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

PIPEMAXP specifies the maximum packet size for pipes 1 to 9.

MXPS[8:0] bits (Maximum Packet Size)

The MXPS[8:0] bits specify the maximum data payload (maximum packet size) for the selected pipe.

Set these bits to the appropriate value for each transfer type based on the USB 2.0 specification. When MXPS[8:0] = 0, do not write to the FIFO buffer or set PID to BUF. These writes have no effect.

DEVSEL[3:0] bits (Device Select)

In host controller mode, the DEVSEL[3:0] bits specify the address of the target device for USB communication. Set up the device address in the associated DEVADDn (n = 0 to 5) register first, and then set these bits to the corresponding value. To set the DEVSEL[3:0] bits to 0x2, for example, first set the address in the DEVADD2 register.

In device controller mode, set these bits to 0x0.

37.2.34 PIPEPERI : Pipe Cycle Control Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x06E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	IFIS	—	—	—	—	—	—	—	—	—	IITV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	IITV[2:0]*1	Interval Error Detection Interval Specifies the interval error detection timing for the selected pipe as the n-th power of 2 of the frame timing	R/W
11:3	—	These bits are read as 0. The write value should be 0.	R/W
12	IFIS	Isochronous IN Buffer Flush 0: Do not flush buffer 1: Flush buffer	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the IITV[2:0] bits while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

PIPEPERI selects whether the buffer is flushed or not when an interval error occurred during isochronous IN transfers, and sets the interval error detection interval for pipes 1 to 9.

IITV[2:0] bits (Interval Error Detection Interval)

To change the IITV[2:0] bits to another value after they are set and USB communication is performed, set the PIPEnCTR.PID[1:0] bits to 00b (NAK) and then set the PIPEnCTR.ACLRM bit to 1 to initialize the interval timer.

The IITV[2:0] bits are not provided for pipes 3 to 5. Write 000b to bit positions of the IITV[2:0] bits associated with pipes 3 to 5.

IFIS bit (Isochronous IN Buffer Flush)

The IFIS bit specifies whether to flush the buffer when the pipe selected in the PIPESEL.PIPESEL[3:0] bits is used for isochronous IN transfers.

In device controller mode when the selected pipe is for isochronous IN transfers, the USBFS automatically clears the FIFO buffer if the USBFS fails to receive the IN token from the USB host within the interval set in theIITV[2:0] bits in terms of frames.

When double buffering is specified (PIPECFG.DBLB = 1), the USBFS only clears the data in the previously used plane.

The USBFS clears the FIFO buffer on receiving the SOF packet immediately after the frame in which the USBFS expected to receive the IN token. Even if the SOF packet is corrupted, the FIFO buffer is cleared at the time the SOF packet is expected to be received by using the internal interpolation function.

When the host controller function is selected, set this bit to 0.

Set this bit to 0 when the selected pipe is not for isochronous transfer.

37.2.35 PIPEnCTR : PIPEn Control Registers (n = 1 to 5)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x070 + 0x2 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	INBUFM	—	—	—	ATREPM	ACLRM	SQCLR	SQSET	SQMON	PBUSY	—	—	—	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID 0 0: NAK response 0 1: BUF response (depends buffer state) 1 0: STALL response 1 1: STALL response	R/W
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	PBUSY	Pipe Busy 0: Pipe n not in use for the transaction 1: Pipe n in use for the transaction	R
6	SQMON	Sequence Toggle Bit Confirmation 0: DATA0 1: DATA1	R
7	SQSET	Sequence Toggle Bit Set*2 Sets the sequence toggle bit for pipe n. 0: Invalid (writing 0 has no effect) 1: Set the expected value for the next transaction to DATA1	R/W*1
8	SQCLR	Sequence Toggle Bit Clear*2 Clears the sequence toggle bit for pipe n. 0: Invalid (writing 0 has no effect) 1: Clear the expected value for the next transaction to DATA0	R/W*1
9	ACLRM	Auto Buffer Clear Mode*3 0: Disable 1: Enable (initialize all buffers)	R/W
10	ATREPM	Auto Response Mode*2 0: Disable auto response mode 1: Enable auto response mode	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W
14	INBUFM	Transmit Buffer Monitor 0: No data to be transmitted is in the FIFO buffer 1: Data to be transmitted is in the FIFO buffer	R
15	BSTS	Buffer Status 0: Buffer access by the CPU disabled 1: Buffer access by the CPU enabled	R

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is read as 0.
Note 2. Only set the ATREPM bit or write 1 to the SQCLR or SQSETbit while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSYbit through the software is not necessary.
Note 3. Only set the ACLRM bit while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting this bit, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00 (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

PIPEnCTR can be set for any pipe selection in the PIPESEL register.

PID[1:0] bits (Response PID)

The PID[1:0]bits specify the response type for the next transaction on the selected pipe.

The default PID[1:0] setting is NAK. Change the PID[1:0] setting to BUF to use the associated pipe for USB transfer. Table 37.9 and Table 37.10 show the basic operations of the USBFS (when there are no errors in the communication packets) based on the PID[1:0] bit setting.

After changing the PID[1:0] setting from BUF to NAK through the software during USB communication on the selected pipe, check that the PBUSY bit is 1 to see if USB transfer on the pipe has actually entered the NAK state. If the USBFS changes the PID[1:0] bits to NAK, checking the PBUSY bit through the software is not necessary.

The USBFS changes the PIPEnCTR.PID[1:0] setting in the following cases:

- The USBFS sets PID to NAK on recognizing completion of the transfer when the selected pipe is in the receiving direction and the PIPECFG.SHTNAK bit for the selected pipe is set to 1 by software
- The USBFS sets PID to STALL (11b) on receiving a data packet with a payload exceeding the maximum packet size of the selected pipe
- The USBFS sets PID to NAK on detecting a USB bus reset in device controller mode
- The USBFS sets PID to NAK on detecting a reception error, such as a CRC error, three consecutive times in host controller mode
- The USBFS sets PID to STALL (11b) on receiving the STALL handshake in host controller mode

To specify the response type, set the PID[1:0] bits as follows:

- To transition from NAK (00b) to STALL, set 10b
- To transition from BUF (01b) to STALL, set 11b
- To transition from STALL (11b) to NAK, set 10b and then 00b
- To transition from STALL to BUF, set 00b (NAK) and then 01b (BUF)

Table 37.9 Operation of the USBFS based on the PID[1:0] setting in host controller mode

PID[1:0] value	Transfer type	Transfer direction (DIR bit)	USBFS operation
00b (NAK)	Does not depend on the setting	Does not depend on the setting	Does not issue tokens
01b (BUF)	Bulk or interrupt	Does not depend on the setting	Issues tokens when the DVSTCTR0.UACT bit is 1 and the FIFO buffer associated with the selected pipe is ready for transmission and reception. Does not issue tokens when the DVSTCTR0.UACT bit is 0 or the FIFO buffer associated with the selected pipe is not ready for transmission or reception.
	Isochronous	Does not depend on the setting	Issues tokens regardless of the status of the FIFO buffer associated with the selected pipe.
10b (STALL) or 11b (STALL)	Does not depend on the setting	Does not depend on the setting	Does not issue tokens.

Table 37.10 Operation of the USBFS based on the PID[1:0] setting in device controller mode (1 of 2)

PID[1:0] value	Transfer type	Transfer direction (DIR bit)	USBFS operation
00b (NAK)	Bulk or interrupt	Does not depend on the setting	Returns NAK in response to the token from the USB host
	Isochronous	Does not depend on the setting	Returns nothing in response to the token from the USB host

Table 37.10 Operation of the USBFS based on the PID[1:0] setting in device controller mode (2 of 2)

PID[1:0] value	Transfer type	Transfer direction (DIR bit)	USBFS operation
01b (BUF)	Bulk	Receiving direction (DIR = 0)	Receives data and returns ACK in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception
	Interrupt	Receiving direction (DIR = 0)	Receives data and returns ACK in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception
	Bulk or interrupt	Transmitting direction (DIR = 1)	Transmits data in response to the token from the USB host if the FIFO buffer associated with the selected pipe is ready for transmission. Otherwise, returns NAK.
	Isochronous	Receiving direction (DIR = 0)	Receives data in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception. Otherwise, discards the data.
	Isochronous	Transmitting direction (DIR = 1)	Transmits data in response to the token from the USB host if the associated FIFO buffer is ready for transmission. Otherwise, transmits a zero-length packet.
10b (STALL) or 11b (STALL)	Bulk or interrupt	Does not depend on the setting	Returns STALL in response to the token from the USB host
	Isochronous	Does not depend on the setting	Returns nothing in response to the token from the USB host

PBUSY bit (Pipe Busy)

The PBUSY bit indicates whether the selected pipe is being used for the current transaction.

The USBFS changes the PBUSY bit from 0 to 1 on start of the USB transaction for the selected pipe, and changes the PBUSY bit from 1 to 0 on completion of one transaction.

Reading the PBUSY bit by software after PID is set to NAK allows you to check whether changing the pipe setting is possible. For details, see section 37.3.4.1. Pipe Control Register Switching Procedures.

SQMON bit (Sequence Toggle Bit Confirmation)

The SQMON bit indicates the expected value of the sequence toggle bit for the next transaction of the selected pipe.

When the selected pipe is not the isochronous transfer type, the USBFS toggles the SQMON flag on successful completion of the transaction. However, the USBFS does not toggle the SQMON flag when a DATA-PID mismatch occurs during transfer in the receiving direction.

SQSET bit (Sequence Toggle Bit Set)

Setting the SQSET bit to 1 through the software causes the USBFS to set DATA1 as the expected value of the sequence toggle bit for the next transaction on the selected pipe. The USBFS clears the SQSET bit to 0.

SQCLR bit (Sequence Toggle Bit Clear)

Setting the SQCLR bit to 1 through the software causes the USBFS to clear the expected value of the sequence toggle bit for the next transaction on the selected pipe to DATA0. The USBFS clears the SQCLR bit to 0.

ACLRM bit (Auto Buffer Clear Mode)

The ACLRM bit enables or disables auto buffer clear mode for the selected pipe. To completely clear the data in the FIFO buffer allocated to the selected pipe, write 1 and then 0 to the ACLRM bit continuously.

Table 37.11 shows the data cleared by writing 1 and 0 to the ACLRM bit continuously and the cases in which this processing is required.

Table 37.11 Data cleared by the USBFS when ACLRM = 1 (1 of 2)

Number	Data cleared by setting the ACLRM bit	Situations requiring data clear
1	All data in the FIFO buffer allocated to the selected pipe (two FIFO buffers in double buffer mode)	When initializing the selected pipe

Table 37.11 Data cleared by the USBFS when ACLRM = 1 (2 of 2)

Number	Data cleared by setting the ACLRM bit	Situations requiring data clear
2	Interval count value when the selected pipe is the isochronous transfer type	When resetting the interval count value
3	Internal flags related to the PIPECFG.BFRE bit	When changing the PIPECFG.BFRE setting
4	FIFO buffer toggle control	When changing the PIPECFG.DBLB setting
5	Internal flags related to the transaction count	When forcing the transaction count function to terminate

ATREPM bit (Auto Response Mode)

The ATREPM bit enables or disables auto response mode for the selected pipe.

This bit can be set to 1 in device controller mode when the selected pipe is the bulk transfer type. When the bit is set to 1, the USBFS responds to the token from the USB host as follows:

- When the selected pipe is set for bulk IN transfers (PIPECFG.TYPE[1:0] = 01b and PIPECFG.DIR = 1):
  a. When the ATREPM bit = 1 and PID = BUF, the USBFS transmits a zero-length packet in response to the IN token.
  b. The USBFS updates the sequence toggle bit (DATA-PID) each time the USBFS receives ACK from the USB host. In a single transaction, the IN token is received, a zero-length packet is transmitted, and then ACK is received. The USBFS does not generate the BRDY or BEMP interrupt.
- When the selected pipe is set for bulk OUT transfers (PIPECFG.TYPE[1:0] = 01b and PIPECFG.DIR = 0):

  When the ATREPM bit = 1 and PID = BUF, the USBFS returns NAK in response to the OUT token and generates an NRDY interrupt.

For USB communication in auto response mode, set the ATREPM bit to 1 while the FIFO buffer is empty. Do not write to the FIFO buffer during USB communication in auto response mode. When the selected pipe uses isochronous transfer, always set this bit to 0.

In host controller mode, always set the ATREPM bit to 0.

INBUFM bit (Transmit Buffer Monitor)

The INBUFM bit indicates the FIFO buffer status for the selected pipe in the transmitting direction.

When the selected pipe is set in the transmitting direction (PIPECFG.DIR = 1), the USBFS sets this bit to 1 when the CPU or DMA/DTC completes writing data to at least one FIFO buffer plane.

The USBFS sets this bit to 0 when the USBFS completes transmission of the data from the FIFO buffer plane to which all the data is written. In double buffer mode (PIPECFG.DBLB = 1), the USBFS sets the INBUFM bit to 0 when the USBFS completes transmission of the data from the two FIFO buffer planes before the CPU or DMA/DTC completes writing data to one FIFO buffer plane.

The INBUFM bit indicates the same value as the BSTS bit when the selected pipe is in the receiving direction (PIPECFG.DIR = 0).

BSTS bit (Buffer Status)

The BSTS bit indicates the FIFO buffer status for the selected pipe.

The meaning of the BSTS bit depends on the PIPECFG.DIR, PIPECFG.BFRE, and DnFIFOSEL.DCLRM settings, as shown in Table 37.12.

Table 37.12 BSTS bit operation

DIR value	BFRE value	DCLRM value	BSTS bit function
0	0	0	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 on completion of data read
		1	Setting prohibited
	1	0	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 when the software sets the BCLR bit in the port control register to 1 after the data read is complete
		1	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 on completion of data read
1	0	0	Sets to 1 when transmit data can be written to the FIFO buffer, and clears to 0 on completion of data write
		1	Setting prohibited
	1	0	Setting prohibited
		1	Setting prohibited

37.2.36 PIPEnCTR : PIPEn Control Registers (n = 6 to 9)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x07A + 0x2 × (n - 6)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	—	—	—	—	—	ACLRM	SQCLR	SQSET	SQMON	PBUSY	—	—	—	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID 0 0: NAK response 0 1: BUF response (depends buffer state) 1 0: STALL response 1 1: STALL response	R/W
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	PBUSY	Pipe Busy 0: Pipe n not in use for the transaction 1: Pipe n in use for the transaction	R
6	SQMON	Sequence Toggle Bit Confirmation 0: DATA0 1: DATA1	R
7	SQSET	Sequence Toggle Bit Set*1 Sets the sequence toggle bit for pipe n. 0: Invalid (writing 0 has no effect) 1: Set the expected value for the next transaction to DATA0	W
8	SQCLR	Sequence Toggle Bit Clear*1 Clears the sequence toggle bit for pipe n. 0: Invalid (writing 0 has no effect) 1: Clear the expected value for the next transaction to DATA0	W
9	ACLRM	Auto Buffer Clear Mode*2 0: Disable 1: Enable (all buffers initialized)	R/W
14:10	—	These bits are read as 0. The write value should be 0.	R/W
15	BSTS	Buffer Status 0: Buffer access disabled 1: Buffer access enabled	R

Note: S-TYPE-3, P-TYPE-3

Note 1. Only write 1 to the SQCLR or SQSET bit while PID is NAK. Before setting these bits, check that the PBUSY bit is 0, and then change the PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00b (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

Note 2. Only set the ACLRM bit while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting this bits, check that the PIPEnCTR.PBUSY bit is 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PID[1:0] bits are changed to 00b (NAK) by the USBFS, checking the PBUSY bit through the software is not necessary.

PID[1:0] bits (Response PID)

The PID[1:0]bits specify the response type for the next transaction of the selected pipe.

The default PID[1:0] setting is NAK. Change the PID[1:0] setting to BUF to use the associated pipe for USB transfer. Table 37.9 and Table 37.10 show the basic operation (when there are no errors in the transmitted and received packets) of the USBFS depending on the PID[1:0] setting.

After changing the PID[1:0] setting from BUF to NAK through the software during USB communication on the selected pipe, check that the PBUSY bit is 1 to see if USB transfer on the selected pipe has actually entered the NAK state. If the USBFS changes the PID[1:0] bits to NAK, checking the PBUSY bit through the software is not necessary.

The USBFS changes the PIPEnCTR.PID[1:0] setting in the following cases:

- The USBFS sets PID to STALL (11b) on receiving a data packet with a payload exceeding the maximum packet size of the selected pipe
- The USBFS sets PID to NAK on detecting a USB bus reset in device controller mode
- The USBFS sets PID to NAK on detecting a reception error, such as a CRC error, three consecutive times in host controller mode
- The USBFS sets PID to STALL (11b) on receiving the STALL handshake in host controller mode

To specify each response type, set the PID[1:0] bits as follows:

- To transition from NAK (00b) to STALL, set 10b
- To transition from BUF (01b) to STALL, set 11b
- To transition from STALL (11b) to NAK, set 10b and then 00b
- To transition from STALL to BUF, set 00b (NAK) and then 01b (BUF)

PBUSY bit (Pipe Busy)

The PBUSY bit indicates whether the selected pipe is being used for the current transaction.

The USBFS changes the PBUSY bit from 0 to 1 on start of the USB transaction for the selected pipe, and changes the PBUSY bit from 1 to 0 on completion of one transaction.

Reading the PBUSY bit by software after PID is set to NAK allows you to check whether changing the pipe setting is possible.

SQMON bit (Sequence Toggle Bit Confirmation)

The SQMON bit indicates the expected value of the sequence toggle bit for the next transaction of the selected pipe.

The USBFS toggles the SQMON bit on successful completion of the transaction. However, the USBFS does not toggle the SQMON bit when a DATA-PID mismatch occurs during transfer in the receiving direction.

SQSET bit (Sequence Toggle Bit Set)

Setting the SQSET bit to 1 through the software causes the USBFS to set DATA1 as the expected value of the sequence toggle bit for the next transaction on the selected pipe. The USBFS sets the SQSET bit to 0.

SQCLR bit (Sequence Toggle Bit Clear)

Setting the SQCLR bit to 1 through the software causes the USBFS to clear the expected value of the sequence toggle bit for the next transaction on the selected pipe to DATA0. The USBFS sets the SQCLR bit to 0.

ACLRM bit (Auto Buffer Clear Mode)

The ACLRM bit enables or disables auto buffer clear mode for the selected pipe. To completely clear the data in the FIFO buffer allocated to the selected pipe, write 1 and then 0 to the ACLRM bit continuously.

Table 37.13 shows the data cleared by writing 1 and 0 continuously to the ACLRM bit and the cases in which this processing is required.

Table 37.13 Data cleared by the USBFS when ACLRM = 1

Number	Data cleared by setting the ACLRM bit	Situations requiring data clear
1	All data in the FIFO buffer allocated to the selected pipe	When initializing the selected pipe
2	Interval count value when the selected pipe is the isochronous transfer type	When resetting the interval count value
3	Internal flags related to the PIPECFG.BFRE bit	When changing the PIPECFG.BFRE setting
4	Internal flags related to the transaction count	When forcing the transaction count function to terminate

BSTS bit (Buffer Status)

The BSTS bit indicates the FIFO buffer status for the selected pipe.

The meaning of the BSTS bit depends on the PIPECFG.DIR, PIPECFG.BFRE, and DnFIFOSEL.DCLRM settings, as shown in Table 37.12.

37.2.37 PIPEnTRE : PIPEn Transaction Counter Enable Register (n = 1 to 5)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x090 + 0x4 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TRENB	TRCLR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	TRCLR	Transaction Counter Clear 0: Invalid (writing 0 has no effect) 1: Clear counter value	R/W
9	TRENB	Transaction Counter Enable 0: Disable transaction counter 1: Enable transaction counter	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Set each bit in PIPEnTRE while PID is NAK. Before setting these bits after changing the PIPEnCTR.PID[1:0] bits for the selected pipe from BUF to NAK, check that the PIPEnCTR.PBUSY bit is 0. However, if the PID[1:0] bits are changed to NAK by the USBFS, checking the PBUSY bit through the software is not necessary.

TRCLR bit (Transaction Counter Clear)

When the TRCLR bit sets to 1, the USBFS clears the value of the transaction counter associated with the selected pipe and then sets the TRCLR bit to 0.

TRENB bit (Transaction Counter Enable)

The TRENB bit enables or disables the transaction counter.

For receiving pipes, setting the TRENB bit to 1 after setting the total number of the packets to be received in the PIPEnTRN.TRNCNT[15:0] bits through the software allows the USBFS to control hardware on having received the number of packets equal to the TRNCNT[15:0] setting, as follows:

- When the PIPECFG.SHTNAK bit is 1, the USBFS changes the PID bits to NAK for the associated pipe on having received the number of packets equal to the TRNCNT[15:0] setting
- When the PIPECFG.BFRE bit is 1, the USBFS asserts the BRDY interrupt on having received the number of packets equal to the TRNCNT[15:0] setting and then reading the last received data

For transmitting pipes, set the TRENB bit to 0.

When the transaction counter is not used, set this bit to 0. When the transaction counter is used, set the TRNCNT[15:0] bits before setting this bit to 1. Set this bit to 1 before receiving the first packet to be counted by the transaction counter.

37.2.38 PIPEnTRN : PIPEn Transaction Counter Register (n = 1 to 5)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x092 + 0x4 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TRNCNT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TRNCNT[15:0]	Transaction Counter When written to, this bit specifies the total packets (number of transactions) to be received by the selected pipe. When read from, when PIPEnTRE.TRENB is 0, this bit indicates the specified number of transactions. When PIPEnTRE.TRENB is 1, this bit indicates the current transaction count.	R/W

Note: S-TYPE-3, P-TYPE-3

The PIPEnTRN registers retain their settings during a USB bus reset.

TRNCNT[15:0] bits (Transaction Counter)

The USBFS increments the value of the TRNCNT[15:0] bits by 1 when all of the following conditions are satisfied on receiving the packet:

- The PIPEnTRE.TRENB bit = 1
- (TRNCNT[15:0] set value ≠ current counter value + 1) on receiving the packet
- The payload of the received packet agrees with the PIPEMAXP.MXPS[9:0] setting

The USBFS clears the value of the TRNCNT[15:0] bits to 0 when any of the following conditions are satisfied:

All of the following conditions are satisfied:

- The PIPEnTRE.TRENB bit = 1
- (TRNCNT[15:0] set value = current counter value + 1) on receiving the packet
- The payload of the received packet agrees with the PIPEMAXP.MXPS[9:0] setting

Both of the following conditions are satisfied:

- The PIPEnTRE.TRENB bit = 1
- The USBFS received a short packet

Both of the following conditions are satisfied:

- The PIPEnTRE.TRENB bit = 1
- The PIPEnTRE.TRCLR bit was set to 1 by software

For transmitting pipes, set the TRNCNT[15:0] bits to 0. When the transaction counter is not used, set the TRNCNT[15:0] bits to 0.

Setting the number of transactions to be transferred to the TRNCNT[15:0] bits is only enabled when the PIPEnTRE.TRENB bit is 0. To set the number of transactions to be transferred, set the TRCLR bit to 1 to clear the current counter value before setting the PIPEnTRE.TRENB bit to 1.

37.2.39 DEVADDn : Device Address n Configuration Register (n = 0 to 5)

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x0D0 + 0x2 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	USBSPD[1:0]		—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
7:6	USBSPD[1:0]	Transfer Speed of Communication Target Device 0 0: Do not use DEVADDn 0 1: Low-speed 1 0: Full-speed 1 1: Setting prohibited	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The DEVADDn register specifies the transfer speed of the peripheral device that is the communication target for pipes 0 to 9.

In host controller mode, set all DEVADDn bits before starting communication to any pipes. Only change the bits in DEVADDn when no valid pipes are using the bit settings. A valid pipe is defined as one that satisfies both of the following conditions:

- The target device of the DEVADDn register is selected in the DEVSEL[3:0] bits
- The PID[1:0] bits are set to BUF for the selected pipe, or the selected pipe is the DCP with the DCPCTR.SUREQ bit set to 1

In device controller mode, set all bits in this register to 0.

USBSPD[1:0] bits (Transfer Speed of Communication Target Device)

TheUSBSPD[1:0] bits specify the USB transfer speed of the target peripheral device. Set these bits to 10b when a full-speed device is connected through the hub. In host controller mode, the USBFS generates packets based on the USBSPD[1:0] setting. In device controller mode, set these bits to 00b.

37.2.40 DPUSR0R : Deep Software Standby USB Transceiver Control/Pin Monitor Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x400

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	DVBSTS0	—	DOVCB0	DOVCA0	—	—	DM0	DP0
Value after reset:	0	0	0	0	0	0	0	0	x	0	x	x	0	0	x	x

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	FIXPHY0	DRPD0	—	RPUE0	SRPC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SRPC0*1	USB Single-ended Receiver Control 0: Disable input through DP and DM inputs 1: Enable input through DP and DM inputs	R/W
1	RPUE0*1	DP Pull-Up Resistor Control 0: Disable DP pull-up resistor 1: Enable DP pull-up resistor	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	DRPD0*1	D+/D- Pull-Down Resistor Control 0: Disable DP/DM pull-down resistor 1: Enable DP/DM pull-down resistor	R/W
4	FIXPHY0	USB Transceiver Output Fix 0: Fix outputs in Normal mode and on return from Deep Software Standby mode 1 1: Fix outputs on transition to Deep Software Standby mode 1	R/W
15:5	—	These bits are read as 0. The write value should be 0.	R/W
16	DP0	USB D+ Input Indicates D+ input signal on the USBFS side	R
17	DM0	USB D- Input Indicates D– input signal on the USBFS side	R
19:18	—	These bits are read as 0. The write value should be 0.	R/W
20	DOVCA0	USB OVRCURA-DS Input*2 Indicates OVRCURA-DS input signal on the USBFS side	R
21	DOVCB0	USB OVRCURB-DS Input*2 Indicates OVRCURB-DS input signal on the USBFS side	R
22	—	The read value is undefined. The write value should be 0.	R/W
23	DVBSTS0	USB VBUS Input Indicates VBUS input signal on the USBFS side	R
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: This register is accessed by the clock of PCLKB/64 to reduce current consumption in Deep Software Standby mode1.
Note 1. Use this bit during operation in Deep Software Standby mode 1. For details, see section 37.3.1.5. Release from Deep Software Standby Mode 1 Because of USB Suspend/Resume Interrupts.
Note 2. OVRCURA or OVRCURB can not be used. Only OVRCURA-DS or OVRCURB-DS can be used in Deep software standby mode1.

SRPC0 bit (USB Single-ended Receiver Control)

The SRPC0 bit controls the D+ and D- inputs of the USB transceiver. In host controller mode, set this bit to 1. In device controller mode, set this bit to 0 when disconnected, set to 1 when suspended. This bit is only valid when the FIXPHY0 bit is 1.

FIXPHY0 bit (USB Transceiver Output Fix)

The FIXPHY0 bit keeps the outputs of the USB transceiver disabled.

37.2.41 DPUSR1R : Deep Software Standby USB Suspend/Resume Interrupt Register

Base address: USBFS = 0x4025_0000
USBFS_NS = 0x5025_0000

Offset address: 0x404

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	DVBINT0	—	DOVRCRB0	DOVRCRA0	—	—	DMINT0	DPINT0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	DVBSE0	—	DOVRCRBE0	DOVRCRAE0	—	—	DMINTE0	DPINTE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DPINTE0	USB DP Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode 1 by DP input 1: Enable recovery from Deep Software Standby mode 1 by DP input	R/W
1	DMINTE0	USB DM Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode 1 by DM input 1: Enable recovery from Deep Software Standby mode by DM input	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	DOVRCRAE0	USB OVRCURA-DS Interrupt Enable/Clear*1 0: Disable recovery from Deep Software Standby mode 1 by OVRCURA-DS input 1: Enable recovery from Deep Software Standby mode 1 by OVRCURA-DS input	R/W
5	DOVRCRBE0	USB OVRCURB-DS Interrupt Enable/Clear*1 0: Disable recovery from Deep Software Standby mode 1 by OVRCURB-DS input 1: Enable recovery from Deep Software Standby mode 1 by OVRCURB-DS input	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	DVBSE0	USB VBUS Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode 1 by VBUS input 1: Enable recovery from Deep Software Standby mode 1 by VBUS input	R/W
15:8	—	These bits are read as 0. The write value should be 0..	R/W
16	DPINT0	USB DP Interrupt Source Recovery 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1 because of DP	R
17	DMINT0	USB DM Interrupt Source Recovery 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1 because of DM input	R
19:18	—	These bits are read as 0. The write value should be 0.	R/W
20	DOVRCRA0	USB OVRCURA-DS Interrupt Source Recovery*1 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1 because of OVRCURA-DS input	R
21	DOVRCRB0	USB OVRCURB-DS Interrupt Source Recovery*1 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1 because of OVRCURB-DS input	R
22	—	This bit is read as 0. The write value should be 0.	R/W
23	DVBINT0	USB VBUS Interrupt Source Recovery 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1 because of VBUS input	R
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: This register is accessed by the clock of PCLKB/64 to reduce current consumption in Deep Software Standby mode1.
Note 1. OVRCURA or OVRCURB can not be used. Only OVRCURA-DS or OVRCURB-DS can be used in Deep Software Standby mode1.

DPINTE0 bit (USB DP Interrupt Enable/Clear)

The DPINTE0 bit enables or disables triggering of recovery from Deep Software Standby mode 1 by the DP input of the USBFS. Writing 0 to this bit while the DPINT0 bit is 1 sets the DPINT0 bit to 0.

DMINTE0 bit (USB DM Interrupt Enable/Clear)

The DMINTE0 bit enables or disables triggering of recovery from Deep Software Standby mode 1 by the DM input of the USBFS. Writing 0 to this bit while the DMINT0 bit is 1 clears the DMINT0 bit to 0.

DOVRCRAE0 bit (USB OVRCURA-DS Interrupt Enable/Clear)

The DOVRCRAE0 bit enables or disables triggering of recovery from Deep Software Standby mode 1 by the OVRCURA-DS input of the USBFS. Writing 0 to this bit while the DOVRCRA0 bit is 1 clears the DOVRCRA0 bit to 0.

DOVRCRBE0 bit (USB OVRCURB-DS Interrupt Enable/Clear)

The DOVRCRBE0 bit enables or disables triggering of recovery from Deep Software Standby mode 1 by the OVRCURB-DS input of the USBFS. Writing 0 to this bit while the DOVRCRB0 bit is 1 clears the DOVRCRB0 bit to 0.

DVBSE0 bit (USB VBUS Interrupt Enable/Clear)

The DVBSE0 bit enables or disables triggering of recovery from Deep Software Standby mode 1 by the VBUS input of the USBFS. Writing 0 to this bit while the DVBINT0 bit is 1 clears the DVBINT0 bit to 0.

DPINT0 bit (USB DP Interrupt Source Recovery)

The DPINT0 bit indicates that the system has returned from Deep Software Standby mode 1 because of the DP input of the USBFS. This recovery is only enabled when the DPINTE0 bit is 1. Writing 0 to the DPINTE0 bit while this bit is 1 clears this bit to 0.

DMINT0 bit (USB DM Interrupt Source Recovery)

The DMINT0 bit indicates that the system has returned from Deep Software Standby mode 1 because of the DM input of the USBFS. This recovery is only enabled when the DMINTE0 bit is 1. Writing 0 to the DMINTE0 bit while this bit is 1 clears this bit to 0.

DOVRCRA0 bit (USB OVRCURA-DS Interrupt Source Recovery)

The DOVRCRA0 bit indicates that the system has returned from Deep Software Standby mode 1 because of the OVRCURA-DS input of the USBFS. This recovery is only enabled when the DOVRCRAE0 bit is 1. Writing 0 to the DOVRCRAE0 bit while this bit is 1 clears this bit to 0.

DOVRCRB0 bit (USB OVRCURB-DS Interrupt Source Recovery)

The DOVRCRB0 bit indicates that the system has returned from Deep Software Standby mode 1 because of the OVRCURB-DS input of the USBFS. This recovery is only enabled when the DOVRCRBE0 bit is 1. Writing 0 to the DOVRCRBE0 bit while this bit is 1 clears this bit to 0.

DVBINT0 bit (USB VBUS Interrupt Source Recovery)

The DVBINT0 bit indicates that the system has returned from Deep Software Standby mode 1 because of the VBUS input of the USBFS. This recovery is only enabled when the DVBSE0 bit is 1. Writing 0 to the DVBSE0 bit while this bit is 1 clears this bit to 0.

37.3 Operation

37.3.1 System Control

This section describes register settings required for initializing the USBFS and controlling power consumption.

37.3.1.1 Setting Data to the USBFS Registers

Setting the SYSCFG.USBE bit to 1 after starting the clock supply (SYSCFG.SCKE bit = 1) enables and starts USBFS operation.

37.3.1.2 Selecting the Controller Function

The USBFS can operate as either a host or device controller.

Use the SYSCFG.DCFM bit to select one of these USBFS functions. The DCFM bit must be changed in the initial settings immediately after a reset or in the D+ pull-up-disabled state (SYSCFG.DPRPU bit = 0) and D+ and D– pull-down-disabled state (SYSCFG.DRPD bit = 0).

37.3.1.3 Controlling the USB Data Bus Using Resistors

The USBFS provides pull-up and pull-down resistors for the D+ and D– lines. Pull these lines up or down by setting the SYSCFG.DPRPU and DRPD bits.

In device controller mode, confirm that connection to the USB host is made, and then set the SYSCFG.DPRPU bit to 1 and pull up the D+ line (in full-speed communication).

When the SYSCFG.DPRPU bit is set to 0 during communication with a PC, the USBFS disables the pull-up resistor of the USB data line, thereby notifying the USB host of disconnection.

In host controller mode, set the SYSCFG.DRPD bit to 1 to pull down the D+ and D– lines.

Table 37.14 USB data bus resistor control

SYSCFG register settings		USB data bus control		
DRPD bit	DPRPU bit	D–	D+	Function
0	0	Open	Open	When resistors not used
0	1	Open	Pull-up	When operating as a device controller at full-speed
1	0	Pull-down	Pull-down	When operating as a host controller
1	1	—	—	Setting prohibited

37.3.1.4 Example External Connection Circuits

Figure 37.2 shows an example OTG connection in the self-powered system. The USBFS controls the pull-up resistor of the D+ line and the pull-down resistor of D+ and D- lines. Select pull-up and pull-down for the lines in the SYSCFG.DPRPU and SYSCFG.DRPD bits. In device controller mode, the pull-up resistor of USB data line is disabled if SYSCFG.DPRPU bit is set to 0 while communicating with the USB host. The USBFS can use this to notify the USB host of a device disconnect.

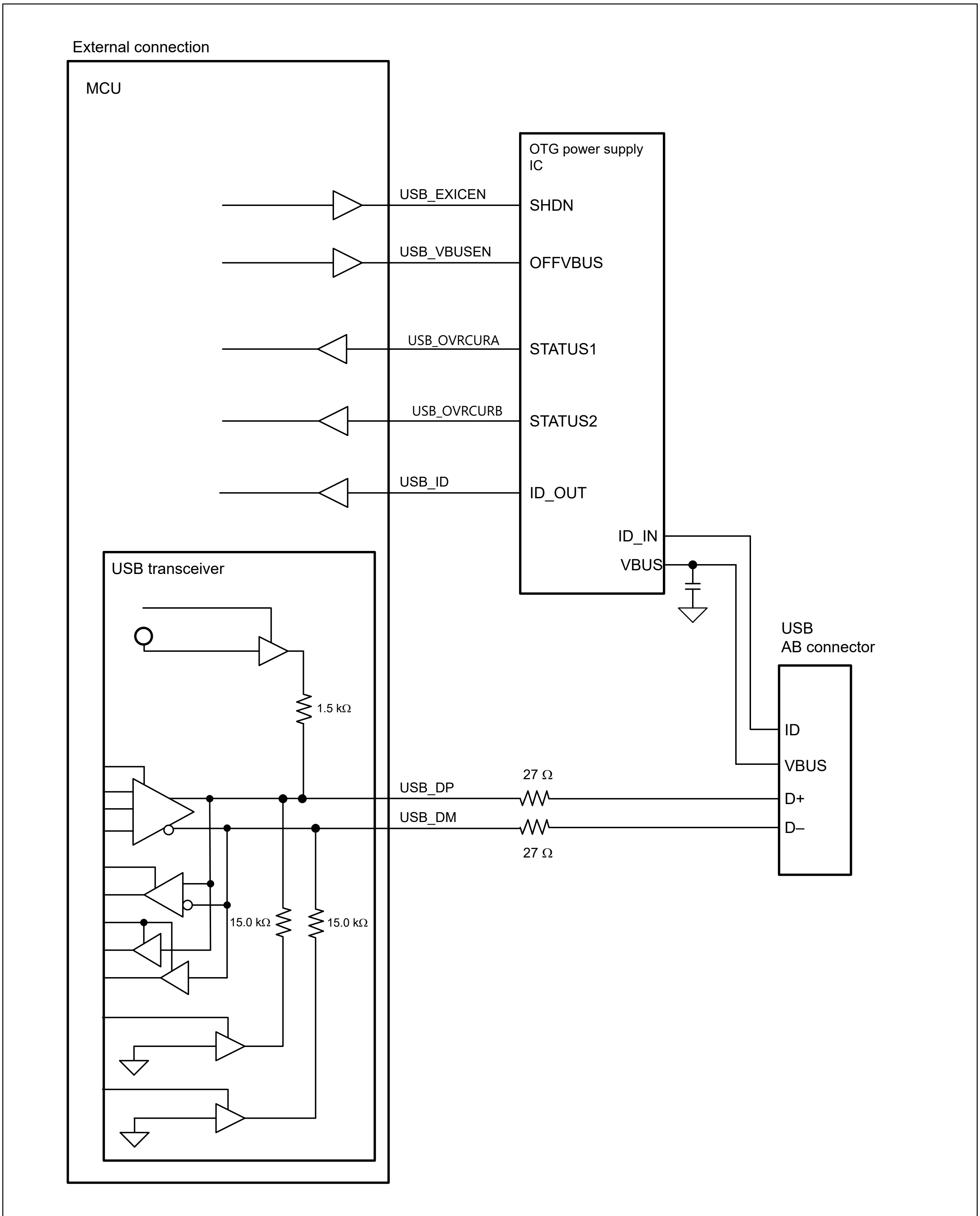


Figure 37.2 Example OTG connection in a self-powered system

Figure 37.3 shows an example device connection in a self-powered system.

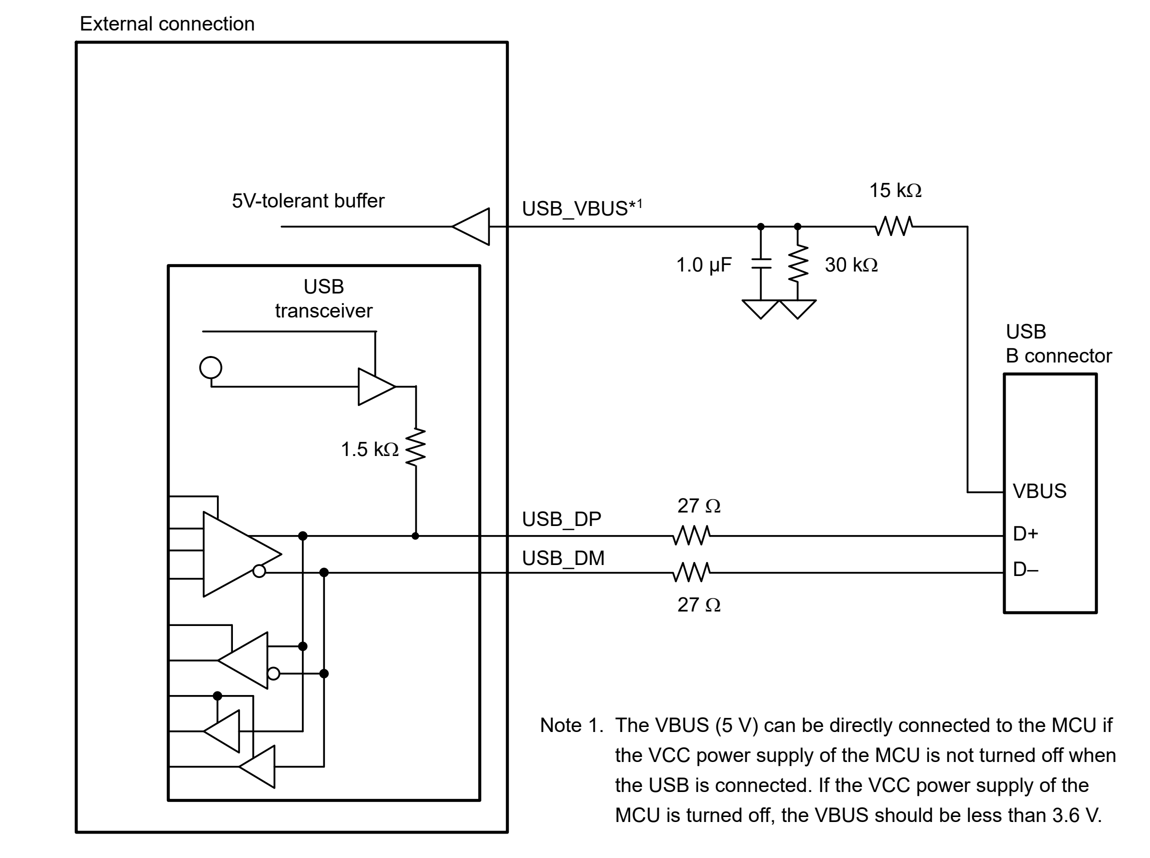


Figure 37.3 Example device connection in a self-powered system

Figure 37.4 shows an example host connection.

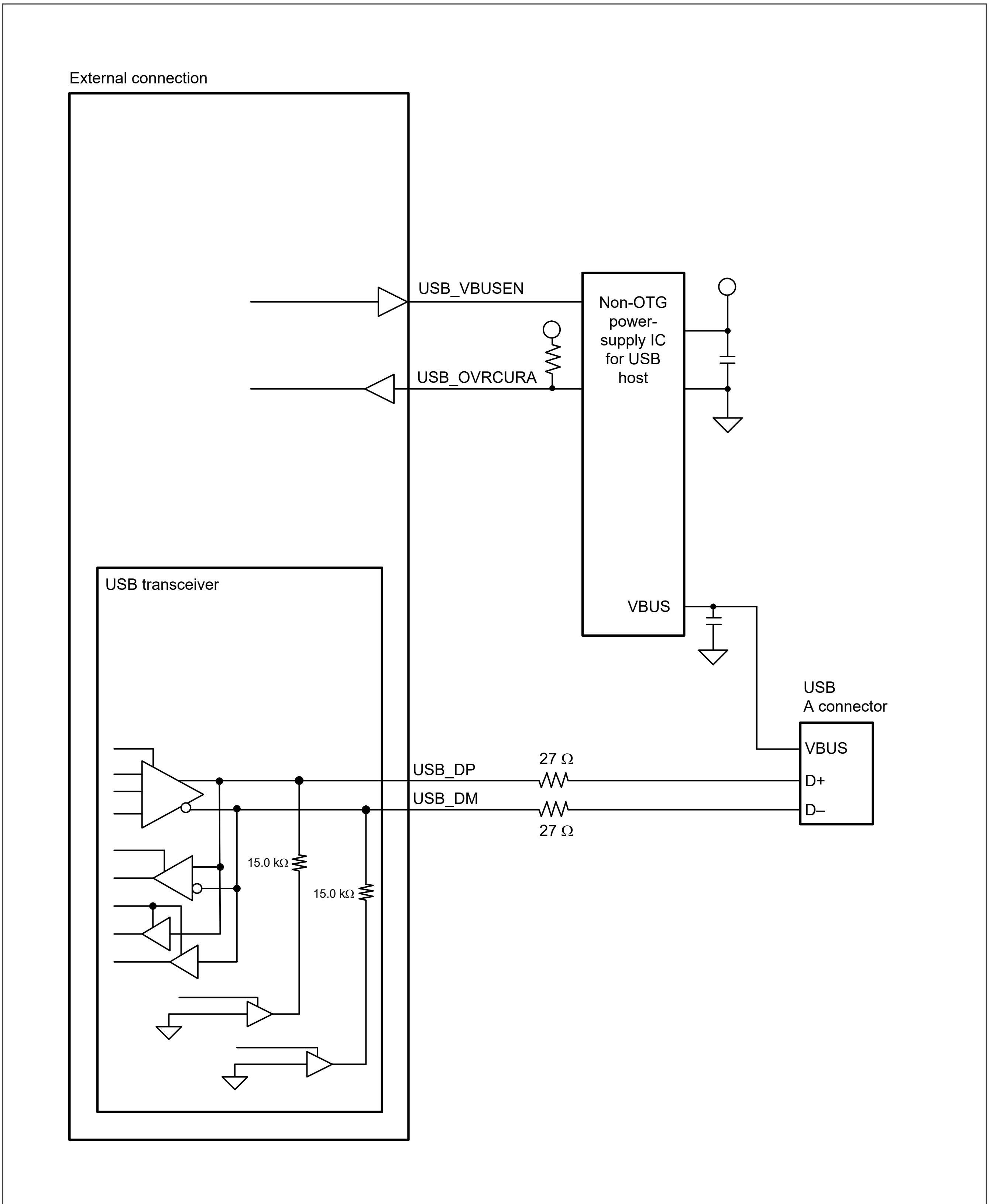


Figure 37.4 Example host connection

Figure 37.5 shows an example device connection in a bus-powered system.

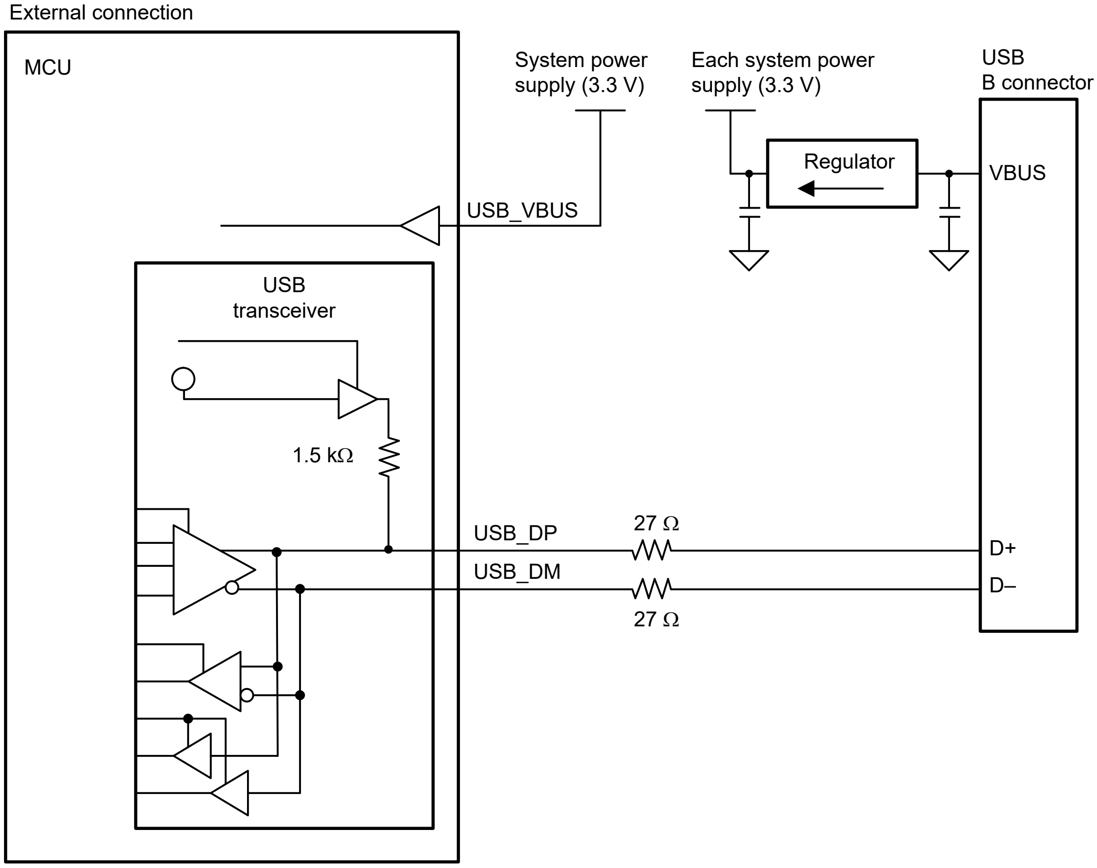


Figure 37.5 Example device connection in a bus-powered state

The examples of external circuits given in this section are simplified circuits, and their operation in every system is not guaranteed.

37.3.1.5 Release from Deep Software Standby Mode 1 Because of USB Suspend/ Resume Interrupts

Deep Software Standby mode1 can be canceled by a USB suspend/resume interrupt. USB suspend/resume interrupts are detected by the USB resume detecting unit, which controls and monitors the USB I/O pins to detect the interrupts.

Figure 37.6 shows a schematic diagram of the connection between the USB resume detecting unit and the USB I/O pins.

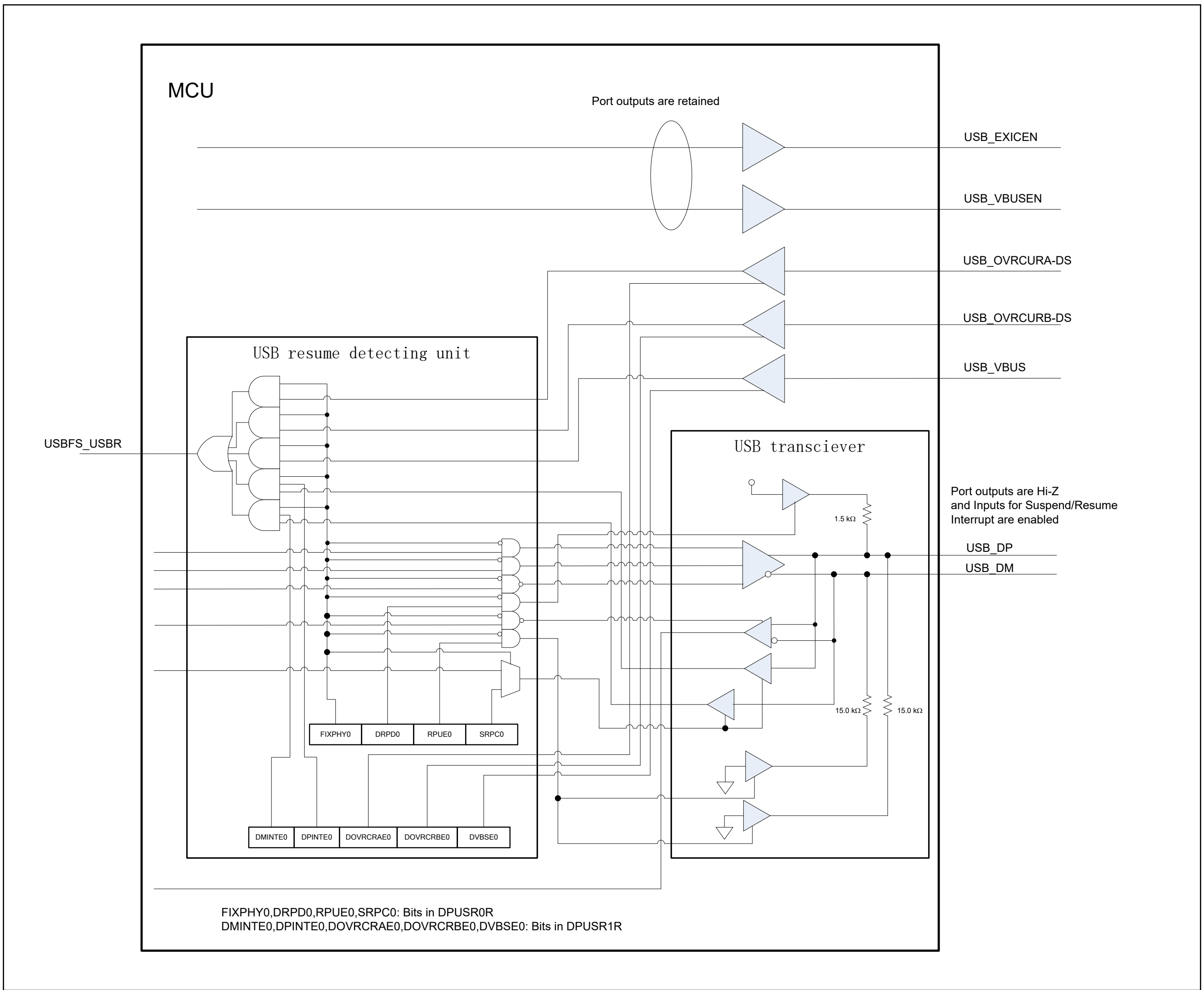


Figure 37.6 Connection between the USB resume detecting unit and the USB I/O pins

Table 37.15 shows the USB suspend and resume interrupt sources and their associated I/O pins.

Table 37.15 USB suspend and resume interrupt sources and their associated I/O pins

USB operating mode	Source	Pin name
Device, OTG	Resume	USB_DP
Host, OTG	Attach or detach	USB_DP, USB_DM
Device	Attach or detach	USB_VBUS
Host	Overcurrent detection	USB_OVRCURA-DS
OTG	Overcurrent detection	USB_OVRCURA-DS, USB_OVRCURB-DS

Figure 37.7 shows the flow for setting the USBFS when entering Deep Software Standby mode 1 from either host or device controller mode. Figure 37.8 shows the flow for setting the USBFS when canceling Deep Software Standby mode 1 from host controller mode. Figure 37.9 shows the flow for setting the USBFS when canceling Deep Software Standby mode 1 from device controller mode.

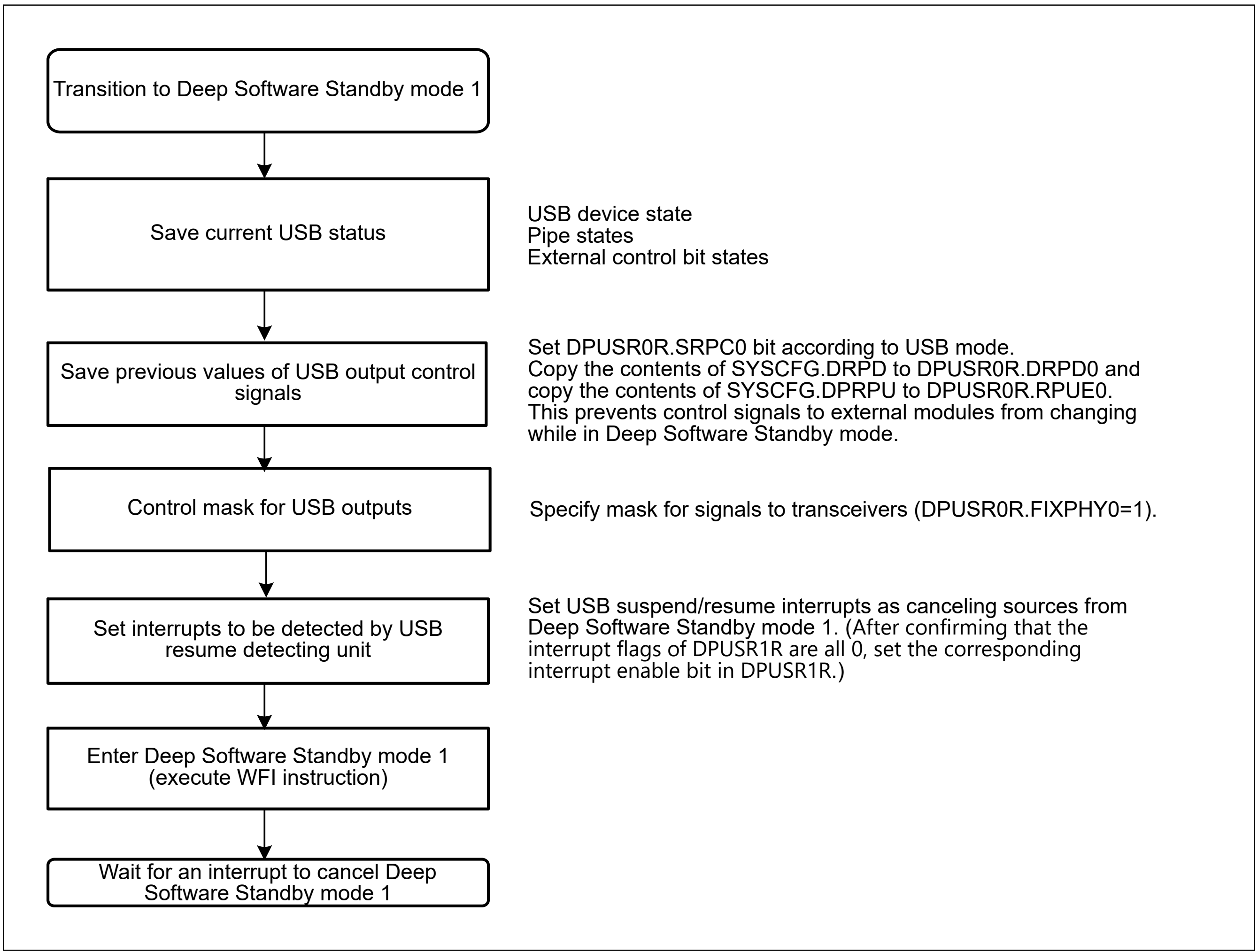


Figure 37.7 USBFS setup flow for transition to Deep Software Standby mode 1 as host or device controller

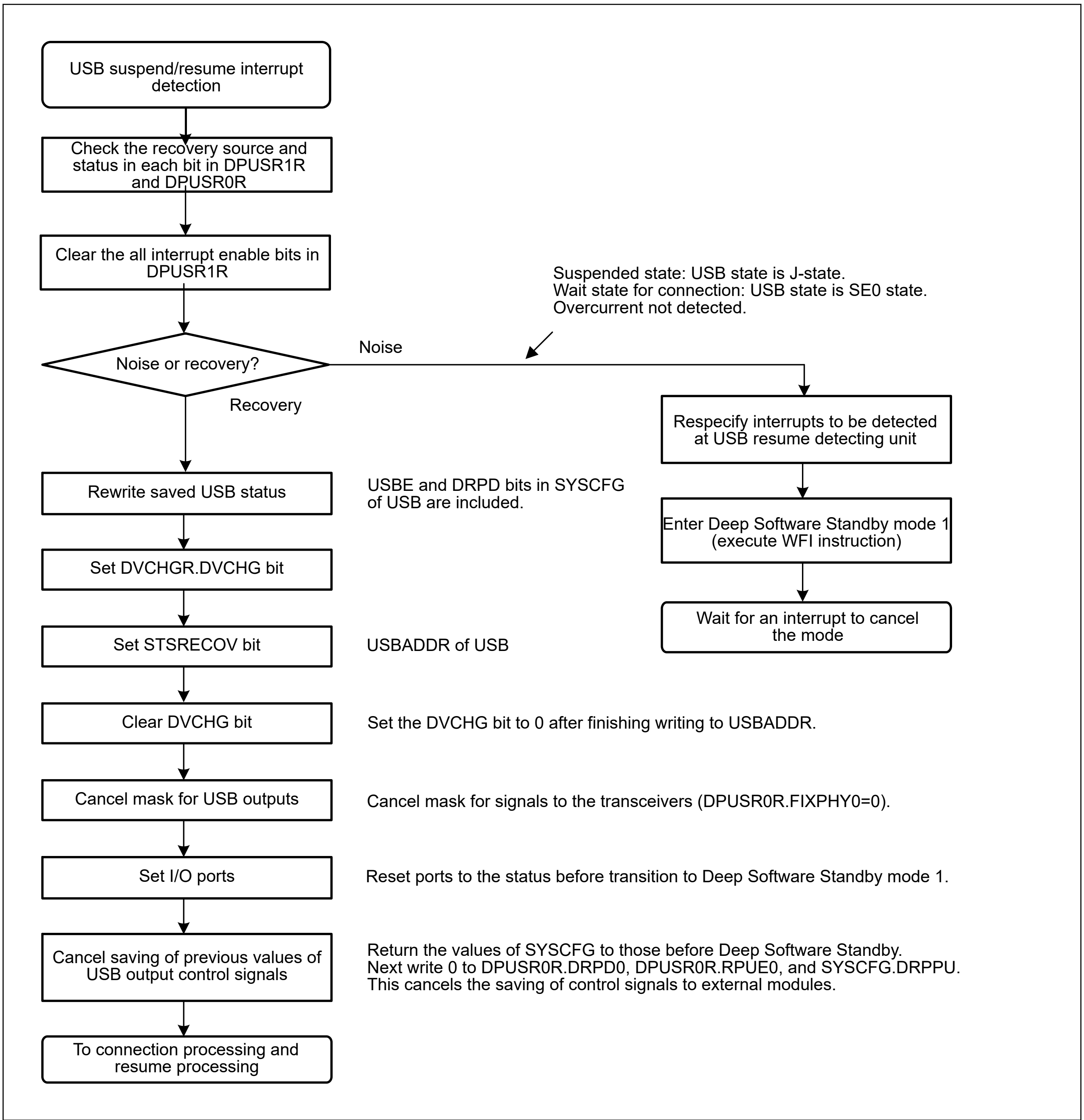


Figure 37.8 USBFS setup flow for canceling Deep Software Standby mode 1 as host controller

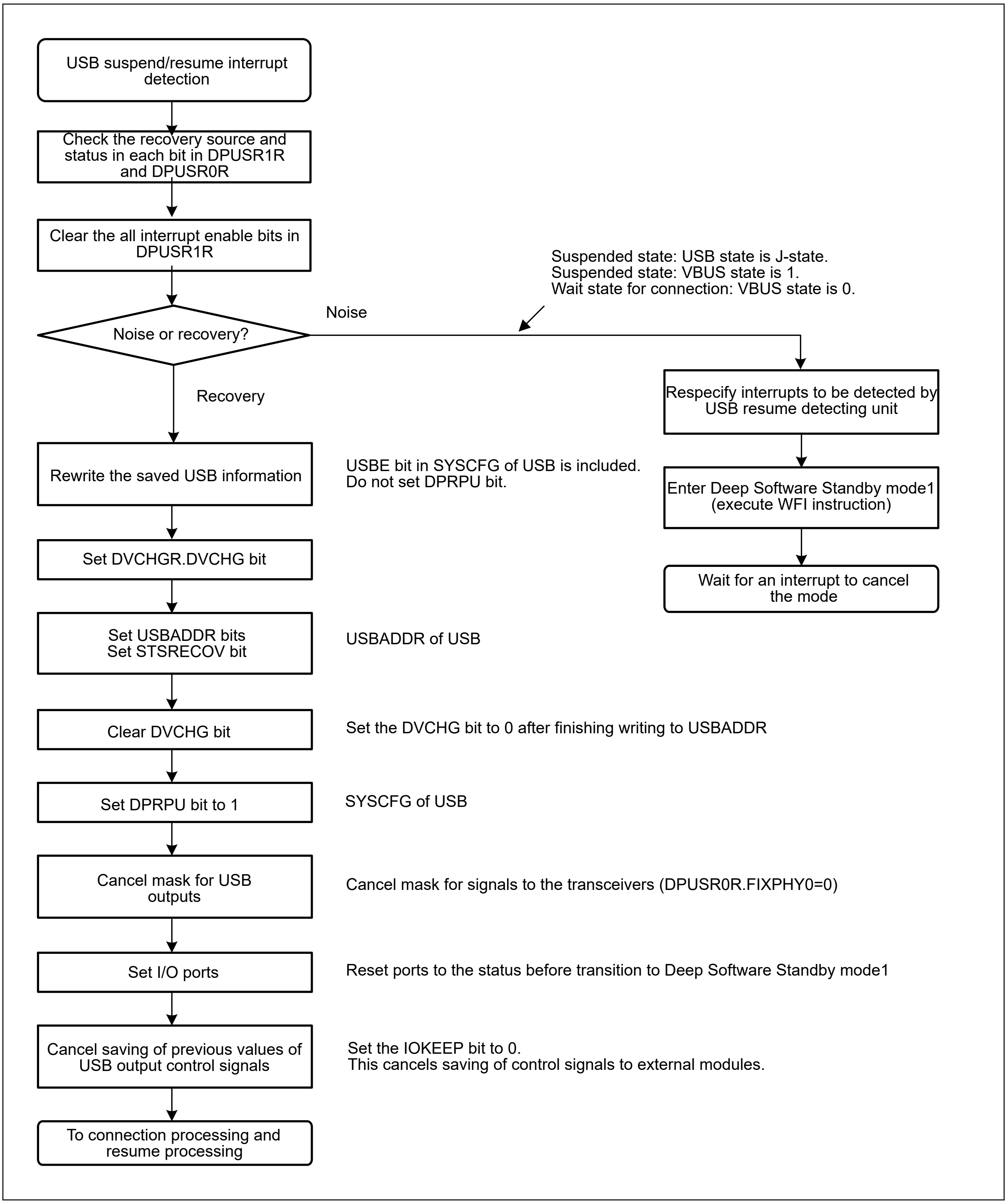


Figure 37.9 USBFS setup flow for canceling Deep Software Standby mode 1 as device controller

37.3.2 Interrupts

Table 37.16 lists the interrupt sources in the USBFS. When an interrupt generation condition is satisfied and the interrupt output is enabled using the associated interrupt enable register, a USBFS interrupt request is issued to the Interrupt Controller Unit (ICU) and an USBFS interrupt is generated.

Table 37.16 Interrupt sources (1 of 2)

Bit to be set to 1	Name	Interrupt source	Applicable controller function	Status flag
VBINT	VBUS interrupt	• A change in the state of the USB_VBUS input pin was detected (low to high or high to low)	Host or device*1	INTSTS0.VBSTS
RESM	Resume interrupt	• A change in the state of the USB bus was detected in the Suspend state (J-state to K-state or J-state to SE0)	Device	—
SOFR	Frame number update interrupt	In host controller mode: • An SOF packet with a different frame number was transmitted In device controller mode: • An SOF packet with a different frame number was received	Host or device	—
DVST	Device state transition interrupt	• One of the following device state transitions was detected: – USB bus reset was detected – Suspend state was detected – SET_ADDRESS request was received – SET_CONFIGURATION request was received	Device	INTSTS0.DVSQ[2:0]
CTRT	Control transfer stage transition interrupt	• A control transfer stage transition was detected because of one of the following: – Setup stage completed – Control write transfer status stage transition occurred – Control read transfer status stage transition occurred – Control transfer completed – Control transfer sequence error occurred	Device	INTSTS0.CTSQ[2:0]
BEMP	Buffer empty interrupt	• The buffer is empty after all FIFO buffer data was transmitted • A packet larger than the maximum packet size was received	Host or device	BEMPSTS.PIPEnBEMP
NRDY	Buffer not ready interrupt	In host controller mode • A STALL response was received from the peripheral device in response to the issued token • The response from the peripheral device in response to the issued token was not received successfully (no response three times consecutively or packet reception error three times consecutively) • An overrun or underrun error occurred during isochronous transfer In device controller mode • NAK was returned for an IN or OUT token while the PID [1:0] bits were set to 01b (BUF) • A CRC error or bit stuffing error occurred during data reception in isochronous transfer • An overrun or underrun occurred during data reception in isochronous transfer	Host or device	NRDYSTS.PIPEnNRDY
BRDY	Buffer ready interrupt	• The buffer is ready (readable or writable state)	Host or device	BRDYSTS.PIPEnBRDY
OVRCR	Overcurrent input change interrupt	• USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB or USB_OVRCURB-DS input pin state change was detected (low to high or high to low)	Host	INTSTS1.OVRCR
BCHG	Bus change interrupt	• USB bus state change was detected	Host or device	SYSSTS0.LNST[1:0]
DTCH	Disconnect detection during full-speed operation	Peripheral device disconnect was detected in full-speed operation	Host	DVSTCTR0.RHST[2:0]
ATTCH	Device connect detection interrupt	• J-state or K-state was detected on the USB bus for 2.5 μs continuously This interrupt can be used to check whether peripheral devices are connected	Host	—

Table 37.16 Interrupt sources (2 of 2)

Bit to be set to 1	Name	Interrupt source	Applicable controller function	Status flag
EOFERR	EOF error detection interrupt	• An EOF error was detected for a peripheral device	Host	—
SACK	Setup normal interrupt	• A setup transaction normal response (ACK) was received	Host	—
SIGN	Setup error interrupt	• A setup transaction error (no response or ACK packet corruption) was detected three consecutive times	Host	—

Note 1. Although this interrupt can be generated in host controller mode, it is not usually used in this mode.

Figure 37.10 shows the circuits related to the USBFS interrupts.

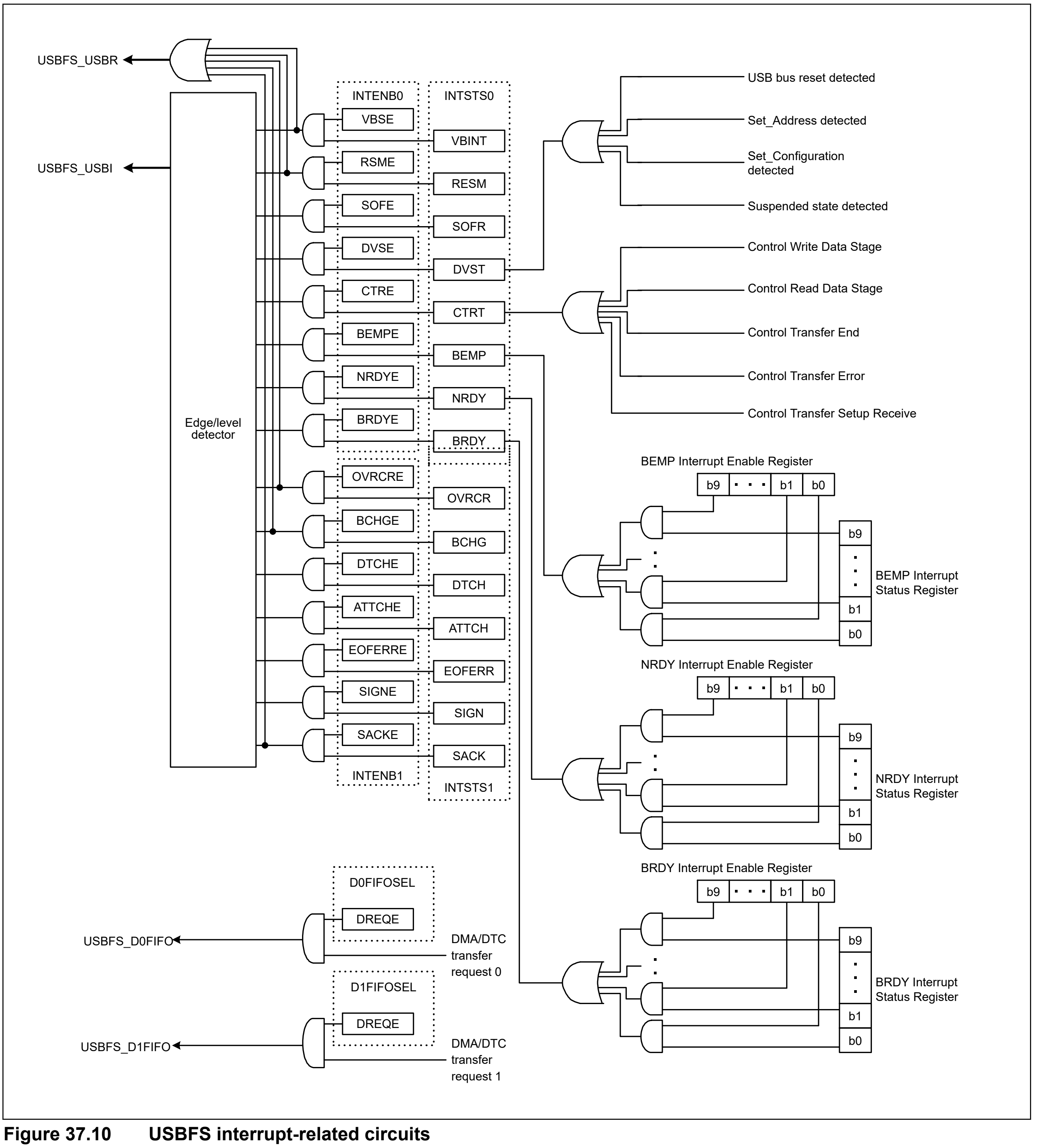


Figure 37.10 USBFS interrupt-related circuits

Table 37.17 shows the interrupts generated by the USBFS.

Table 37.17 USBFS interrupts

Interrupt name	Interrupt status flag	DTC activation	DMAC activation	Priority
USBFS_D0FIFO	DMA transfer request 0	Possible	Possible	High
USBFS_D1FIFO	DMA transfer request 1	Possible	Possible	↑
USBFS_USBI	VBUS interrupt, resume interrupt, frame number update interrupt, device state transition interrupt, control transfer stage transition interrupt, buffer empty interrupt, buffer not ready interrupt, buffer ready interrupt, overcurrent input change interrupt, bus change interrupt, disconnect detection interrupt during full-speed operation, device connect detection interrupt, EOF error detection interrupt, normal setup operation interrupt, and setup error interrupt	Not possible	Not possible	Low
USBFS_USBR*1	VBUS interrupt, resume interrupt, overcurrent input change interrupt and setup error interrupt	Not possible	Not possible	—

Note 1. Software Standby mode can be canceled. And Deep Software Standby mode 1 also can be canceled, but only dedicated pin (OVRCURA-DS, OVRCURB-DS) can be used for overcurrent pins.

37.3.3 Interrupt Descriptions

37.3.3.1 BRDY Interrupt

The BRDY interrupt is generated in both host and device controller modes. This section describes the conditions in which the USBFS sets the associated bit in BRDYSTS to 1. Under these conditions, the USBFS generates a BRDY interrupt if the software has set the bit in BRDYENB associated with the given pipe to 1 and the INTENB0.BRDYE bit to 1.

The conditions for generating and clearing the BRDY interrupt depend on the SOFCFG.BRDYM and PIPECFG.BFRE settings for each pipe as follows:

(1) When SOFCFG.BRDYM = 0 and PIPECFG.BFRE = 0

With these settings, the BRDY interrupt indicates that the FIFO port is accessible.

On any of the following conditions, the USBFS generates an internal BRDY interrupt request trigger and sets the BRDYSTS.PIPEnBRDY bit associated with the selected pipe to 1.

For transmitting pipes

- When the DIR bit is changed from 0 to 1 by software
- When packet transmission is complete for a pipe while write-access from the CPU to the FIFO buffer for the pipe is disabled (when the BSTS bit is read as 0)
- When one FIFO buffer is empty on completion of writing data to the other FIFO buffer in double buffer mode
- No request trigger is generated until completion of writing data to the currently-written FIFO buffer even if transmission to the other FIFO buffer is complete
- When the hardware flushes the buffer of the pipe for isochronous transfers
- When 1 is written to the PIPEnCTR.ACLRM bit, which causes the FIFO buffer to transition from the write-disabled to write-enabled state

No request trigger is generated for the DCP, that is, during data transmission for control transfers.

For receiving pipes

- When packet reception is successfully complete, enabling the FIFO buffer to be read while read-access from the CPU to the FIFO buffer for the given pipe is disabled (when the BSTS bit is read as 0). No request trigger is generated for transactions in which a DATA-PID mismatch has occurred.
- When one FIFO buffer is read-enabled on completion of reading data from the other FIFO buffer in double buffer mode. No request trigger is generated until completion of reading data from the currently-read FIFO buffer, even if reception by the other FIFO buffer is complete.

In device controller mode, the BRDY interrupt is not generated in the status stage of control transfers. The PIPEBRDY interrupt status of the selected pipe can be set to 0 by writing 0 to the associated PIPEnBRDY bit through software. In this case, the other PIPEBRDY bit should be set to 1.

Clear the BRDY status before accessing the FIFO buffer.

(2) When SOFCFG.BRDYM = 0 and PIPECFG.BFRE = 1

With these settings, the USBFS generates a BRDY interrupt on completion of reading all data for a single transfer using the receiving pipe, and sets the bit in BRDYSTS associated with the pipe to 1.

On any of the following conditions, the USBFS determines that the last data for a single transfer was received.

- When a short packet including a zero-length packet is received
- When the PIPEn transaction counter register (PIPEnTRN) is used and the number of packets specified in the PIPEnTRN.TRNCNT[15:0] bits are completely received

When the data is completely read after any of these conditions is satisfied, the USBFS determines that all data for a single transfer is completely read.

When a zero-length packet is received while the FIFO buffer is empty, the USBFS determines that all data for a single transfer is completely read when the FRDY bit in the FIFO port control register is 1 and the DTLN[8:0] bits are 0. In this case, to start the next transfer, write 1 to the BCLR bit in the associated port control register through the software. With these settings, the USBFS does not detect a BRDY interrupt for the transmitting pipe.

The PIPEBRDY interrupt status of a pipe can be set to 0 by writing 0 to the associated BRDYSTS.PIPEnBRDY bit through the software. In this case, 1 must be written to the PIPEBRDY bits for the other pipes.

In this mode, do not change the PIPECFG.BFRE bit setting until all data for a single transfer is processed. When it is necessary to change the PIPECFG.BFRE bit before completion of processing, all FIFO buffers for the pipe must be cleared using the PIPEnCTR.ACLRM bit.

(3) When SOFCFG.BRDYM = 1 and PIPECFG.BFRE = 0

With these settings, the BRDYSTS.PIPEnBRDY values are linked to the BSTS bit setting for each pipe. In other words, the BRDY interrupt status bits (PIPEBRDY) are set to 1 or 0 by the USB depending on the FIFO buffer status.

For transmitting pipes

The BRDY interrupt status bits are set to 1 when the FIFO buffer is ready for write access, and are set to 0 when it is not ready. The BRDY interrupt is not generated for the DCP in the transmitting direction even when it is ready for write access.

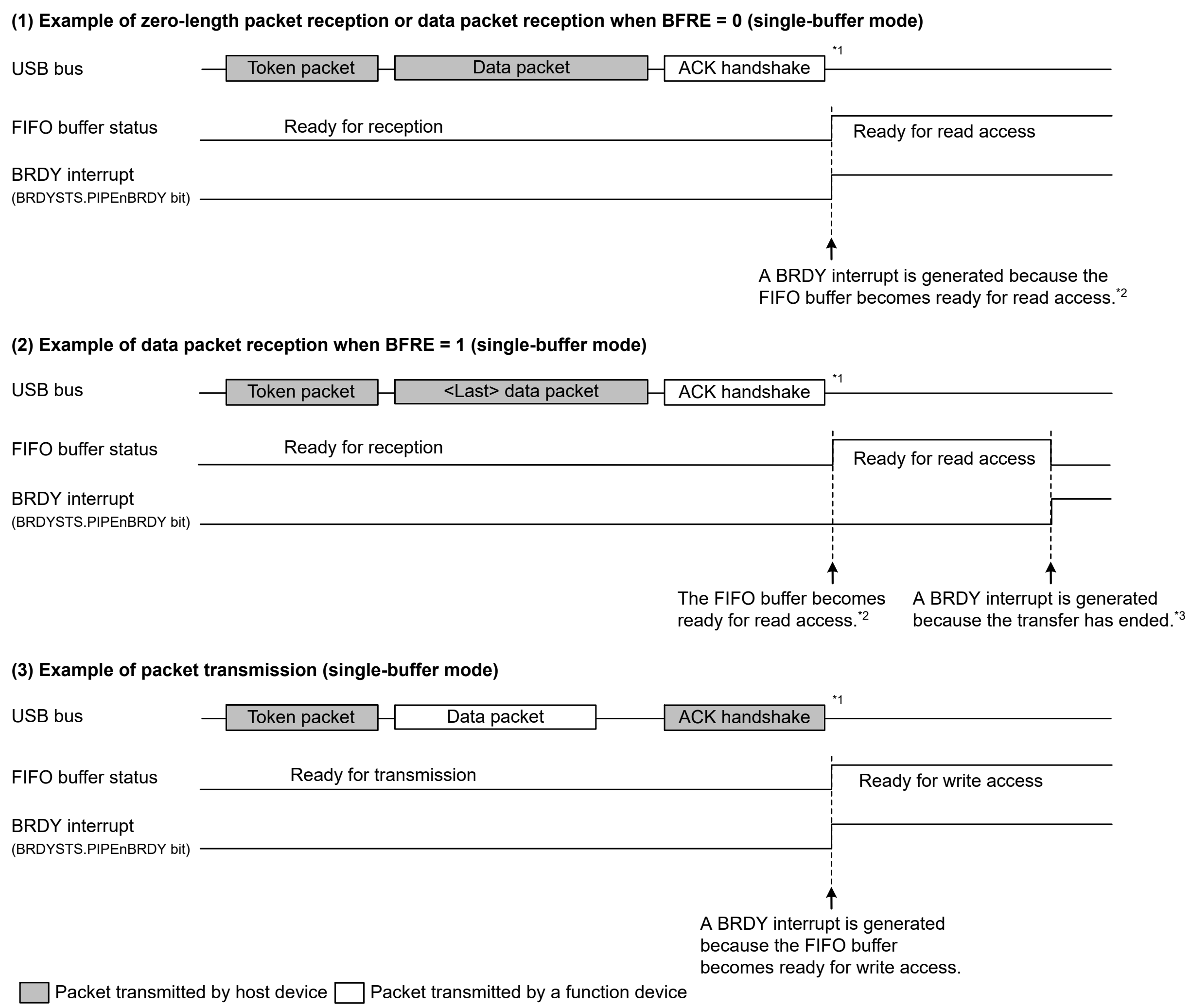
For receiving pipes

The BRDY interrupt status bits set to 1 when the FIFO buffer is ready for read access, and set to 0 when all data is read (not ready for read access).

When a zero-length packet is received while the FIFO buffer is empty, the associated bit is set to 1 and the BRDY interrupt is continuously generated until the software writes 1 to BCLR. With this setting, the PIPEnBRDY bit cannot be set to 0 by software.

When the SOFCFG.BRDYM bit is set to 1, set the PIPECFG.BFRE bit for all pipes to 0.

Figure 37.11 shows the timing of BRDY interrupt generation.



Note 1. The ACK handshake is not used in isochronous transfers.
Note 2. The FIFO buffer becomes ready for read access under the following condition:
When a packet is received while no data remains unread in the FIFO buffer in the CPU.
Note 3. A transfer ends under either of the following conditions:
1. When a short packet including a zero-length packet is received.
2. When the number of packets specified in the transaction counter are received.

Figure 37.11 Timing of BRDY interrupt generation

The condition for clearing the INTSTS0.BRDY bit depends on the SOFCFG.BRDYM bit setting, as shown in Table 37.18.

Table 37.18 Conditions for clearing the BRDY bit

BRDYM bit	Condition for clearing BRDY bit
0	The USBFS clears the BRDY bit to 0 when all bits in BRDYSTS are set to 0 by software.
1	The USBFS clears the BRDY bit to 0 when the BSTS bits for all pipes have cleared to 0.

37.3.3.2 NRDY Interrupt

On generating an internal NRDY interrupt request for the pipe whose PID bits are set to BUF by software, the USBFS sets the associated PIPEnNRDY bit in NRDYSTS to 1. If the associated bit in NRDYENB is set to 1 by software, the USBFS sets the INTSTS0.NRDY bit to 1 and generates a USBFS interrupt.

This section describes the conditions in which the USBFS generates the internal NRDY interrupt request for a given pipe.

The internal NRDY interrupt request is not generated during setup transaction execution in host controller mode. During setup transactions in host controller mode, the SACK or SIGN interrupt is detected.

The internal NRDY interrupt request is not generated during status stage execution of the control transfer in device controller mode.

(1) In host controller mode

For transmitting pipes

On any of the following conditions, the USBFS detects an NRDY interrupt:

- For isochronous transfer pipes, when the time to issue an OUT token comes while there is no data to be transmitted in the FIFO buffer. In this case, the USBFS transmits a zero-length packet following the OUT token and sets the associated NRDYSTS.PIPEnNRDY bit and the FRMNUM.OVRN bit to 1.
- During communications other than setup transactions on pipes not used for isochronous transfers, when any combination of the following two cases occur three consecutive times:
  - No response is returned from the peripheral device (when timeout is detected before detection of the handshake packet from the peripheral device
  - An error is detected in the packet from the peripheral device. In this case, the USBFS sets the associated PIPEnNRDY bit to 1 and changes the associated PID[1:0] setting for the pipe to NAK
- During communications other than setup transactions, when the STALL handshake is received from the peripheral device. In this case, the USBFS sets the associated PIPEnNRDY bit to 1 and changes the PID[1:0] setting for the associated pipe to STALL (11b).

For receiving pipes

- For isochronous transfer pipes, when the time to issue an IN token comes but there is no space available in the FIFO buffer. In this case, the USBFS discards the received data for the IN token and sets the PIPEnNRDY bit associated with the pipe and the OVRN bit to 1. When a packet error is detected in the received data for the IN token, the USBFS also sets the FRMNUM.CRCE bit to 1.
- For non-isochronous transfer pipes, when any combination of the following two cases occur three consecutive times:
  - No response is returned from the peripheral device for the IN token issued by the USBFS (when timeout is detected before detection of the DATA packet from the peripheral device)
  - An error is detected in the packet from the peripheral device. In this case, the USBFS sets the associated PIPEnNRDY bit to 1 and changes the associated PID[1:0] setting for the pipe to NAK
- For isochronous transfer pipes, when no response is returned from the peripheral device for the IN token (when timeout is detected before detection of the DATA packet from the peripheral device) or an error is detected in the packet from the peripheral device. In this case, the USBFS sets the PIPEnNRDY bit associated with the pipe to 1. The PID[1:0] setting for the pipe is not changed.
- For isochronous transfer pipes, when a CRC error or a bit stuffing error is detected in the received data packet. In this case, the USBFS sets the PIPEnNRDY bit associated with the pipe and the CRCE bit to 1.
- When the STALL handshake is received. In this case, the USBFS sets the PIPEnNRDY bit associated with the pipe to 1 and changes the PID[1:0] setting for the associated pipe to STALL.

(2) In device controller mode

For transmitting pipes

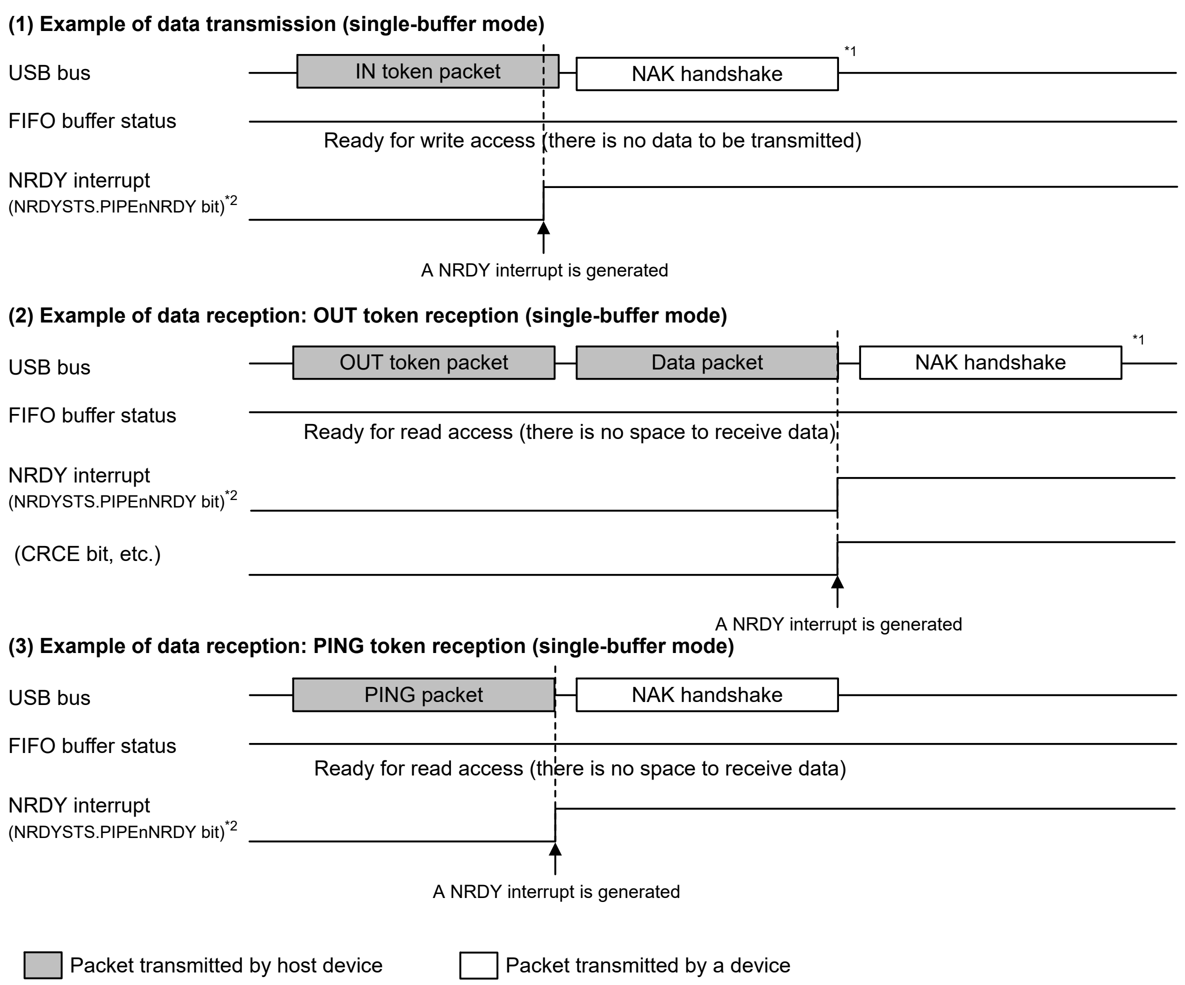
- When an IN token is received while there is no data to be transmitted in the FIFO buffer. In this case, the USBFS generates a NRDY interrupt request on reception of the IN token and sets the NRDYSTS.PIPEnNRDY bit to 1. For an isochronous transfer pipe in which an interrupt is generated, the USBFS transmits a zero-length packet and sets the FRMNUM.OVRN bit to 1.

For receiving pipes

- When an OUT token is received but there is no space available in the FIFO buffer. For an isochronous transfer pipe in which an interrupt is generated, the USBFS generates a NRDY interrupt request on reception of the OUT token and sets the PIPEnNRDY bit to 1 and OVRN bit to 1. For a non-isochronous transfer pipe in which an interrupt is generated, the USBFS generates a NRDY interrupt request when a NAK handshake is transferred after the data following the OUT token is received, and sets the PIPEnNRDY bit to 1. The NRDY interrupt request is not generated during retransmission because of a DATA-PID mismatch. In addition, the NRDY interrupt request is not generated if an error occurs in the DATA packet.

- For isochronous transfer pipes, when a token is not received successfully within an interval frame. In this case, the USBFS generates an NRDY interrupt request when the SOF is received, and sets the PIPEnNRDY bit to 1.

Figure 37.12 shows the timing of NRDY interrupt generation in device controller mode.



Note 1. The handshake is not used in isochronous transfers.
Note 2. The CRCE and OVRN bits change only while the target pipe is set to isochronous transfers.

Figure 37.12 Timing of NRDY interrupt generation in device controller mode

37.3.3.3 BEMP Interrupt

On detecting a BEMP interrupt for the pipe whose PID bits are set to BUF by software, the USBFS sets the associated BEMPSTS.PIPEnBEMP bit to 1. If the associated bit in BEMPENB is set to 1 by software, the USBFS sets the INTSTS0.BEMP bit to 1 and generates a USBFS interrupt. This section describes the conditions in which the USBFS generates an internal BEMP interrupt request.

(1) For transmitting pipes

When the FIFO buffer of the associated pipe is empty on completion of transmission, including zero-length packet transmission, and in single buffer mode, an internal BEMP interrupt request is generated simultaneously with the BRDY interrupt for a non-DCP pipe. The internal BEMP interrupt request is not generated in any of the following conditions:

- When the CPU or DMA/DTC has already started writing data to the FIFO buffer of the CPU on completion of transmitting data from one FIFO buffer in double buffer mode

- When the buffer is cleared (emptied) by setting the PIPEnCTR.ACLRM or the BCLR bit to 1 in the port control register
- When an IN transfer (zero-length packet transmission) is performed during the control transfer status stage in device controller mode

(2) For receiving pipes

When a successfully-received data packet size exceeds the specified maximum packet size. In this case, the USBFS generates a BEMP interrupt request, sets the associated BEMPSTS.PIPEnBEMP bit to 1, discards the received data, and changes the associated PID[1:0] setting for the pipe to STALL (11b). The USBFS returns no response in host controller mode, and returns STALL response in device controller mode.

The internal BEMP interrupt request is not generated in any of the following conditions:

- When a CRC error or a bit stuffing error is detected in the received data
- When a setup transaction is being performed:
  - Writing 0 to the BEMPSTS.PIPEnBEMP bit clears the status
  - Writing 1 to the BEMPSTS.PIPEnBEMP bit has no effect

Figure 37.13 shows the timing of BEMP interrupt generation in device controller mode.

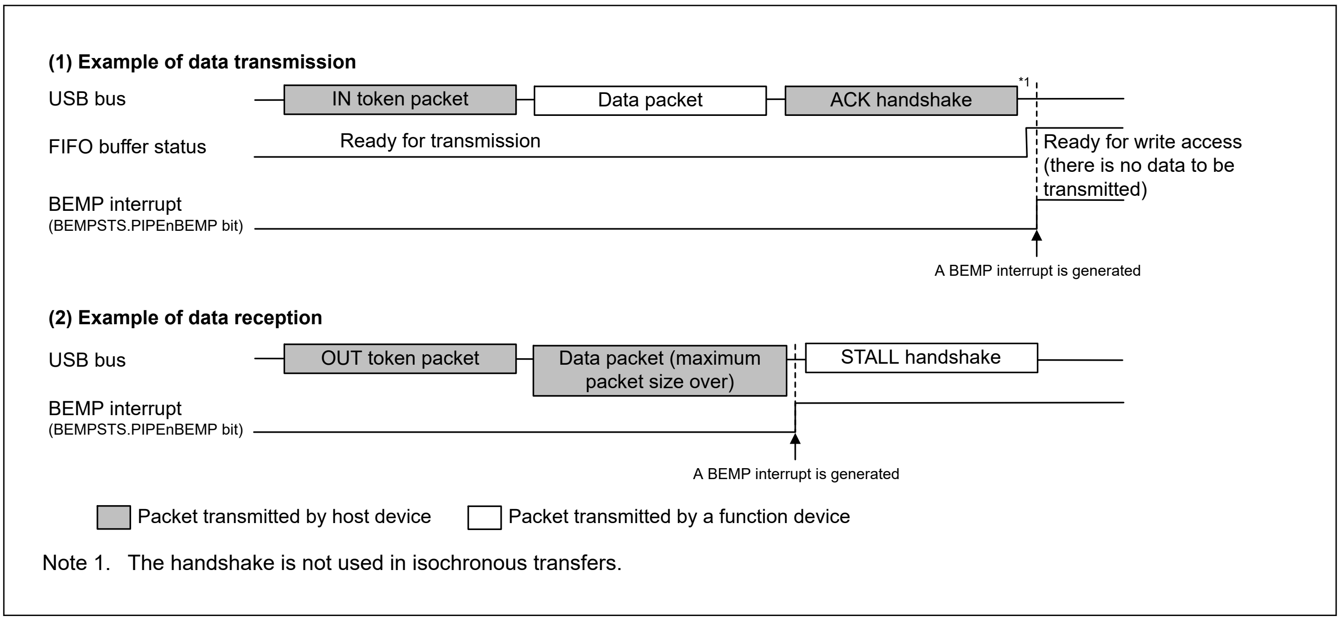


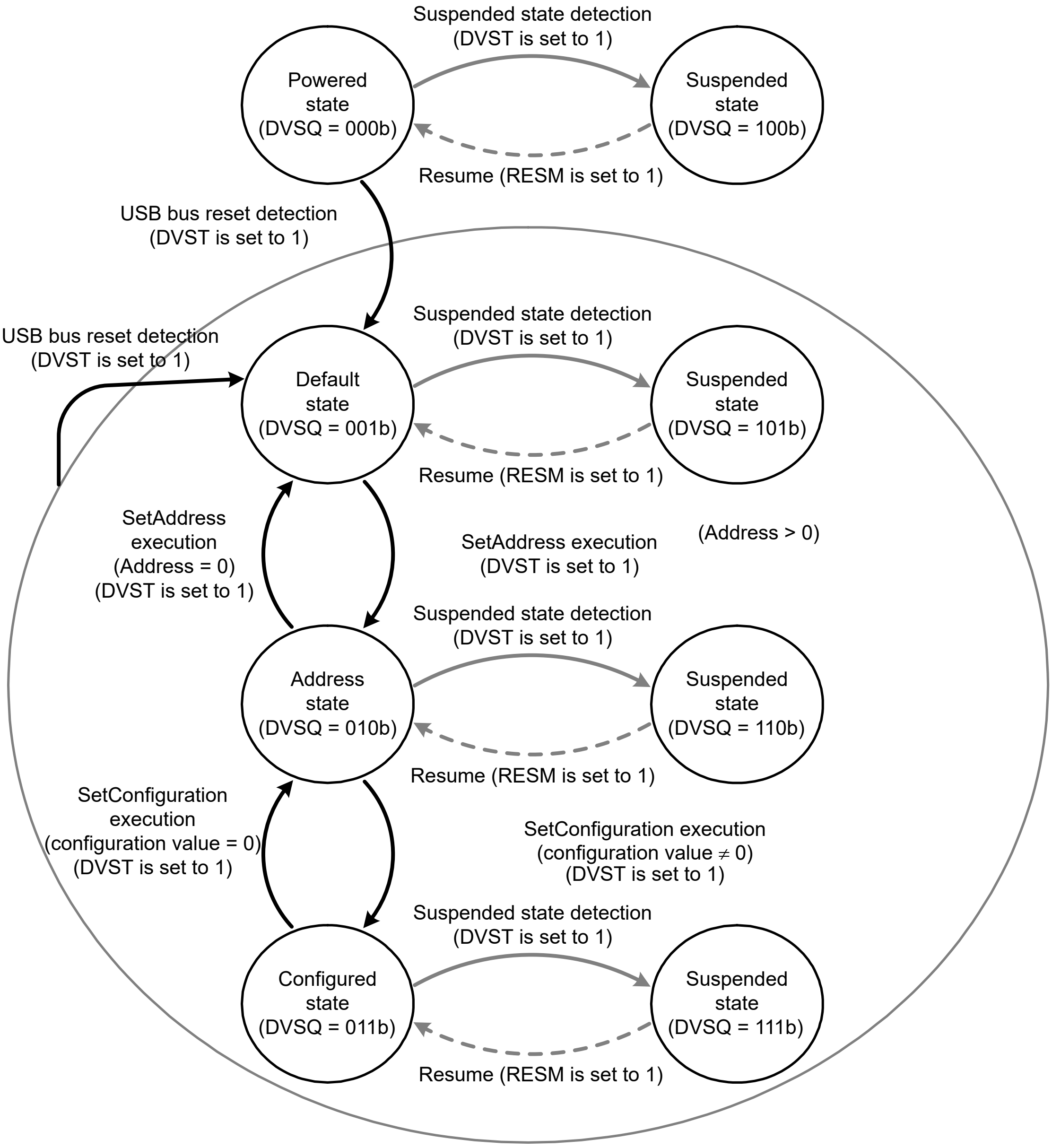
Figure 37.13 Timing of BEMP interrupt generation in device controller mode

37.3.3.4 Device State Transition Interrupt (Device Controller Mode)

Figure 37.14 shows a diagram of the USBFS device state transitions. The USBFS controls device states and generates device state transition interrupts. However, recovery from the Suspend state (resume signal detection) is detected by means of the resume interrupt. Device state transition interrupts can be enabled or disabled independently in INTENB0. Devices whose states have changed can be checked in the INTSTS0.DVSQ[2:0] bits.

When a transition is made to the default state, a device state transition interrupt is generated after a USB bus reset is detected.

The USBFS controls device states, and device state transition interrupts can be generated, only in device controller mode.



Note: For a transition indicated by a solid line, the DVST bit is set to 1.
For a resume indicated by a dashed line, the RESM bit is set to 1.

Figure 37.14 Device state transitions

37.3.3.5 Control Transfer Stage Transition Interrupt (Device Controller Mode)

Figure 37.15 shows a diagram of the control transfer stage transitions of the USBFS. The USBFS controls the control transfer sequence and generates control transfer stage transition interrupts. Control transfer stage transition interrupts can be enabled or disabled independently in INTENB0. Transfer stages that have transitioned can be checked in the INTSTS0.CTSQ[2:0] bits.

Control transfer stage transition interrupts are generated only in device controller mode. This section describes control transfer sequence errors. If an error occurs, the DCPCTR.PID[1:0] bits are set to 1xb (STALL response).

(1) Control read transfer errors

- An OUT token is received but no data is transferred in response to the IN token at the data stage
- An IN token is received at the status stage

- A data packet with DATAPID = DATA0 is received at the status stage

(2) Control write transfer errors

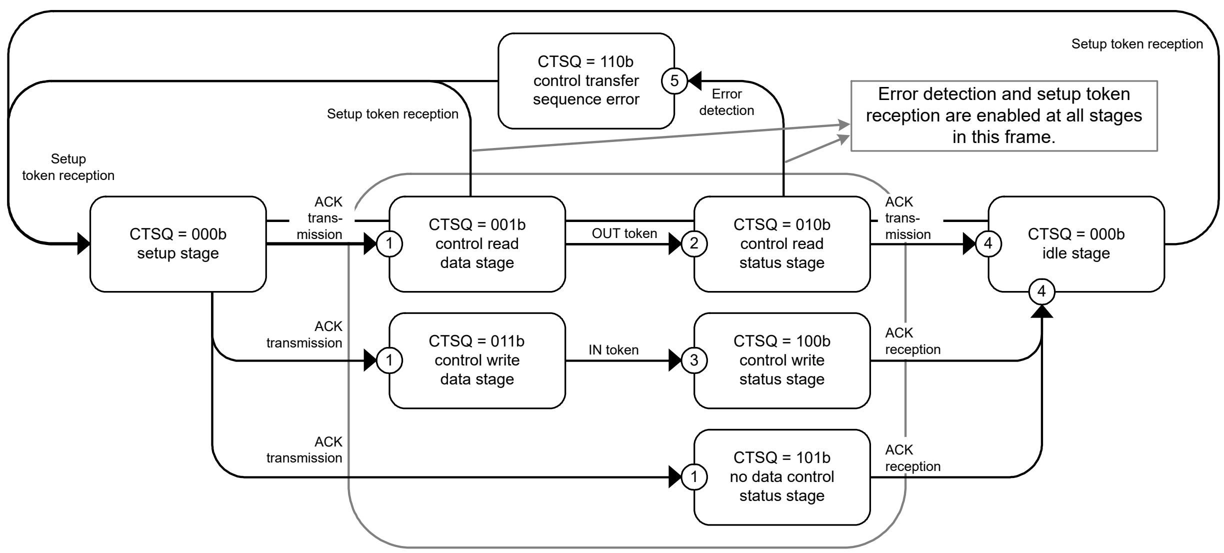
- An IN token is received but no ACK is returned in response to the OUT token at the data stage
- A data packet with DATAPID = DATA0 is received as the first data packet at the data stage
- An OUT token is received at the status stage

(3) Control write no data transfer errors

- An OUT token is received at the status stage

At the control write transfer data stage, if the receive data length exceeds the wLength value of the USB request, it is not recognized as a control transfer sequence error. At the control read transfer status stage, packets other than zero-length packets are received by an ACK response and the transfer ends normally.

When a CTRT interrupt occurs in response to a sequence error (INTSTS0.CTRT = 1), the CTSQ[2:0] = 110b value is saved until the CTRT bit is set to 0, clearing the interrupt status. While CTSQ[2:0] = 110b is being saved, no CTRT interrupt for ending the setup stage is generated, even if a new USB request is received. The USBFS saves the setup stage completion status, and it generates a CTRT interrupt after the interrupt status is cleared by software.



Note: CTRT interrupts
1. Setup stage end
2. Control read transfer status stage transition
3. Control write transfer status stage transition
4. Control transfer end
5. Control transfer sequence error

Figure 37.15 Control transfer stage transitions

37.3.3.6 Frame Update Interrupt

In host controller mode, an interrupt is generated when the frame number is updated.

In device controller mode, an SOFR interrupt is generated when the frame number is updated. The USBFS updates the frame number and generates an SOFR interrupt if it detects a new SOF packet during full-speed operation.

37.3.3.7 VBUS Interrupt

When the USB_VBUS pin level changes, a VBUS interrupt is generated. The level of the USB_VBUS pin can be checked with the INTSTS0.VBSTS bit. Whether the host controller is connected or disconnected can be confirmed using the VBUS interrupt. If the system is activated with the host controller connected, the first VBUS interrupt is not generated, because there is no change in the USB_VBUS pin level.

37.3.3.8 Resume Interrupt

In device controller mode, a resume interrupt is generated when the device state is the Suspend state and the USB bus state has changed (from J-state to K-state, or from J-state to SE0). Recovery from the Suspend state is detected by means of the resume interrupt.

In host controller mode, no resume interrupt is generated. Use the BCHG interrupt to detect a change in the USB bus state.

37.3.3.9 OVRCR Interrupt

An OVRCR interrupt is generated when the USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB or USB_OVRCURB-DS pin level has changed. The levels of the USB_OVRCURA, USB_OVRCURA-DS, USB_OVRCURB and USB_OVRCURB-DS pins can be checked in the SYSSTS0.OVCMON[1:0] flags. The external power supply IC can check whether overcurrent is detected using the OVRCR interrupt.

For OTG connections, the OVRCR interrupt allows you to check whether a change is detected in the VBUS comparator.

37.3.3.10 BCHG Interrupt

A BCHG interrupt is generated when the USB bus state has changed. The BCHG interrupt can be used to detect whether a peripheral device is connected and can also be used to detect a remote wakeup in host controller mode. The BCHG interrupt is generated in both host and device controller modes.

37.3.3.11 DTCH Interrupt

A DTCH interrupt occurs when a USB bus disconnect is detected in host controller mode. The USBFS detects bus disconnects in compliance with the USB 2.0 specification.

On interrupt detection, all pipes in which communications are being carried out for the relevant port must be terminated by software. The pipes enter the wait state for a bus connection to the port, waiting for an ATTCH interrupt to occur. Regardless of the value set in the associated interrupt enable bit, the USBFS hardware:

- Sets the DVSTCTR0.UACT bit for the port in which the DTCH interrupt is detected to 0
- Puts the port in which the DTCH interrupt occurred into the idle state

37.3.3.12 SACK Interrupt

A SACK interrupt is generated when an ACK response for the transmitted setup packet is received from the peripheral device in host controller mode. The SACK interrupt can be used to confirm that the setup transaction is successfully complete.

37.3.3.13 SIGN Interrupt

A SIGN interrupt is generated when an ACK response for the transmitted setup packet is not correctly received from the peripheral device three consecutive times in host controller mode. The SIGN interrupt can be used to detect no ACK response transmitted from the peripheral device or corruption of an ACK packet.

37.3.3.14 ATTCH Interrupt

An ATTCH interrupt is generated when J-state or K-state of the full-speed signal level is detected on the USB port for 2.5 µs in host controller mode. To be more specific, an ATTCH interrupt is detected on any of the following conditions:

- When K-state, SE0, or SE1 changes to J-state, and J-state continues 2.5 µs
- When J-state, SE0, or SE1 changes to K-state, and K-state continues 2.5 µs

37.3.3.15 EOFERR Interrupt

An EOFERR interrupt occurs when the USBFS detects that communication is not complete at the EOF2 timing defined in the USB 2.0 specification.

On interrupt detection, all pipes in which communications are being carried out for the relevant port must be terminated by software, and the port must be re-enumerated. Regardless of the value set in the associated interrupt enable bit, the USBFS hardware:

- Sets the DVSTCTR0.UACT bit for the port in which the EOFERR interrupt is detected to 0
- Puts the port in which the EOFERR interrupt is generated into the idle state

37.3.4 Pipe Control

Table 37.19 lists the pipe settings for the USBFS. USB data transfer is performed through logical pipes that the software associates with endpoints. The USBFS provides 10 pipes that are used for data transfer. Set up the pipes based on your system specifications.

Table 37.19 Pipe settings

Register name	Bit name	Setting	Notes
DCPCFG PIPECFG	TYPE	Transfer type	Pipes 1 to 9: Settable
	BFRE	BRDY interrupt mode	Pipes 1 to 5: Settable
	DBLB	Double buffer select	Pipes 1 to 5: Settable
	DIR	Transfer direction select	IN or OUT settable
	EPNUM	Endpoint number	Pipes 1 to 9: Settable A value other than 0000b must be set when the pipe is used.
	SHTNAK	Selects disabled state for pipe when transfer ends	Pipes 1 and 2: Settable only for bulk transfers Pipes 3 to 5: Settable
DCPMAXP PIPEMAXP	DEVSEL	Device select	Referenced only in host controller mode.
	MXPS	Maximum packet size	Compliant with the USB 2.0 specification.
PIPEPERI	IFIS	Buffer flush	Pipes 1 and 2: Settable only for isochronous transfers Pipes 3 to 9: Setting disabled
	IITV	Interval counter	Pipes 1 and 2: Settable only for isochronous transfers Pipes 3 to 5: Setting disabled Pipes 6 to 9: Settable only in host controller mode
DCPCTR PIPEnCTR	BSTS	Buffer status	For the DCP, receive buffer status and transmit buffer status are switched with the ISEL bit.
	INBUFM	IN buffer monitor	Available only for pipes 1 to 5.
	SUREQ	Setup request	Settable only for the DCP and controlled in host controller mode
	SUREQCLR	SUREQ clear	Settable only for the DCP and controlled in host controller mode
	ATREPM	Auto response mode	Pipes 1 to 5: Settable only in device controller mode
	ACLRM	Auto buffer clear	Pipes 1 to 9: Settable
	SQCLR	Sequence clear	Clears the data toggle bit
	SQSET	Sequence set	Sets the data toggle bit
	SQMON	Sequence monitor	Monitors the data toggle bit
	PBUSY	Pipe busy status	—
	PID	Response PID	See section 37.3.4.6. Response PID.
PIPEnTRE	TRENB	Transaction counter enable	Pipes 1 to 5: Settable
	TRCLR	Current transaction counter clear	Pipes 1 to 5: Settable
PIPEnTRN	TRNCNT	Transaction counter	Pipes 1 to 5: Settable

37.3.4.1 Pipe Control Register Switching Procedures

The following bits in the pipe control registers can be changed only when USB communication is prohibited (PID = NAK).

Do not change the following registers and bits when USB communication is enabled (PID = BUF):

- Bits in DCPCFG and DCPMAXP

- SQCLR and SQSET bits in DCPCTR
- Bits in PIPECFG, PIPEMAXP, and PIPEPERI
- ATREPM, ACLRM, SQCLR, and SQSET bits in PIPEnCTR
- Bits in PIPEnTRE and PIPEnTRN

To set these bits when USB communication is enabled (PID = BUF):

1. A request to change the bits in the pipe control register occurs.
2. Set the PID[1:0] bits associated with the pipe to NAK.
3. Wait until the associated PBUSY bit clears to 0.
4. Set the bits in the pipe control register.

The following bits in the pipe control registers can be changed only when the selected pipe information is not set in the CURPIPE[3:0] bits in CFIFOSEL, D0FIFOSEL, and D1FIFOSEL.

Do not set the following registers when the CURPIPE[3:0] bits are set:

- Bits in DCPCFG and DCPMAXP
- Bits in PIPECFG, PIPEMAXP and PIPEPERI

To change pipe information, you must set the CURPIPE[3:0] bits in the port select registers to a pipe other than the one to be changed. For the DCP, the buffer must be cleared using the BCLR bit in the Port Control Register after the pipe information is changed.

37.3.4.2 Transfer Types

The PIPECFG.TYPE[1:0] bits specify the following transfer types for each pipe:

- DCP: No setting is necessary (fixed at control transfer)
- Pipes 1 and 2: Set to bulk or isochronous transfer
- Pipes 3 to 5: Set to bulk transfer
- Pipes 6 to 9: Set to interrupt transfer

37.3.4.3 Endpoint Number

The PIPECFG.EPNUM[3:0] bits are used to set the endpoint number for each pipe. The DCP is fixed at endpoint 0. The other pipes can be set from endpoint 1 to 15.

- DCP: No setting is necessary (fixed at endpoint 0)
- Pipes 1 to 9: Select and set the endpoint numbers from 1 to 15 so that the combination of the PIPECFG.DIR and EPNUM[3:0] bits is unique

37.3.4.4 Maximum Packet Size Setting

Specify the maximum packet size for each pipe in the DCPMAXP.MXPS[6:0] and PIPEMAXP.MXPS[9:0] bits. The DCP and pipes 1 to 5 can be set to any of the maximum pipe sizes defined in the USB 2.0 specification. For pipes 6 to 9, the maximum packet size is 64 bytes. Set the maximum packet size as follows before starting a transfer (PID = BUF):

- DCP: Set to 8, 16, 32, or 64
- Pipes 1 to 5: Set to 8, 16, 32, or 64 for bulk transfers
- Pipes 1 and 2: Set between 1 and 256 for isochronous transfers
- Pipes 6 to 9: Set between 1 and 64

37.3.4.5 Transaction Counter for Pipes 1 to 5 in the Receiving Direction

When the specified number of transactions is complete in the data packet receiving direction, the USBFS recognizes that the transfer ended. Two transaction counters are provided: one is the PIPEnTRN register, which specifies the number of transactions to be executed, and the other is the current counter, which internally counts the number of executed transactions. If the PIPECFG.SHTNAK bit is set to 1, when the current counter value matches the specified number of transactions, the associated PIPEnCTR.PID[1:0] bits are set to NAK and the subsequent transfer is disabled. The transactions can be counted again from the beginning by initializing the current counter of the transaction counter function through the PIPEnTRE.TRCLR bit. The data read from PIPEnTRN differs depending on the PIPEnTRE.TRENB setting as follows:

- The TRENB bit = 0: Specified transaction counter value can be read
- The TRENB bit = 1: Current counter value indicating the internally counted number of executed transactions can be read

The following constraints apply when working with the TRCLR bit:

- If the transactions are being counted and PID = BUF, the current counter cannot be cleared
- If there is any data left in the buffer, the current counter cannot be cleared

37.3.4.6 Response PID

Specify the response PID for each pipe in the PID[1:0] bits in DCPCTR and PIPEnCTR. This section describes the USBFS operation with different response PID settings.

(1) Software response PID settings in host controller mode

Select the response PID to specify the execution of transactions as follows:

- NAK setting: Using pipes is disabled and no transactions are executed
- BUF setting: Transactions are executed based on the FIFO buffer state:
  - OUT direction: An OUT token is issued if the FIFO buffer contains transmit data.
  - IN direction: An IN token is issued if the FIFO buffer is not full and can receive data.
- STALL setting: Using pipes is disabled and no transactions are executed

Note: Use the DCPCTR.SUREQ bit to execute setup transactions for the DCP.

(2) Software response PID settings in device controller mode

Select the response PID to respond as follows to transactions from the host:

- NAK setting: A NAK response is returned to all generated transactions
- BUF setting: A response is returned to transactions based on the FIFO buffer
- STALL setting: A STALL response is returned to all generated transactions

Note: For setup transactions, an ACK response is always returned, regardless of the PID[1:0] bits setting, and the USB request is stored in the register.

Sections (3) and (4) describe situations in which the USBFS writes to the PID[1:0] bits because of specific transaction results.

(3) Hardware response PID settings in host controller mode

- NAK setting: PID = NAK is set in the following cases, and issuing of tokens is automatically stopped:
  - When a non-isochronous transfer is performed and an NRDY interrupt is generated

    (For details, see section 37.3.3.2. NRDY Interrupt.)
  - If a short packet is received when the PIPECFG.SHTNAK bit is set to 1 for bulk transfers
  - If transaction counting ends when the SHTNAK bit is set to 1 for bulk transfers
- BUF setting: The USBFS does not write this setting.
- STALL setting: PID = STALL is set in the following cases, and issuing of tokens is automatically stopped:
  - When STALL is received in response to a transmitted token
  - When a received data packet exceeds the maximum packet size

(4) Hardware response PID settings in device controller mode

- NAK setting: PID = NAK is set in the following cases, and a NAK response is returned to transactions:
  - When the setup token is received normally (DCP only)
  - If transaction counting ends or a short packet is received when the PIPECFG.SHTNAK bit is set to 1 for bulk transfers
- BUF setting: There is no BUF writing by the USBFS.
- STALL setting: PID = STALL is set in the following cases, and a STALL response is returned to transactions:
  - When a received data packet exceeds the maximum packet size
  - When a control transfer sequence error is detected (DCP only)

37.3.4.7 Data PID Sequence Bit

The USBFS automatically toggles the sequence bit in the data PID when data is transferred successfully in the control transfer data stage, bulk transfer, and interrupt transfer. The sequence bit of the next data PID to be transmitted can be confirmed with the SQMON bit in DCPCTR and PIPEnCTR. When data is transmitted, the sequence bit toggles on ACK handshake reception. When data is received, the sequence bit toggles on ACK handshake transmission. The SQCLR bit in DCPCTR and the SQSET bit in PIPEnCTR can be used to change the data PID sequence bit.

In device controller mode when control transfers are used, the USBFS automatically sets the sequence bit for stage transitions. DATA1 is returned when the setup stage ends. The sequence bit is not referenced and PID = DATA1 is returned in the status stage. Therefore, no software settings are required. However, in host controller mode when control transfers are used, the sequence bit must be set by software for the stage transitions.

For ClearFeature requests for transmission or reception, the data PID sequence bit must be set by software in both host and device controller modes.

37.3.4.8 Response PID = NAK Function

The USBFS provides a function for disabling pipe operation (PID response = NAK) when the final data packet of a transaction is received. The USBFS automatically distinguishes this based on reception of a short packet or the transaction counter. Enable this function by setting the PIPECFG.SHTNAK bit to 1.

When the double buffer mode is being used for the FIFO buffer, using this function enables reception of data packets in transfer units. If pipe operation is disabled, the software must enable the pipe again (PID response = BUF).

The response PID = NAK function can be used only for bulk transfers.

37.3.4.9 Auto Response Mode

For bulk transfer pipes (1 to 5), when the PIPEnCTR.ATREPM bit is set to 1, a transition is made to auto response mode. During an OUT transfer (PIPECFG.DIR = 0), OUT-NAK mode is invoked, and during an IN transfer (DIR = 1), null auto response mode is invoked.

37.3.4.10 OUT-NAK Mode

For bulk OUT transfer pipes, NAK is returned in response to an OUT token, and an NRDY interrupt is output when the PIPEnCTR.ATREPM bit is set to 1. To transition from normal mode to OUT-NAK mode, specify OUT-NAK mode while pipe operation is disabled (PID[1:0] = 00b for NAK response). Next enable pipe operation (PID[1:0] = 01b for BUF response), on which OUT-NAK mode becomes valid. If an OUT token is received immediately before pipe operation is disabled, the token data is normally received, and an ACK is returned to the host.

To transition from OUT-NAK mode to normal mode, cancel OUT-NAK mode while pipe operation is disabled (NAK). Next enable pipe operation (BUF). In normal mode, reception of OUT data is enabled.

37.3.4.11 Null Auto Response Mode

For bulk IN transfer pipes, zero-length packets are continuously transmitted when the PIPEnCTR.ATREPM bit is set to 1.

To transition from normal mode to null auto response mode, specify null auto response mode while pipe operation is disabled (response PID = NAK). Next enable pipe operation (response PID = BUF) on which null auto response mode becomes valid. Before setting null auto response mode, check that PIPEnCTR.INBUFM = 0, because the mode can be set only when the buffer is empty. If the INBUFM bit is 1, empty the buffer using the PIPEnCTR.ACLRM bit. Do not write data from the FIFO port while a transition to null auto response mode is being made.

To transition from null auto response mode to normal mode, keep pipe operation disabled (response PID = NAK) for the period of the zero-length packet transmission (about 10 µs) before canceling the null auto response mode. In normal mode, data can be written from the FIFO port, so packet transmission to the host is enabled by enabling pipe operation (response PID = BUF).

37.3.5 FIFO Buffer

The USBFS provides a FIFO buffer for data transfers, and it manages the memory area used for each pipe. The FIFO buffer has two states depending on whether the access right is assigned to the system (CPU side) or the USBFS (SIE side).

(1) Buffer status

Table 37.20 and Table 37.21 show the buffer status in the USBFS. The FIFO buffer status can be confirmed using the DCPCTR.BSTS and PIPEnCTR.INBUFM bits. The transfer direction for the FIFO buffer can be specified in either the PIPECFG.DIR or CFIFOSEL.ISEL bit (when DCP is selected).

The INBUFM bit is valid for pipes 1 to 5 in the transmitting direction.

When a transmitting pipe uses double buffering, the software can read the BSTS bit to monitor the FIFO buffer status on the CPU side and the INBUFM bit to monitor the FIFO buffer status on the SIE side. When write access to the FIFO port by the CPU or DMA/DTC is slow and the buffer empty status cannot be determined using the BEMP interrupt, the software can use the INBUFM bit to confirm the end of transmission.

Table 37.20 Buffer status indicated in the BSTS bit

ISEL or DIR	BSTS	FIFO buffer status
0 (receiving direction)	0	There is no received data, or data is being received. Reading from the FIFO port is disabled.
0 (receiving direction)	1	There is received data, or a zero-length packet is received. Reading from the FIFO port is allowed. When a zero-length packet is received, reading is not possible and the buffer must be cleared.
1 (transmitting direction)	0	Transmission has not completed. Writing to the FIFO port is disabled.
1 (transmitting direction)	1	Transmission is complete. CPU write is allowed.

Table 37.21 Buffer status indicated in the INBUFM bit

DIR	INBUFM	FIFO buffer status
0 (receiving direction)	Invalid	Invalid
1 (transmitting direction)	0	Transmission is complete. There is no data waiting to be transmitted.
1 (transmitting direction)	1	The FIFO port has written data to the buffer. There is data to be transmitted.

37.3.6 FIFO Buffer Clearing

Table 37.22 shows the methods for clearing the FIFO buffer. The FIFO buffer can be cleared using BCLR bit in the port control register, DnFIFOSEL.DCLRM, or the PIPEnCTR.ACLRM bit.

Single or double buffering can be selected for pipes 1 to 5 in the PIPECFG.DBLB bit.

Table 37.22 Buffer clearing methods (1 of 2)

FIFO buffer clearing mode	Clearing FIFO buffer on the CPU side	Mode for automatically clearing the FIFO buffer after reading the specified pipe data	Auto buffer clear mode for discarding all received packets
Register used	CFIFOCTR DnFIFOCTR	DnFIFOSEL	PIPEnCTR

Table 37.22 Buffer clearing methods (2 of 2)

FIFO buffer clearing mode	Clearing FIFO buffer on the CPU side	Mode for automatically clearing the FIFO buffer after reading the specified pipe data	Auto buffer clear mode for discarding all received packets
Bit used	BCLR	DCLRM	ACLRM
Clearing condition	Cleared by writing 1	1: Mode valid 0: Mode invalid	1: Mode valid 0: Mode invalid

(1) Auto buffer clear mode function

The USBFS discards all received data packets if the PIPEnCTR.ACLRM bit is set to 1. If a correct data packet is received, the ACK response is returned to the host controller. The auto buffer clear mode function can only be set in the FIFO buffer reading direction.

Setting the ACLRM bit to 1 and then to 0 clears the FIFO buffer of the selected pipe regardless of the access direction. An access cycle of at least 100 ns is required for the internal hardware sequence processing between ACLRM = 1 and ACLRM = 0.

37.3.7 FIFO Port Functions

Table 37.23 shows the settings for the FIFO port functions. In write access, writing data until the maximum packet size is reached automatically enables transmission of the data. To enable transmission before the maximum packet size is reached, set the BVAL flag in the port control register to end writing. To send a zero-length packet, use the BCLR bit to clear the buffer, and then set the BVAL flag to end writing.

In reading, reception of new packets is automatically enabled when all data is read. Data cannot be read when a zero-length packet is received (DTLN[8:0] = 0), so the buffer must be cleared with the BCLR bit. The length of the receive data can be confirmed in the DTLN[8:0] bits in the port control register.

Table 37.23 FIFO port function settings

Register name	Bit name	Description
CFIFOSEL, DnFIFOSEL (n = 0, 1)	RCNT	Selects DTLN[11:0] read mode
	REW	FIFO buffer rewind (re-read, rewrite)
	DCLRM	Automatically clears receive data for a specified pipe after the data is read (only for DnFIFO)
	DREQE	Enables DMA/DTC transfers (only for DnFIFO)
	MBW	FIFO port access bit width
	BIGEND	Selects FIFO port endian
	ISEL	FIFO port access direction (only for DCP)
	CURPIPE	Selects the current pipe
CFIFOCTR, DnFIFOCTR (n = 0, 1)	BVAL	Ends writing to the FIFO buffer
	BCLR	Clears the FIFO buffer on the CPU side
	DTLN	Checks the length of receive data

(1) FIFO port selection

Table 37.24 shows the pipes that can be selected with the different FIFO ports. The pipe to be accessed must be selected in the CURPIPE[3:0] bits in the port select register. After the pipe is selected, the software must check whether the written value can be read correctly from the CURPIPE[3:0] bits. (If the previous pipe number is read, it indicates that the USBFS is modifying the pipe.) Next, the software checks that the FRDY bit in the port control register is 1.

In addition, the software must specify the bus width to be accessed in the MBW bit in the port select register. The FIFO buffer access direction conforms to the PIPECFG.DIR setting. For the DCP only, the ISEL bit in the port select register determines the direction.

Table 37.24 FIFO port access by pipe

Pipe	Access method	Ports that can be used
DCP	CPU access	CFIFO port register
Pipes 1 to 9	CPU access	● CFIFO port register ● D0FIFO/D1FIFO port register
	DMA/DTC access	D0FIFO/D1FIFO port register

(2) REW bit

It is possible to temporarily stop access to a pipe currently being accessed, access a different pipe, and then continue processing for the first pipe again. The REW bit in the port select register is used for this processing.

If a pipe is selected in the CURPIPE[3:0] bits in the port select register with the REW bit set to 1, the pointer used for reading from and writing to the FIFO buffer is reset, and reading or writing can be carried out from the first byte. If a pipe is selected with 0 set for the REW bit, data can be read and written in continuation from the previous selection, without the pointer being reset.

To access the FIFO port, the software must check that the FRDY bit in the port control register is 1 after selecting a pipe.

37.3.8 DMA Transfers (D0FIFO and D1FIFO Ports)

(1) Overview of DMA transfers

For pipes 1 to 9, the FIFO port can be accessed using the DMAC. When buffer access for a pipe targeted for DMA transfer is enabled, a DMA transfer request is issued.

Select the unit of transfer to the FIFO port in the DnFIFOSEL.MBW bit, and select the pipe targeted for the DMA transfer in the DnFIFOSEL.CURPIPE[3:0] bits. Do not change the selected pipe during the DMA transfer.

(2) DnFIFO auto clear mode (D0FIFO and D1FIFO port reading direction)

If 1 is set in the DnFIFOSEL.DCLRM bit, the USBFS automatically clears the FIFO buffer of the selected pipe when reading of data from the FIFO buffer is complete.

Table 37.25 shows the packet reception and FIFO buffer clearing processing by software for each of the settings.
As shown in the table, the buffer clearing conditions depend on the value set in the PIPECFG.BFRE bit. Using the DnFIFOSEL.DCLRM bit eliminates the need for the buffer to be cleared by software in any situation that requires buffer clearing. This enables DMA transfers without involving software.

The DnFIFO auto clear mode can only be set in the FIFO buffer reading direction.

Table 37.25 Packet reception and FIFO buffer clearing processing by software

Buffer status when packet is received	Register setting			
	DCLRM = 0		DCLRM = 1	
	BFRE = 0	BFRE = 1	BFRE = 0	BFRE = 1
Buffer full	No clearing required	No clearing required	No clearing required	No clearing required
Zero-length packet reception	Clearing required	Clearing required	No clearing required	No clearing required
Normal short packet reception	No clearing required	Clearing required	No clearing required	No clearing required
Transaction count end	No clearing required	Clearing required	No clearing required	No clearing required

37.3.9 Control Transfers Using the DCP

The DCP is used for data transfers in the control transfer data stage. The FIFO buffer of the DCP is a 64-byte single buffer with a fixed area for both control reads and control writes. The FIFO buffer can be accessed only through the CFIFO port.

37.3.9.1 Control Transfers in Host Controller Mode

(1) Setup stage

The USQREQ, USBVAL, USBINDX, and USBLENG registers are used to transmit USB requests for setup transactions. Writing the setup packet data to the registers and then writing 1 to the DCPCTR.SUREQ bit transmits the specified data for the setup transaction. On completion of the transaction, the SUREQ bit clears to 0. Do not change these USB request registers while SUREQ = 1.

When an attached function device is detected, the software must issue the first setup transaction for the device using this sequence with the DCPMAXP.DEVSEL[3:0] bits cleared to 0 and the DEVADD0.USBSPD[1:0] bits set appropriately.

When an attached function device is shifted to the Address state, the software must issue setup transactions using this sequence with the assigned USB address set in the DEVSEL[3:0] bits and the bits in DEVADDn corresponding to the specified USB address set appropriately. For example, when PIPEMAXP.DEVSEL[3:0] = 0010b, make appropriate settings in DEVADD2. When PIPEMAXP.DEVSEL[3:0] = 0101b, make appropriate settings in DEVADD5.

When the setup transaction data is sent, an interrupt request is generated based on the response from the peripheral device (SIGN or SACK bit in INTSTS1). This interrupt request allows the software to check the setup transaction result.

A DATA0 data packet (USB request) for the setup transaction is always transmitted regardless of the status of the DCPCTR.SQMON bit.

(2) Data stage

The data stage is used to transfer data using the DCP FIFO buffer.

Before accessing the DCP FIFO buffer, specify the access direction in the CFIFOSEL.ISEL bit. Specify the transfer direction in the DCPCFG.DIR bit.

For the first data packet of the data stage, the data PID must be transferred as DATA1. Set data PID = DATA1 in the DCPCTR.SQSET bit and set the PID bits = BUF. Completion of data transfer is detected using the BRDY or BEMP interrupt.

For control write transfers, when the number of data bytes to be sent is an integer multiple of the maximum packet size, the software must send a zero-length packet at the end.

(3) Status stage

The status stage is used for zero-length packet data transfers in the reverse direction of the data stage. As in the data stage, data is transfered using the DCP FIFO buffer. Transactions are executed using the same procedure as the data stage.

Data packets in the status stage must be transmitted and received with the data PID set to DATA1 using the DCPCTR.SQSET bit.

When a zero-length packet is received, check the receive-data length in the CFIFOCTR.DTLN[8:0] bits after a BRDY interrupt is generated, and then clear the FIFO buffer using the BCLR bit.

37.3.9.2 Control Transfers in Device Controller Mode

(1) Setup stage

The USBFS sends an ACK response to a normal setup packet for the USBFS. The USBFS operates in the setup stage as follows:

On receiving a new setup packet, the USBFS sets the following bits:

- Sets the INTSTS0.VALID bit to 1
- Sets the DCPCTR.PID[1:0] bits to NAK
- Sets the DCPCTR.CCPL bit to 0

When the USBFS receives a data packet following a setup packet, it stores the USB request parameters in USBREQ, USBVAL, USBINDX, and USBLENG.

Before performing the response processing for a control transfer, set the VALID flag to 0. When the VALID bit = 1, PID = BUF cannot be set, and the data stage cannot be terminated.

Using the VALID bit function, the USBFS can suspend a request being processed when it receives a new USB request during a control transfer and return a response to the latest request.

In addition, the USBFS automatically detects the direction bit (bmRequestType bit [8]) and the request data length (wLength) in the received USB request. It distinguishes between control read transfers, control write transfers, and no-data control transfers, and it controls stage transitions. For an incorrect sequence, a sequence error occurs in the control transfer stage transition interrupt, and the interrupt is reported to the software. For a diagram of the stage control by the USBFS, see Figure 37.15.

(2) Data stage

The DCP must be used to execute data transfers for received USB requests. Before accessing the DCP FIFO buffer, specify the access direction in the CFIFOSEL.ISEL bit.

If the transfer data is larger than the size of the DCP FIFO buffer, execute the data transfer using the BRDY interrupt for control write transfers and the BEMP interrupt for control read transfers.

(3) Status stage

Control transfers are terminated by setting the DCPCTR.CCPL bit to 1 while the DCPCTR.PID[1:0] bits are set to BUF.

After this setting is made, the USBFS automatically executes the status stage based on the data transfer direction determined at the setup stage. The procedure is as follows:

- For control read transfers

  The USBFS receives a zero-length packet from the USB host and transmits an ACK response.
- For control write transfers and no-data control transfers

  The USBFS transmits a zero-length packet and receives an ACK response from the USB host.

(4) Control transfer auto response function

The USBFS automatically responds to a correct SET_ADDRESS request. If any of the following errors occurs in the SET_ADDRESS request, a response from the software is necessary.

- bmRequestType is not 0x00: Any transfer other than a control write transfer
- wIndex is not 0x00: Request error
- wLength is not 0x00: Any transfer other than a no-data control transfer
- wValue is larger than 0x7F: Request error
- INTSTS0.DVSQ[2:0] are 011b (Configured state): Control transfer of a device state error

For all requests other than the SET_ADDRESS request, a response is required from the corresponding software.

37.3.10 Bulk Transfers (Pipes 1 to 5)

The FIFO buffer usage (single/double buffer setting) is configurable for bulk transfers. The USBFS provides the following functions for bulk transfers:

- BRDY interrupt function (PIPECFG.BFRE bit), see section 37.3.3.1. BRDY Interrupt
- Transaction count function (PIPEnTRE.TRENB, TRCLR, and PIPEnTRN.TRNCNT[15:0] bits), see section 37.3.4.5. Transaction Counter for Pipes 1 to 5 in the Receiving Direction
- Response PID = NAK function (PIPECFG.SHTNAK bit), see section 37.3.4.8. Response PID = NAK Function
- Auto response mode (PIPEnCTR.ATREPM bit), see section 37.3.4.9. Auto Response Mode

37.3.11 Interrupt Transfers (Pipes 6 to 9)

In device controller mode, the USBFS performs interrupt transfers based on the timing dictated by the host controller.

In host controller mode, the software can set the timing for issuing tokens using the interval counter.

37.3.11.1 Interval Counter for Interrupt Transfers in Host Controller Mode

Specify the transaction interval for interrupt transfers in the PIPEPERI.IITV[2:0] bits. The USBFS issues interrupt transfer tokens based on this interval.

(1) Counter initialization

The USBFS initializes the interval counter under the following conditions:

- Power-on reset

  This initializes the IITV[2:0] bits.

- FIFO buffer initialization using the PIPEnCTR.ACLRM bit:

  This does not initialize the IITV[2:0] bits, but does initialize the count value. Setting the PIPEnCTR.ACLRM bit to 0 starts counting from the value set in IITV[2:0].

The interval counter is not initialized in the following case:

- USB bus reset or USB suspended

  The IITV[2:0] bits are not initialized. Setting 1 to the DVSTCTR0.UACT bit starts counting from the value saved before entering the USB bus reset state or USB suspend state.

(2) Operation when tokens cannot be transmitted or received even on token generation

No token is generated in the following cases even at token generation time. In these cases, the USBFS tries to execute the transaction in the next interval.

- When the PID is set to NAK or STALL
- When the FIFO buffer is full at token transmit time in the receiving (IN) direction
- When there is no data to be transmitted in the FIFO buffer at token transmit time in the transmitting (OUT) direction

37.3.12 Isochronous Transfers (Pipes 1 and 2)

The USBFS provides the following functions for isochronous transfers:

- Notification of isochronous transfer error
- Interval counter specified in the PIPEPERI.IITV[2:0] bits
- Isochronous IN transfer data setup control (IDLY function)
- Isochronous IN transfer buffer flush function specified in the PIPEPERI.IFIS bit

37.3.12.1 Error Detection in Isochronous Transfers

The USBFS provides a function for detecting the errors described in this section, so that when errors occur in isochronous transfers, they can be controlled by software. Table 37.26 and Table 37.27 show the priority order for errors detected by the USBFS and the associated interrupts.

PID errors

- The PID value of the received packet is invalid.

CRC errors and bit stuffing errors

- A CRC error is found in a received packet or the bit stuffing is illegal.

Maximum packet size exceeded

- The data size of the received packet exceeds the specified maximum packet size.

Overrun and underrun errors

In host controller mode:

- The FIFO buffer is full at token transmit time in the IN (receiving) direction
- There is no data to be sent in the FIFO buffer at token transmit time in the OUT (transmitting) direction

In device controller mode:

- There is no data to be sent in the FIFO buffer at token receive time in the IN (transmitting) direction
- The FIFO buffer is full at token receive time in the OUT (receiving) direction

Interval errors

In device controller mode, the following cases are treated as an interval error:

- Failure to receive an IN token in the interval frame during an isochronous IN transfer
- Failure to receive an OUT token in the interval frame during an isochronous OUT transfer

Table 37.26 Error detection for token transmission and reception

Detection priority	Error	Generated interrupt and status
1	PID error	No interrupts are generated in either host or device controller mode (ignored as a corrupted packet)
2	CRC or bit stuffing error	No interrupts are generated in either host or device controller mode (ignored as a corrupted packet)
3	Overrun or underrun error	An NRDY interrupt is generated to set the FRMNUM.OVRN bit to 1 in both host and device controller modes. In device controller mode, a zero-length packet is transmitted in response to an IN token. No data packets are received in response to OUT token.
4	Interval error	An NRDY interrupt is generated in device controller mode. No interrupt is generated in host controller mode.

Table 37.27 Error detection for data packet reception

Detection priority	Error	Generated interrupt and status
1	PID error	No interrupts are generated (ignored as a corrupted packet)
2	CRC or bit stuffing error	An NRDY interrupt is generated and the FRMNUM.CRCE bit sets to 1 in both host and device controller modes
3	Maximum packet size exceeded error	A BEMP interrupt is generated and the PID[1:0] bits set to STALL in both host and device controller modes

37.3.12.2 DATA-PID

In device controller mode, the USBFS responds to a received PID as follows:

(1) IN direction

- DATA0: Transmitted as data packet PID
- DATA1: Not transmitted
- DATA2: Not transmitted
- mDATA: Not transmitted

(2) OUT direction

- DATA0: Received normally as data packet PID
- DATA1: Received normally as data packet PID
- DATA2: Packets ignored
- mDATA: Packets ignored

37.3.12.3 Interval Counter

The isochronous transfer interval can be set in the PIPEPERI.IITV[2:0] bits. In device controller mode, the interval counter enables the functions as shown in Table 37.28. In host controller mode, the USBFS generates the token issuance timing, and the interval counter operation is the same as that for interrupt transfers.

Table 37.28 Interval counter functions in device controller mode

Transfer direction	Function	Conditions for detection
IN	Transmit buffer flush	Failure to receive an IN token successfully in the interval frame during an isochronous IN transfer.
OUT	Notification of no reception of token	Failure to receive an OUT token successfully in the interval frame during an isochronous OUT transfer.

The interval count is performed when an SOF is received or for interpolated SOFs, so the isochronism can be maintained even if an SOF is corrupt. The frame interval can be set to 2^{IITV} frames.

(1) Counter initialization in device controller mode

The USBFS initializes the interval counter under the following conditions:

- Power-on reset

  This initializes the PIPEPERI.IITV[2:0] bits.
- FIFO buffer initialization using the ACLRM bit

  This does not initialize the IITV[2:0] bits, but does initialize the count value.

After the interval counter is initialized, the interval count starts under one of the following conditions when a packet is transferred successfully:

- An SOF is received after data is transmitted in response to an IN token when PID = BUF
- An SOF is received after data is received in response to an OUT token when PID = BUF

The interval counter is not initialized in the following conditions:

- When the PID[1:0] bits are set to NAK or STALL

  This does not stop the interval timer. The USBFS attempts the transaction in the next interval.
- When the USB bus is reset or USBFS is suspended

  This does not initialize the IITV[2:0] bits. When an SOF is received, the interval counter starts counting from the value set before SOF was received.

(2) Interval counting and transfer control in host controller mode

The USBFS controls the interval between token issuance operations based on the PIPEPERI.IITV[2:0] bits settings. Specifically, the USBFS issues a token for a selected pipe once every 2^{IITV} frames.

The USBFS starts counting the token issuance interval at the frame following the frame in which the PID[1:0] bits are set to BUF by software.

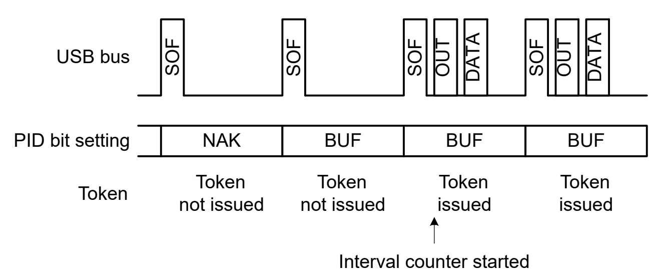


Figure 37.16 Token issuance when IITV = 0

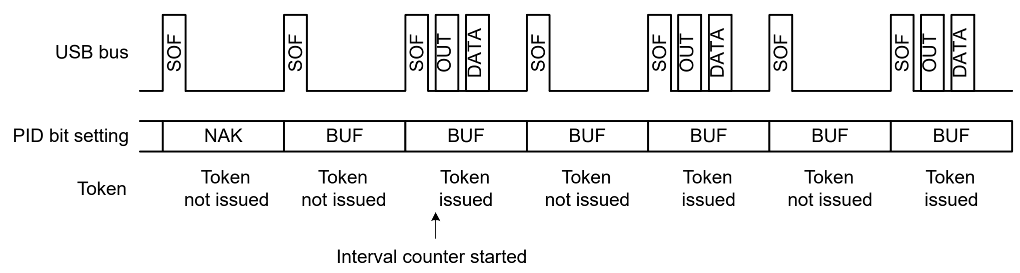


Figure 37.17 Token issuance when IITV = 1

When the selected pipe is set for isochronous transfers, the USBFS performs the following operation in addition to controlling the token issuance interval. The USBFS issues a token even when the NRDY interrupt generation condition is satisfied.

When the selected pipe is for isochronous IN transfers

The USBFS generates an NRDY interrupt when the USBFS issues an IN token but does not successfully receive a packet from a peripheral device (no response or packet error).

The USBFS sets the FRMNUM.OVRN bit to 1, generating an NRDY interrupt, when the time to issue an IN token occurs while the USBFS cannot receive data because the FIFO buffer is full, because the CPU or DMAC/DTC is too slow in reading data from the FIFO buffer.

When the selected pipe is for isochronous OUT transfers

The USBFS sets the OVRN bit to 1, generating an NRDY interrupt and transmitting a zero-length packet, when the time to issue an OUT token comes while there is no data to be transmitted in the FIFO buffer, or because the CPU or DMAC/DTC is too slow in writing data to the FIFO buffer.

The token issuance interval is reset on any of the following conditions:

- When the USBFS is reset through a reset pin

  This initializes the IITV[2:0] bits.
- When the PIPEnCTR.ACLRM bit is set to 1 by software

(3) Interval counting and transfer control in device controller mode

When the selected pipe is for isochronous OUT transfers

The USBFS generates an NRDY interrupt when it fails to receive a data packet within the interval set in the PIPEPERI.IITV[2:0] bits.

The USBFS also generates an NRDY interrupt when it fails to receive data because of a CRC error or other errors contained in the data packet or because the FIFO buffer is full.

The NRDY interrupt is generated on SOF packet reception. Even if the SOF packet is corrupted, internal interpolation allows the interrupt to be generated when the SOF packet is received. However, when the IITV bits are set to a value other than 0, the USBFS generates an NRDY interrupt on receiving an SOF packet for every interval after interval counting starts.

When the PID[1:0] bits are set to NAK by software after starting the interval timer, the USBFS does not generate an NRDY interrupt on receiving an SOF packet.

The timing for starting interval counting depend on the IITV[2:0] setting as follows:

- When the IITV[2:0] bits = 0:

  Interval counting starts at the next frame after the software changes the PID[1:0] bits of the selected pipe to BUF.
- When the IITV[2:0] bits ≠ 0:

  Interval counting starts on completion of successful reception of the first data packet after the PID[1:0] bits for the selected pipe are changed to BUF.

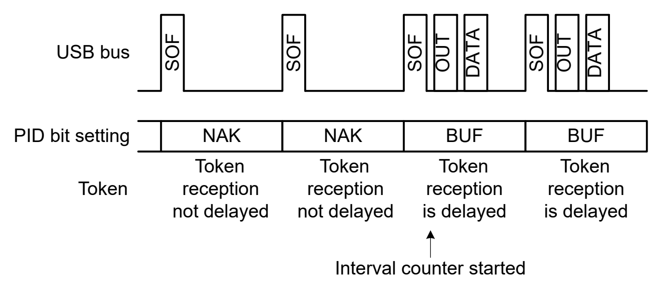


Figure 37.18 Relationship between frames and expected token reception when IITV[2:0] = 0

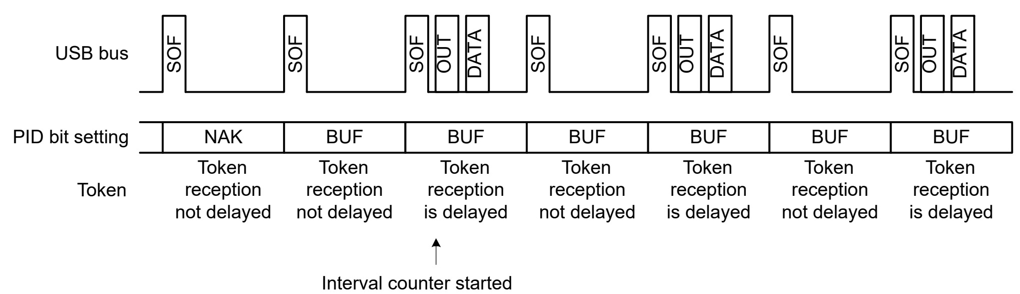


Figure 37.19 Relationship between frames and expected token reception when IITV[2:0] ≠ 0

When the selected pipe is for isochronous IN transfers

The PIPEPERI.IFIS bit must be 1 for this use case. When IFIS = 0, the USBFS transmits a data packet in response to a received IN token regardless of the PIPEPERI.IITV[2:0] setting.

When IFIS is 1 and there is data to be transmitted in the FIFO buffer, the USBFS clears the FIFO buffer when it fails to receive an IN token in the frame at the interval set in the IITV[2:0] bits.

The USBFS also clears the FIFO buffer when it fails to receive an IN token successfully because of a bus error, such as a CRC error, contained in the IN token.

The FIFO buffer is cleared on SOF packet reception. Even if the SOF packet is corrupted, the internal interpolation allows the FIFO buffer to be cleared when the SOF packet is received.

The timing to start interval counting depends on the IITV[2:0] setting, as with OUT transfers.

The interval is counted on any of the following conditions in device controller mode:

- When a hardware reset is applied to the USBFS (which also sets the IITV[2:0] bits to 000b)
- When the PIPEnCTR.ACLRM bit is set to 1 by software
- When the USBFS detects a USB bus reset

(4) Transmit data setup for isochronous transfers in device controller mode

With isochronous data transmission using the USBFS in device controller mode, after data is written to the FIFO buffer, a data packet can be transmitted in the first frame after the SOF packet is detected. This isochronous transfer transmit data setup function can identify the frame that started transmission.

When the double buffering is used, transmission is only enabled for the buffer where data writing was completed first, even after the data write to both buffers is complete. Accordingly, even if multiple IN tokens are received, only the one packet of FIFO buffer data is transmitted.

When the FIFO buffer is ready to transmit data when an IN token is received, the data is transferred and a normal response is returned. However, if the FIFO buffer cannot transmit data, a zero-length packet is transmitted and an underrun error occurs.

Figure 37.20 shows an example transmission using the isochronous transfer transmission data setup function when IITV = 0 (every frame) is set.

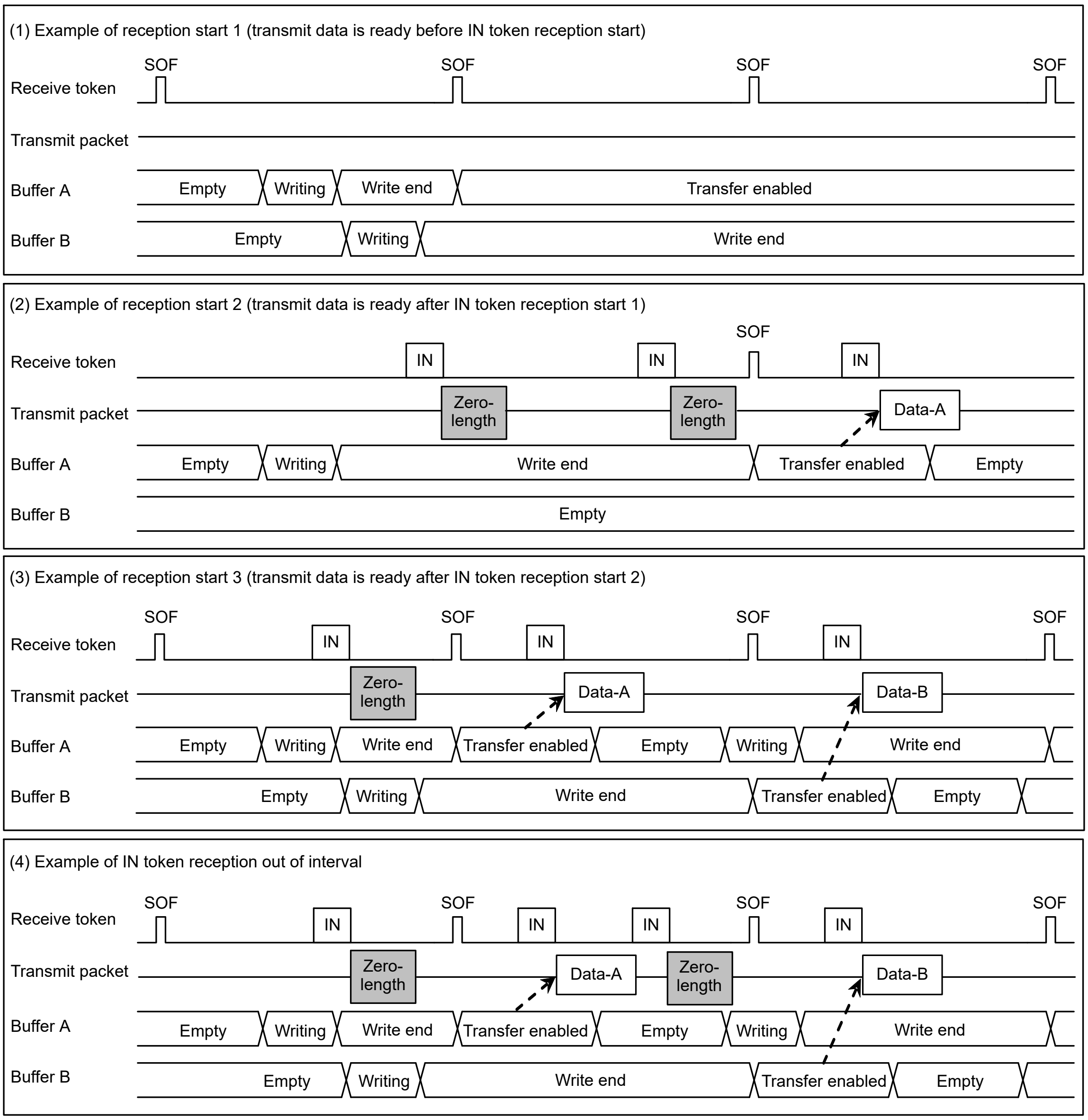


Figure 37.20 Example data setup operation

(5) Transmit buffer flush for isochronous transfers in device controller mode

In device controller mode during isochronous data transmission, if the USBFS receives an SOF packet for the next frame without receiving an IN token in the interval frame, it operates as if the IN token is corrupt and clears the buffer that is enabled for transmission, putting that buffer in the writing enabled state.

When double buffering is used and writing to both buffers is complete, the cleared FIFO buffer is assumed to be the one where the data was transmitted in the interval frame, and transmission is enabled for the FIFO buffer that was not cleared on SOF packet reception.

The timing of the buffer flush function depends on the PIPEPERI.IITV[2:0] setting as follows:

- When IITV = 0:

  The buffer flush operation starts from the first frame after the pipe is enabled.
- When IITV ≠ 0:

  The buffer flush operation starts after the first normal transaction.

Figure 37.21 shows an example buffer flush. When an unanticipated token is received before the interval frame, the USBFS sends the write data or a zero-length packet as an underrun error, depending on the data setup status.

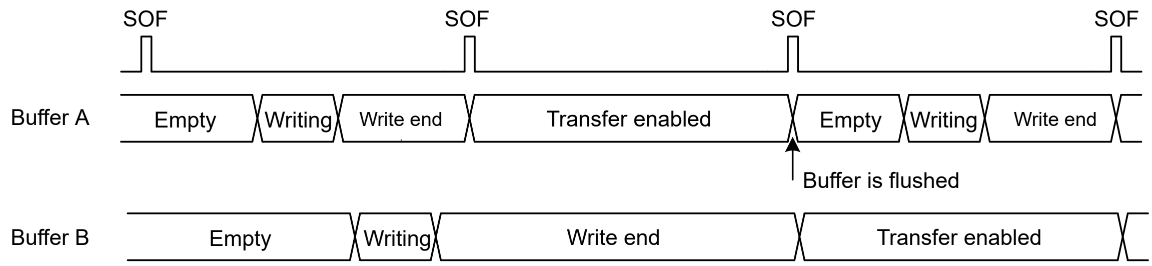


Figure 37.21 Example buffer flush operation

Figure 37.22 shows an example interval error occurrence. There are five types of interval errors, as shown in the figure. An interval error occurs at timing (A), and the buffer flush function is activated.

If an interval error occurs during an IN transfer, the buffer flush function is activated. If it occurs during an OUT transfer, an NRDY interrupt is generated. Use the FRMNUM.OVRN bit to distinguish between this and NRDY interrupts triggered by received packet errors and overrun errors.

For tokens that are shaded in the figure, responses are returned based on the FIFO buffer status.

- IN direction:
  - If the buffer is ready to transfer data, the data is transferred and a normal response is returned
  - If the buffer is not ready to transfer data, a zero-length packet is transmitted and an underrun error occurs
- OUT direction:
  - If the buffer is ready to receive data, the data is received and a normal response is returned
  - If the buffer is not ready to receive data, the received data is discarded and an overrun error occurs

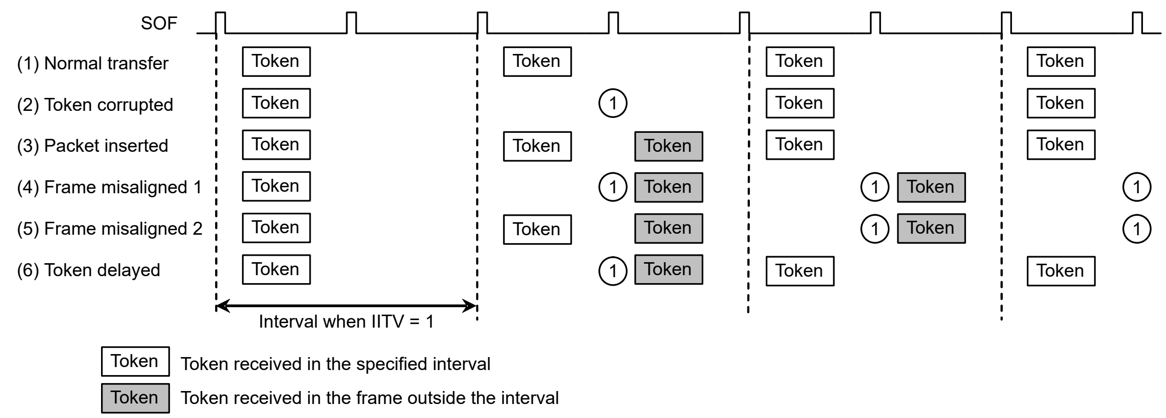


Figure 37.22 Example interval error occurrence when IITV = 1

37.3.13 SOF Interpolation Function

In device controller mode, if packet reception is disabled at intervals of 1 ms because the SOF packet is corrupted or missing, the USBFS interpolates the SOF. SOF interpolation begins when the USBE and SCKE bits in SYSCFG are set to 1 and an SOF packet is received.

The interpolation function is initialized under the following conditions:

- MCU reset
- USB bus reset
- Suspend state detection

The SOF interpolation operates as follows:

- The interpolation function is not activated until an SOF packet is received
- When the first SOF packet is received, interpolation is performed by counting 1 ms on the 48-MHz internal clock
- When the second and subsequent SOF packets are received, interpolation is performed at the previous reception interval
- Interpolation is not performed in the Suspend state or on reception of a USB bus reset

The USBFS supports the following functions controlled by SOF packet reception. These functions operate normally with SOF interpolation if the SOF packet is missing:

- Updating of the frame number
- SOFR interrupt timing
- Isochronous transfer interval count

If an SOF packet is missing during full-speed operation, the FRMNUM.FRNM[10:0] bits are not updated.

37.3.14 Pipe Schedule

37.3.14.1 Conditions for Generating Transactions

In host controller mode and when the DVSTCTR0.UACT bit is set to 1, the USBFS generates transactions under the conditions shown in Table 37.29.

Table 37.29 Conditions for generating transactions

Transaction	Conditions for generation				
	DIR	PID	IITV0	Buffer state	SUREQ
Setup	—*1	—*1	—*1	—*1	1 setting
Control transfer data stage, status stage, bulk transfer	IN	BUF	Invalid	Receive area exists	—*1
	OUT	BUF	Invalid	Transmit data exists	—*1
Interrupt transfer	IN	BUF	Valid	Receive area exists	—*1
	OUT	BUF	Valid	Transmit data exists	—*1
Isochronous transfer	IN	BUF	Valid	*2	—*1
	OUT	BUF	Valid	*3	—*1

Note 1. An em dash (—) in the table indicates that the condition is unrelated to the generating of tokens. “Valid” indicates that, for interrupt transfers and isochronous transfers, a transaction is generated only in transfer frames that are based on the interval counter. “Invalid” indicates that a transaction is generated regardless of the interval counter.
Note 2. This indicates that a transaction is generated regardless of whether there is a receive area. If there is no receive area, however, the received data is discarded.
Note 3. This indicates that a transaction is generated regardless of whether there is any data to be transmitted. If there is no data to be transmitted, however, a zero-length packet is transmitted.

37.3.14.2 Transfer Schedule

This section describes the transfer scheduling within a frame of the USBFS. After the USBFS sends an SOF, the transfer is carried out in the following sequence:

1. Execution of periodic transfers:

   A pipe is searched for in the order of pipe 1 → pipe 2 → pipe 6 → pipe 7 → pipe 8 → pipe 9, and then if there is a pipe for which an isochronous or interrupt transfer transaction can be generated, the transaction is generated.

2. Setup transactions for control transfers:

   The DCP is checked, and if a setup transaction is possible, it is sent.

3. Execution of bulk transfers, control transfer data stages, and control transfer status stages:

   A pipe is searched for in the order of DCP → pipe 1 → pipe 2 → pipe 3 → pipe 4 → pipe 5, and then if there is a pipe for which a transaction for a bulk transfer, a control transfer data stage, or a control transfer status stage can be generated, the transaction is generated.

   When a transaction is generated, processing moves to the next pipe transaction regardless of whether the response from the peripheral device is ACK or NAK. If there is time for transfer within the frame, step 3 is repeated.

37.3.14.3 Enabling USB Communication

Setting the DVSTCTR0.UACT bit to 1 initiates an SOF transmission, and transaction generation is enabled. Setting the UACT bit to 0 stops SOF transmission and the Suspend state is invoked. If the UACT setting is changed from 1 to 0, processing stops after the next SOF is sent.

37.4 Usage Notes

37.4.1 Settings for the Module-Stop State

USBFS operation can be disabled or enabled using Module Stop Control Register B (MSTPCRB). The USBFS is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see section 11, Low Power Mode.

37.4.2 Clearing the Interrupt Status Register on Canceling Software Standby Mode

Because the input buffer is always enabled in Software Standby mode, an unexpected interrupt might occur under the following conditions:

- When the interrupt is enabled in Normal mode
- When the interrupt is disabled in Software Standby mode
- When the input level of the pin that cancels software standby mode is changed in Software Standby mode

These conditions might cause the associated interrupt flag in the Interrupt Status Register to set unexpectedly. After the MCU cancels the Software Standby mode, the unexpected interrupt might be sent to the interrupt controller. To avoid this, always clear the INTSTS0 and INTSTS1 registers in the canceling sequence.

37.4.3 Clearing the Interrupt Status Register after Setting Up the Port Function

The input buffer is disabled before the PmnPFS.PSEL and PmnPFS.PMR port is set up, so the internal signal is fixed high or low. The input buffer is enabled after the port is set so that the external pin state is propagated to the MCU. An unexpected interrupt might occur at this time, causing the VBINT and OVRCR bits in INTSTS0 and INTSTS1, or other interrupt status flags to set to 1. To avoid a malfunction, always clear the INTSTS0 and INTSTS1 registers after setting up the port.

37.4.4 Setting Up the Port Function Before Setting USB Function

USB_DP, USB_DM pins are compatible with P814, P815 I/O ports. So before setting USBFS registers to enable USBFS function, set register bit P814PFS.PMR and P815PFS.PMR to 1.